

Neuro 210 Brain Plasticity

Numbered Transcript

NEUR 210 · Dr. David Eagleman

Contents

Lecture 1. The Malleable Mind

Lecture 2. Maps in the Brain

Lecture 3. New Sensory Frontiers

Lecture 4. Motor Mastery

Lecture 5. The Science of Skill

Lecture 6. Prediction Potentials

Lecture 7. Adapting with Age

Lecture 8. Learning and Memory

Lecture 1: The Malleable Mind

At a glance — Eagleman introduces the malleable mind and the brain's plasticity.

- 1.1 OK, how many of you have ever seen a baby zebra get born? Or a baby horse? How many of you seen something like that? OK, a few of you. All right, so what you might have noticed is that, in about 45 minutes, it wobbles to its little legs and it starts running around, right? But how many of you have ever seen a little baby Homo sapiens get born? OK, you might notice that they're not moving around in 45 minutes. It takes them a really long time to start walking around. So why is that?
- 1.2 Well, this is what we're going to talk about, because it turns out, what Mother Nature developed with humans was a trick, which is dropping the brain into the world half-baked and letting the world wire it up. And that's the notion that we're going to be talking about. What we're going to cover is a, is, we're going to get a deep understanding of what brain plasticity is and what the consequences are. OK, so my goal for this first lecture is to give you a sense of how to think about the brain. And then in successive lectures, we're going to dive into more detail.
- 1.3 And so when you look at the animal kingdom, what you find is plasticity everywhere. So if you have a dog, you know that you can teach the dog some tricks and it can learn a few things, but it's really limited. It's not able to do all of the things that we are able to do. And the reason for that is because of what this course is about. This thing is very special about humans. So when you think about the human brain, you can't think about it as hardware and you can't think about it as software. It's what I'm going to call "liveware." And the technical term for this that we use in the field is brain plasticity or neuroplasticity.
- 1.4 But I'll tell you the reason that I don't use that term all the time. Plasticity is because it was coined by William James because he was impressed with plastic manufacturing. That's where the word comes from. So you take plastic and you can melt this material down and make something like a toy out of it. And it'll hold that shape, it'll hold onto that shape, and that's what we love about plastic. And so, William James thought, "Look, with the brain, you're doing something, it has experiences, it changes the shape of the brain, and then it holds onto that shape." So he used the term plasticity.
- 1.5 But the fact is, we might be beyond being impressed with plastic toys at this point, because what you've got is a massive forest of neurons underneath the hood there. We'll be talking about some details about that. But every second of your life, this is changing. This is moving and growing. Every second of your life, your brain is changing. You've got neurons. Those are the specialized cell-type of the brain. They're seeking, they're plugging, they're changing their strength of connections. They're unplugging. And this is what's going on every moment for you. And so that's why I think we need a new term beyond plasticity.

- 1.6 Whether intentionally or not, the term plasticity, we usually think about it as, OK, something obtains a shape and it stays there. But if you think about something like a city—we happen to be in Miami right now, so take the city of Miami. You know, it's constantly evolving. It's constantly in flux. Cities always are. How they, where they build their buildings and their new roads, and how they're changing their immigration policies, and how they're deciding taxation and whatever. Cities are constantly evolving. And that's what's going on with your brain, too. That's why we're going to use the term liveware.
- 1.7 Now, one way that we can really see what is special about livewiring is to just look at an airplane window. So you're flying, you look out the window, you see a forest below you. What you can notice is that the creatures in the forest are doing the same thing that they have been doing for the last 100,000 years. I mean, they're just living their same lives and eating and mating and sleeping and whatever. But you arrive at your destination city, and it looks like a motherboard has risen from the Earth because of one species. There's one species that's doing something different from all the others.
- 1.8 And we have taken over every corner of the Earth, and we have gotten off the Earth, and we've invented skyscrapers and symphonies, and we've looked with ever increasing detail into the functioning of our own brain. So there's something really different about us. And the question is, what is it? We've outcompeted every species on the planet. If you take something like a Komodo dragon and move it from Indonesia to snowy Canada or something, it's not going to survive there. And so the question is, what is it about us that allows us to move all around? What makes humans stand out?
- 1.9 So, there are a few things that make us stand out. It has a little bit to do with thumbs and larynx and so on, but the bigger thing has to do with the structure of the human brain. So when we zoom in on the brains of our neighbors in the animal kingdom, one of the things we see is, it's not just the size that matters. One of the issues is that we've had this massive expansion of the cortex, which is the wrinkly outer bit on the outside. And one of the consequences of this is that we have much more distance in between the input and the output, and that allows many more possibilities. Computationally speaking, when we have some input, we don't have to just be reflexive. We can think through lots of different possibilities.
- 1.10 So if you throw some food in front of a rat, it's going to eat the food. But you throw food in front of a human, the human says, "Well, I'm on a diet," and so on. And so, we have lots of possibilities because we have so much cortex that allows us to generate a lot of rich possibilities. We also have a much larger prefrontal cortex. That's the bit behind the forehead there. And this is the part of the brain that allows you to generate what-ifs and to think about potential futures before you take an action. So part of the human standout is what it is because of these changes in brain anatomy, just this explosion of the cortex.

- 1.11** The other thing to note is that our brains are really big. Compared to our body size, we have unbelievably large brains. So the human brain is about 3 pounds, and it has 86 billion neurons. And the thing to note here is that every neuron in your head is as complicated as the city of Miami. Every neuron has the entire human genome in it. It's trafficking millions of proteins in very complicated cascades. And each one of these is connected to about 10,000 of its neighbors, which means what you're looking at is something like 200 trillion connections in the brain. So this is the kind of thing that bankrupts our language. We have no way to even talk about a system this size, except we know that this is you. All your hopes, your dreams, your aspirations, the agony, the ecstasy. It's all happening right there.
- 1.12** And how do we know that? It's because if you damage even a very small part of your brain, that can change you entirely. You know, if you damage another part of your body, like you hurt your pinky or something, you're sad about that, but you're no different as a person. But you damage an equivalently sized little chunk of brain, it can change your decision-making, your risk aversion, your capacity to name colors or name animals or understand music. Any of this stuff can change just based on very small damage to the brain. And that's how we know.
- 1.13** Given this number of connections in the brain, if you were to take just a cubic millimeter of brain tissue, you have as many connections there as you have stars in the Milky Way Galaxy, just in a cubic millimeter. So it's an enormously complex system. But what we are going to talk about in this course is that it's not just the number of parts that make the brain interesting, it's the way that those parts interact. So when you look at a textbook on the brain, if you just pick up a textbook on the brain, you'll see that everything is neatly mapped out. You've got this region devoted to this task, and this area exists for vision, and this for hearing, and that for decision-making and so on.
- 1.14** But what we're going to see is that this approach is actually very limited. And this is why I'll sometimes refer, I'll sometimes make an analogy of the brain as a globe. So when you're a young student and you learn about history, you think, OK, all the borders of the countries are there, and that's the way it has to be. But as you get older, if you become a historian, you realize, wow, there's a lot of happenstance involved in that. And if you re-ran history just a slightly different way, all these country borders would look very different. If this would-be king had died in infancy, or this corn pestilence hadn't happened, or this warship had sunk at the beginning of the battle or whatever, then everything would be different and the borders would be different.
- 1.15** That's the right way to think about the brain, too. It's a very fluid system, and sometimes we get used to the way that we see it. We think, "Oh, that must be the way it has to be." But it doesn't. And we're going to see many examples of that in this course. To understand why there's this dynamism in the system, we have to dive down to the level of the neurons. So we're going to spend some time here. Now, as a side note, when we look at videos like this, they're not terribly accurate. Can somebody tell

me why? Someone have a guess why this is not so accurate? The distance, right. It's because what these videos show is lots of space in between the neurons. And the reason they have to do that is because otherwise you wouldn't see anything. It would just be everything all smushed together.

- 1.16** When we look at pictures of neurons that have been stained so that you can see the neurons, we usually see things like this, which makes it look like there's distance between these. But the reason that we see this is because, when people come up with staining techniques for neurons, the chemicals that are really useful are those that impregnate just a few of the neurons and not the rest. So in other words, the staining chemical has a very low probability of getting into a neuron.
- 1.17** Why is that useful? Because then you can see a few of them. If you had a chemical that got into all the neurons, you wouldn't see anything at all because everything is packed so closely together. You can only resolve the distance in between neurons with electron microscopy. That's how packed everything is in the brain. So, back to this. It's actually like 100 times denser than what you see in a video like this. But I think this is important for us to look at videos like this because this is the right way to think about the brain, is think about it like a forest. You've got this unbelievably dense forest that goes on and on. I did a calculation watching this video of how long it would take you to get from the, across the human brain at this pace. So, give me a, give me a guess, how long it would take you to get across the human brain if you were moving at this pace through it.
- 1.18** One second, an hour, a week. 9.5 years is how long it would take you to get through the human brain. It's really stunning, right? That's how large a system we're talking about. And the thing to understand is that each one of these neurons has its own life. Each one of these neurons is like a person that's striving to survive and do the right things and make the right connections and have the right dating life and so on. That's what's going on here. And it would take—that's the size of the system that we're talking about here.
- 1.19** So, here's a dish that I'm just going to show you here. This is just seven or eight neurons plated out on a dish. And you can see the way that they scramble and they live and they compete and they move around. Instead of just seeing this with 7 or 8 neurons, try to picture this with 80 neurons or 800 or 8 billion or 80 billion neurons, to give you a sense of what is happening under the hood. OK.
- 1.20** Now, the other thing that I'm going to emphasize every lecture is that everything that you know and believe and think, all your memories, your culture, your family, your neighborhood, all of that is stored in here, is stored in the configuration of these neurons. And so, you know, you're adjusting the strengths of these connections. As I said, things unplug and replug. This is the central principle, though, that we're going to be talking about, is that brains absorb the world around them. So, your world is contained in those 3 pounds.

- 1.21** Now, in order to really understand this, we need to start by understanding how brains are really different from other analogies that we might use—for example, our digital computers. So brains are super different from your computer. They are extremely flexible in ways that I think will surprise you. I'll give you an example. So, some years ago there was a clinical case of a 44-year-old man who, normal IQ, had some leg pain. So he went to his doctor to figure out, "OK, what's going on with my leg?" So the doctor couldn't figure it out and he said, "Why don't we get a brain scan, just make sure everything looks normal?" So what I'm going to show you first is just what a normal brain scan looks like. So, what this is, is a slice down almost the middle of the brain and you're just seeing what a normal brain looks like.
- 1.22** The thing to pay attention to is the one that's labeled number three. That's just a space, a volume that's empty in the brain, called the lateral ventricle. It's not actually empty. You have fluid flowing through there, cerebrospinal fluid. Anyway, that's what a normal brain looks like. This is what this guy's brain looked like. The lateral ventricle had too much pressure in it. It's called hydrocephalus, and it had squished his brain up against the sides of the skull. Now, the interesting thing is, he had probably had this from the time he was a kid. So his brain was always squished up against the side of his skull. Nonetheless, he was married, he had a career, he had kids, he had a normal life. Now this is very weird material. This is not like the kind of stuff we know how to make in Silicon Valley. If I run over my laptop with my truck, it's not going to function. But liveware is a very different kind of beast.
- 1.23** So I'll give you another example of this. Where I grew up in Albuquerque, New Mexico, there was a young boy named Matthew who ended up having seizures, epileptic seizures. His parents tried to figure out what to do, and Matthew was having these seizures more and more commonly, to the point where it was multiple times an hour. And so his parents took him around to all the doctors. They ended up flying to other places. And what they finally figured out was that he had something called Rasmussen's encephalitis, which is a form of inflammation that affects an entire hemisphere of the brain.
- 1.24** And so the doctors finally told the parents, "Look, there's actually no choice here, except for one thing. There's only one thing that you can do here." Does anyone know what you do with encephalitis like this? It's worse than a lobotomy. What you do is what's called a hemispherectomy, where you remove an entire half of the brain. So for any of you who might have kids, you can imagine the horror for a parent having to take a child in for a surgery like this. But you know what? Hundreds of kids go through this every year, and they're perfectly fine. You can remove half the brain, and the kid is fine.
- 1.25** They tend to have a slight limp on the other side of their body, because this side of the brain controls that side of the body. But otherwise they can do math tests, vocabulary tests. They can get through life just fine. Now, this is quite stunning because, again, I can't take my cell phone and rip out half the circuitry and say, oh, it's still running fine.

So there's something very different about liveware, and that's the key. So that's by way of introduction.

- 1.26** Now, the second principle that I want to tell you about is about the, how to think about this in terms of nature and nurture when we're thinking about liveware. So I'm going to be talking this whole course about brain plasticity and the malleability of the circuitry and what can change it. But I want to be clear at the beginning that brains are not born as blank slates. Instead, they come pre-equipped with lots of expectations. Just as an example, you know, when you look at a baby chicken, this is just moments after hatching, it starts wobbling onto its little legs and it moves around and so on. In its environment, it doesn't have months or years to figure out how to move around.
- 1.27** And when you look at little human infants, they do all kinds of things that are cute. Like, if you stick your tongue out at a human infant, it will imitate you and stick its tongue out. I don't know if any of you have babies. You might have seen this kind of thing before. But this is actually a very sophisticated maneuver because the baby has to see what's going on visually and somehow translate that into the motor active, sticking its tongue out. What this means is that, you know, for a neuroscientist looking at this, we think, OK, look, there's some very sophisticated pre-programming that's going on here. It's not like it's a completely blank slate. This is a feat that requires a sophisticated ability to turn visual input into motor action. So it's not that we have to learn everything.
- 1.28** OK, now, can anyone tell me who this is? Yes, exactly. This is Crick and Watson. And in 1953, Francis Crick burst into the Eagle and Child pub in England and said, "We have discovered the secret of life." And this was one of the great pub-crashing moments of science. And they had just figured out the structure of DNA. It was a really big deal. It was such a big deal because the DNA that we have inside of us, this codes for how to build the brain and the body.
- 1.29** By the way, does anyone know if you wrote down ACTG, if you wrote that into books, all, you know, your human genome, how high a stack of books would that be? You have a guess? No, not to the moon. No, it would be as high as the Washington Monument, as high as the Washington Monument. And that's what you have inside every cell in your body, is that amount of information. So it was a great and very important thing. And so this gives the general architecture of your brain and body. But Crick, who I worked with as a postdoc, he was not quite correct that they discovered the secret of life.
- 1.30** What they had discovered was actually half the secret to life because the other half is everything around us. All of the experiences you have, your culture, your neighborhood, your language, your every person you meet—this is what actually wires up your brain to make you who you are. So, the way to think about it is that what DNA builds is a machine that rewrites its own circuitry to reflect the world that it's in and to optimize its efficiency in it. We'll see a lot more about that later.

- 1.31** We're not, you know, totally pre-programmed and we're not a blank slate. The way to think about this correctly is that you have genes and you have your environment, and these interact. So you drop into the world with whatever genes you've got, you have whatever experiences you have. That's what makes you you. And then given your current situation, that'll cause you to behave a particular way, and that feeds back to your experience. And what you can guess is that this makes brains very different from one another. Meaning, you know, you each drop into the world with slightly different genetics. You all have slightly different experiences. And so brains move off of these trajectories and become as different as if you look around a room and you look at the faces in the room. Faces are very different from one another. Same with brains.
- 1.32** I want to give you a quick example of gene environment interactions. I'm not going to show that many graphs in this course, but I'll just give one quickly, which is, there was a scientist, Caspi, who was interested in understanding how the serotonin transporter, a little protein, was related to depression, to your chances of getting depression. And of course, because you inherit half your genetics from your mother and half from your father, you can end up with two flavors of this. You can either end up with the short version twice or the long version twice, or short and a long. I'm skipping the details because it doesn't matter, but the point is you can have these three different flavors of serotonin transporters that you end up with.
- 1.33** Now the question is, is the serotonin transporter related to depression? The answer is, he interviewed a bunch of people. After taking their genetics, he interviewed lots of people on what had happened in their life, on the bad, like, really stressful experiences that had happened to them. You know, big experiences like a car accident or the death of a loved one, things like that. And what he found was that the number of stressful events correlates with your probability of presenting with major depression. But it depends on the genes that you're carrying. So if you are carrying the short-short version, then with the number of stressful life events, your probability of presenting with depression goes up much more strongly than if you have, let's say, the long-long version is what you're carrying.
- 1.34** So what does this mean? It means that your probability of presenting with depression is neither just the genes you come to the table with or your life experiences, but it's the intersection of those two. It's which genes you're carrying with and what happens to you in your life. OK? And so that's the key thing about thinking about life experiences and the genes that you happen to be carrying. I just want you to keep this in mind through the rest of the class, is that we are going to be talking entirely about the remarkable adaptability of the brain and how it adapts to its environment and to the requirements that are presented to it. But that doesn't mean we're born into the world as blank slates. Instead, we come pre-equipped with expectations, and these things interact with one another.

- 1.35** OK, so we're back to this issue about brains absorbing the world around them. So I want to dive into that. So imagine that you were born 30,000 years ago. Exact same DNA. You slide out of the womb, but you're in a different world 30,000 years ago. And I want you to think about, would you be you, or would you be someone different? So you've got the same DNA. And the question is, OK, would you look at the beautiful stars in the sky and think about flora and fauna, and think about how you're going to do the hunt, and worry about whether it's going to rain at night when you're trying to sleep outdoors or whatever?
- 1.36** It's a trick question because you wouldn't be you at all. Even with the same DNA, you would be such a different person, the reason being that all of your experiences are different. Everything that you've been exposed to. First of all, of course, you'd speak a different language and so on, but all of your experiences about society and culture and which arbitrary deities you believe in, and all that stuff would be so different. You wouldn't be you. You would have the same DNA, but you wouldn't be the same person. Let me give you another example of what I mean here. Imagine I were to show you two rectangles with different colored pigments in it. Probably wouldn't mean anything to you, right? Like, if you were this you from 30,000 years ago or something, it doesn't have any meaning inherently.
- 1.37** But if I show you these rectangles here, that has meaning to you. Whatever it is for each person here, it has meaning to you, but that's just because of your experience in your life. If you were a dog and you looked at these, you would just see rectangles with colors on them. It wouldn't mean anything. So the thing I want you to appreciate about all this forest of neurons in your skull is that these rectangles are meaningless. They accrue meaning only because of the brain plasticity that you have.
- 1.38** And by the way, it's because of the way that we absorb the world and remix it that the history of invention and innovation looks the way it does. We'll come back to this point, but, you know, if you look at the evolution of anything, vehicles, helmets, bicycles, whatever, cell phones, everything evolves from what came right before it. Now, what is always so striking about this is somebody could have thought of this later helmet, you know, at the beginning, or someone could have thought of making the, you know, the Tesla right after, you know, Henry Ford made his Model T. Someone could have, but no one does, right?
- 1.39** Because the way we actually function in the world is we absorb the world around us and then we remix what's in our heads. We bend and break and blend it and come up with new ideas. And that's what you see reflected in the history of innovation all around us. OK? Because what we're doing is constantly remixing the world.
- 1.40** Now, I want to get into a little more detail, which is that what brains do, they don't just adjust themselves with everything coming in, they adjust themselves to the tasks at hand. So your forest of neurons is responding to the needs around you, the needs and challenges of your life. So if you work on a farm, if you grew up on a farm, you're going

to absorb the details of tractors and seasons and seeds and things like that. If you grow up in the big city, you're going to absorb the details of bus routes and elevators and fire escapes and stuff like that. What the brain does is absorb what matters to you.

1.41 So I'll just give you an analogy. Think about a college campus. Often what happens with a new campus is that the architects will build all the buildings, but they won't lay down the paths yet. And what they do is they see where the students actually go. And when they start laying down those paths, then the architects end up paving that later. The key is that the paths don't get carved randomly by the students. They get carved based on what is relevant to the tasks at hand, where the students need to go. And this is what brains do.

1.42 I'll give you an analogy here, which is what happened in Japan after World War II. They had been fighting wars for 40 years in Japan. And this was what they were trained as a society to do. They had all these military engineers who were very talented because they had been fighting from the Russo-Japanese war through World War I, World War II. That's all they'd been doing, is designing these fighter planes and things like that. And what happened is, after the fatigue of war and the atomic bombs and so on, Japan was done. War was over. And they had to readjust. They had to figure out what to do with all these military engineers.

1.43 So what they did was, they set themselves to particular tasks, like building this train, this high-speed train, which looks like a fighter plane, by the way. But what happened is, Japan adjusted the talents that it had to the needs at hand, to the tasks at hand. They took all these military engineers who'd been working on these great fighter jets and tanks and whatever, and had them design wheels and axles and train shapes and stuff like that. And we will see, this is what brains do. They adjust themselves to the tasks that are important, to the tasks that sit in front of them.

1.44 So the brain is chronically adjusting itself to those tasks. And the question is, why? Why does it sculpt itself like Japan did? Why does it sculpt itself to what is required? The first answer is speed. When it adjusts its own circuitry, what it can do is accomplish things much faster. So essentially think about it this way. You've got, you know, your consciousness, which is very creative and so on, but takes a long time to figure things out. And then you've got, with a lot of practice, you burn things into the circuitry. You actually make it part of the hardware of the brain, and then you can do things much faster. I'll give you some examples about that.

1.45 But the second thing which goes hand in hand with this is energy efficiency. Once you make something part of the hardware, you can run that. You can really run that fast. So think about, I don't know if any of you play soccer. When you were an amateur soccer player, you know, you're constantly getting the ball stolen from you, and everything to you is a blur. There's, you know, elbows and knees and people running. You don't really get what's going on that well. Once you become a pro, then you're able to do things fast and efficient and you're able to run around with the ball and score goals and, you

know, just by watching and looking for particular signals, who's coming at you, when, why, what they're going to do, and so on.

- 1.46** And so what's going on is, you have taken the goal of soccer, you've taken the mission of soccer, and you've burned it down into the circuitry of the brain. You've actually made the brain a soccer-playing machine so that you can do things fast and efficient. Just to drive this point, I'll give you another example. Some years ago I met the world champion cup-stacker . So this kid is 10 years old. I don't know if you've seen cup-stacking, but it's this whole routine that you do where you stack and you unstack and whatever. It's pretty impressive. This kid can do it in 3.5 seconds. So it's blindingly fast.
- 1.47** So anyway, I decided to compete against him. I had never done cup-stacking before. I decided to compete against him while we were both wearing electroencephalography, EEG, so we could measure what was going on in our brains. Now, obviously, he beat me handily at this, but the question is, because he's so good at this, you might think that his brain is screaming with activity as he's doing this, right? But it's exactly the opposite. His brain is quiet. His brain is doing very little, while my brain is screaming off with activity because I'm like the amateur soccer player trying to figure out what happens next, whereas he's burning into his circuitry so that he is fast and efficient.
- 1.48** One of the first places this was studied was with Tetris players. Scientists trained a group of adolescent girls who had never played Tetris on playing the game and had them play it for three months. And then they compared them to a group of girls who had not played Tetris. And what you find, without going into detail, is that all these areas in blue here are the areas that showed greater efficiency. In other words, in the girls who had trained, they were much more efficient at it. And in red, you actually had physical changes, a thicker cortex. And we'll see more of that in later lectures, about how you get physical structural changes in the brain.
- 1.49** But again, what you see here, with all these blue areas, is that more expertise leads not only to, you know, much faster playing, but also more energy efficiency, because you turn the software of trying to figure out, "What the heck am I going to do with this shape?" directly into the hardware. That's what we see, is that brains adjust themselves.
- 1.50** I want to move to the next principle now, which is that because brains arrive half-baked, they are expecting a world that has particular sorts of inputs in it. For example, a world with love and touch and language around you and things like that. And what we find sometimes are these natural, cruel experiments that happen, for example, when a child is massively deprived or neglected or abused.
- 1.51** So one example is from a girl named Genie who was found in Los Angeles. She had been so severely abused by her parents who had mental illness, and the father couldn't stand noise and didn't want her to make any noise. And what they did was they kept her locked in a dark closet for essentially her whole childhood and usually constrained

in a sleeping bag. And so she was beaten and starved and restrained, and she was found at the age of 13. And there were physical consequences, like, she was only 59 pounds and 54 inches tall, but there were also cognitive consequences because of this. She couldn't speak any language. She had only a few words. She could say no, and she could say, stop it. And that was all she could say. She couldn't chew solid food. She wasn't toilet-trained. She couldn't focus her eyes past 12 feet.

- 1.52** So a whole team of psychologists came in once she was discovered and rescued. They all worked with her and spent all this time and love and attention with her, trying to train her, for example, with language. But it turns out, it was too late. And we're going to see more about that later. But it turns out, there are these critical windows where your brain has to learn particular things like language. And once it misses that window, once you have missed the window, and I'm not talking about learning a second language, I'm talking about understanding what language is, how you can use words to identify things in the outside world, the rules of grammar, things like that. Once that window is passed, it's too late. We'll see that again in class seven.
- 1.53** Cases like this, with horrible abuse of a child—unfortunately, this comes up every year where something like this happens. And it's from these cruel natural experiments that we learn that deprivation interferes with development. In other words, when a brain drops into the world, there's expectations that certain things are going to happen. And when you look at Genie's brain, what you see is that, with extreme neglect, the brain doesn't even develop normally.
- 1.54** And in fact, in the Romanian orphanages, what happened after the fall of Ceausescu, there were tens of thousands of children who ended up in these orphanages because their parents had been killed. And so the staff was totally overwhelmed. And so what they did is they said, look, the right thing to do is don't talk to the kids and don't touch the kids, because if you do that, they'll become clingy. And so what they did is, they would feed the kids and so on, but they wouldn't speak to them or touch them. And all these kids grew up with real cognitive deficits as a result.
- 1.55** And again, this is the kind of thing that teaches us, OK, what brains need is a whole. You know, there's a whole expectation. Unlike the Komodo dragon, who drops into the world and can just go do his thing, a child needs lots of input from the world. So when Mother Nature made this very flexible circuitry, she had to assume that the starting input would be OK. And this leads to the overarching principle for today's lecture, which is that your DNA doesn't build this fixed hardware for the brain. What yours does is build a system that adapts to the world around it, and it's expecting the right sorts of inputs, OK?
- 1.56** So we're going to wrap up lecture one, and I want you to think about this issue of who you would be if you were born 30,000 years ago. What would be the same and what would be different, given all of the world that you have absorbed? So the question is, would you be you? And to the extent that that's not true, that is the degree to which

your brain has been shaped by the world around you. And this is what we're going to come to understand in great depth in this course.

- 1.57** And it's because of the study of brain plasticity that my favorite quotation is from Martin Heidegger, who said, "Every man is born as many men and dies as a single one." And you probably have a sense of what this quotation means, but by the end of this course, I promise that you'll understand this very deeply, the optionality with which we come into the world and the way the world shapes us by its experiences. So that's what brain plasticity is about, and that's what we're going to master in the coming lectures.
- 1.58** OK, so what we saw is that we are very different from all of our nearest neighbors in the animal kingdom in terms of having taken over the world. And it's all because of brain plasticity. We don't drop into the world pre-programmed, but instead we come flexibly into the world such that, by the time you are a little child, you can come to learn everything that all the Homo sapiens before you have figured out. You can learn that and then springboard off the top of that. That's very different from what your dog does, or the Komodo dragon, or the horse or the chicken, who just come into the world and they say, "OK, I got it. Eat, mate, swim," whatever they're doing. And they just do that one thing. And that's why we have become such a runaway species.
- 1.59** We also saw that your DNA sets up your brain and body. But for the rest of this course, we won't be talking about the pre-programmed stuff. We're just going to be talking about how the DNA has set you up so that you are an adaptive system. And finally, we saw that this strategy of Mother Nature's is a little bit of a gamble, because if you're not getting the right input from the world, the system does not develop optimally. OK, so that's it. And I'll take any questions.
- 1.60** All right, so I've been following along Iain McGilchrist's concept of, like, the hemispheric differentiation between the right and the left. I'm wondering, with the child that had to get his—what hemisphere was taken out? Do you know?
- 1.61** Yes. So, great question. With a hemispherectomy, you can actually remove either hemisphere. And it turns out that this idea of there being a big difference between the left and the right hemisphere is a little bit illusory. The two hemispheres are mostly alike. They're almost entirely alike. And the only difference, really, is that the left hemisphere tends to do finer motor movement, which is why it's involved in speech, and the right hemisphere is involved in sort of broader, what are called grosser, motor movements. But turns out, the most amazing thing about the hemispherectomy is that you can remove either one. And the other one says, "Whoa, gosh, I better make sure that I wire up all the functions over here."
- 1.62** Sorry. So the question is essentially about consciousness and how that relates to your brain. And how is it that— you mentioned that, OK, if we were dropped in the world, like, a million years from or 30,000, but, you know, a long time ago, would we be completely different? But how does that work with consciousness, being that it seems

like our consciousness is able to separate itself almost from our brain and being able to look at ourselves in a way that doesn't really make sense with, if ourselves were just part of our neurology?

- 1.63** OK, cool question. So the key thing about consciousness is that it emerges from the operation of these 86 billion neurons. Your consciousness is a function of that. Now, how do we know that? Well, first of all, it's easy for you to change your consciousness. All you need to do is pour some ethanol over your mucosal membranes, and then you'll be drunk, and your consciousness will be different, right? You take psychedelic drugs, and now you're talking with silver leprechauns about the secrets of the cosmos. You hit your head and you lose consciousness. You get some disease state that does something to the neurons, and your consciousness changes.
- 1.64** So that's how we know that consciousness is a function of this very delicate balance of what's going on with these 86 billion neurons. And, in fact, the reason I'm using the word delicate or fragile is because I'm always struck by this. You know, when you look at something like a psychedelic drug, it's just some molecule that binds to receptors on neurons, and it changes the flow through the system by, you know, 5% or something, and yet completely changes your consciousness. It's because what that illustrates to me is how fragile the system is.
- 1.65** And Mother Nature goes through a lot of trouble to keep consciousness just in this very tight range. Because in that range, I hypothesize, that's when we can work together and build cities and skyscrapers and symphonies and so on. But if everyone were in this slightly different range, then it would never work, and our species would never have made it. So, point is, consciousness emerges from the exact functioning of what's happening under the hood.
- 1.66** My question is on efficiency. You had done, you presented those two experiments with the cups. Is there an average on how quickly the brain adapts to a new task at hand? I think about businesses, employers hiring new people to their team. Is there a specific timeframe where a new test becomes, you achieve a level of efficiency?
- 1.67** Yeah, great question. You might have heard of the 10,000-hour rule. So that number is totally made up, but it's, you know, it's approximately. What it indicates is that you have to put a lot of work into digging things down into the brain. And we're going to see more about this in lecture five. We're going to see this in great detail, actually. But there are things that you can cognitively know. Like, let's say a kid who doesn't yet know how to ride a bicycle. Let's say you explain exactly how a bicycle—you still need to spend the hours doing the thing and learning the balance, all of which becomes unconscious. For example, you can ride a bicycle, but presumably you can't articulate to me how you do that. So that's where this 10,000-hour rule comes from. But what we'll see in lecture five is that there are even some more subtleties to it—as in, it has to be relevant to you for your brain to even make the changes and encode it.

- 1.68 You were mentioning about our brain giving us windows of maybe learning a new language, or does that apply to just languages, or—?
- 1.69 Great question, we'll come back to this. But it turns out that the idea of learning what language is has to be done in the first few years. You can learn other languages at any point in your life. You can learn second languages and third languages. But learning how to speak language has to happen at the beginning. And by the way, there are all kinds of subtleties even within that. We will talk about this in a later lecture. But something like accent, you know, if you move to a new country before about the age of 13, you won't have any accent in the language of that country. But if you move after 13, you typically do retain an accent. So there are many aspects to language.

Lecture 2: Maps in the Brain

At a glance — Maps in the brain — how the brain figures out the world.

- 2.1** OK, now we're going to talk about how the brain, which is perched in silence and darkness, figures out anything about the inputs from the body. So we're going to start in 1951, when the Canadian neurosurgeon, Wilder Penfield, was putting electrodes into the brains of patients who were undergoing neurosurgery. An electrode is just a thin little wire. And it turns out that, when somebody has a hole in their skull because they're getting a surgery, you can stick a wire into the brain. Does anyone know why you can do that? It turns out, the brain doesn't have any pain receptors. So the brain takes care of registering the pain from the entire body, but it itself has no pain receptors. So you can poke around in and do whatever you want there.
- 2.2** So he was sticking electrodes in. And what you can do is listen very carefully when there's an electrical signal. So neurons communicate with each other via these electrical signals. They host these things called action potentials, where they have a spike in the voltage across their membrane. And then these action potentials travel down the axon of the neuron and cause the release of neurotransmitter, and that causes the next neurons to spike. So we don't have to know any more details than that, other than there's a lot of electrical activity going on in the brain because each one of these 86 billion neurons is popping off with these spikes.
- 2.3** OK, so the point is, Wilder Penfield and others could measure when there was spiking going on in neurons by having an electrode just sitting next to it and hearing what was going on there. What you do when you eavesdrop on neurons like that, if anyone's ever heard this, you hear essentially [noise]. Each neuron is doing its own little speaking like that. And one of the things we're still trying to do, actually, is decode the language of what neurons are talking about. But in any case, what you can also do with an electrode is you can put some electricity in there. You can add a little bit of electrical current to a neuron and cause the neuron to fire.
- 2.4** And what he realized is that, as he passed the electrode into different spots from here to here to here, he realized that if you, let's say, zapped some electricity, that would cause the patient's bicep to contract. And then if he moved it over just a little bit, maybe it caused the shoulder to contract. And if he moved it over just a little bit more, it would cause the neck muscle to contract. And what he realized was that, bizarrely and amazingly, there was a map of the body in the brain. The brain actually knew what the body looked like.
- 2.5** So there's actually two maps going on. One is a map of all the muscles that you're moving around. That's called the motor or somata motor map. The other is somata sensory. By the way, somata just means body. So the sensory map is detecting all the touch and things from parts of the body. So just to orient us, if you wear headphones,

that's where these two strips are, right underneath where you'd wear headphones. One is all the information going out to the body to drive it. The other is all the information coming in from the body. But what's fascinating is, you have these maps, and what you might notice is that the size of things on the map aren't represented by their actual size. They're represented by the importance of them.

2.6 So for example, on the somatosensory cortex, the areas that have more receptors and sensitivities, like your lips or your fingertips, those have bigger representations. They have more real estate in the map because there's lots of information coming from them. Whereas, let's say, your knees, your thighs, or something, they don't have that much information in somatosensory cortex. And likewise, with the motor cortex going out, the things that you can drive with more detail, like your fingers or even your tongue or things like, because you need to move the tongue around a lot when you're speaking, those all have more real estate devoted to them. So what Penfield discovered is this map of the body and the brain. And this was a surprise because the brain, as I said, is locked in silence and darkness.

2.7 So how does this happen? So the obvious answer is genetics. Somehow, genetically, you've got this map of the body. But it turns out, as we're going to see, that's not the right answer. It's not genetically pre-programmed. Penfield called this the homunculus. Does anyone know what the homunculus stands for? Little person, exactly. Little man. And so that's what he called it. And the question is, how does the homunculus come about? Now, it turns out that the answer to this came about some decades later, and it was predicated on some experiments that were done with monkeys in Silver Spring, Maryland. So these came to be known as the Silver Spring monkeys.

2.8 The researcher there was looking at what happens if you sever a nerve in the monkey, such that it can't use its arm or can't feel from its arm. How does the brain deal with that? And how can you speed up rehabilitation of that? So this researcher was doing that. There was a young man who came into his lab who pretended to be an interested student, but in fact was there from this budding young organization called the People for the Ethical Treatment of Animals. And so what he did, this young man, is he staged some photographs to make it look really bad for the monkeys. And the police ended up coming in and shutting down the lab for inadequate veterinary care. So this moved to the court system, and this was argued about in courts for years, and eventually the charges were overturned.

2.9 But the point I want to make is that during this whole time, these monkeys were just living and having a nice life, and they weren't being studied for a very long time. And so what happened is, nine years later, one of the monkeys became terminally ill and was going to die. And so the research team asked the court, "Look, can we examine what's going on in this monkey's brain? Because nine years ago, we severed this nerve, and no one actually knows what happens to the brain." And that's when they discovered that this map of the body changes. In other words, the monkey's representation of its body,

which before represented the entire monkey body, now represented the body, but without the arm, because the arm, the nerve was severed.

- 2.10** So what does this mean? It means it's not genetically pre-programmed. It means that the brain is assessing all the data coming in from the body, and if it's not getting data from the arm, then as far as it's concerned, there is no arm. And the map readjusts itself. So this happens to humans all the time. I don't know if any of you have been to London and seen the statues of Admiral Lord Horatio Nelson, who festoons all these museums. And in Trafalgar Square, it's his statue on the big pedestal there. But what most people don't notice when they see these is that he's actually missing his right arm. And the reason is because [he was] shot by a musket ball during one of his naval battles, and they amputated it right there on the ship.
- 2.11** Admiral Nelson came up with a whole hypothesis about the afterlife because he said that he could still feel his arm. This is called phantom limb pain, which we'll talk about later. But as a result, he said, "Oh, there must be an afterlife where you have a whole body." It turns out that's probably incorrect, but we'll talk about why he was feeling that. The interesting thing is, what do we know about what happened to Nelson's maps inside his brains or the maps of any amputee? The answer is, when you have, let's say, you still have your arm, this is represented in the somatosensory cortex, where the color-coding here shows where the face is represented and the upper arm and the hand and forearm. And what happens when you lose the hand and forearm is the somatosensory cortex readjusts, and that's what you see on the right there.
- 2.12** This is the same thing that happens in monkeys and in humans, which we can measure very easily nowadays. You have this dynamic readjustment all the time of what the brain thinks the body is doing. And so the principle that we get here is that your brain's maps are not predetermined like people originally guessed, but instead, they're constantly adapting to the input. And to tie this back to the last lecture, remember how I said that brains are not blank slates? So how do you think the brain map originally gets laid down? Because in everybody, it essentially looks the same in terms of where the body is. Let's say, if you have all your limbs, it looks about the same.
- 2.13** The answer is the genetics set down the rough thing about, OK, here's going to be the feet and here's the head and so on. The genetics set that down. But from there, it's all about dynamic adaptation to what your body is actually doing. Now, I want to return to a side question, which we'll return to in lecture 8, but this is the question of why did Lord Nelson still feel like his hand was there? We know this is called phantom limb pain, and it sounds like, from some nods I saw here, everyone knows that, when you lose an arm, people often still feel the pain from their arm. They think it's still there. They can feel it. The answer has to do with, and this is the part we'll come back to, it has to do with the fact that the primary somatosensory cortex, the part I showed a picture of, that's changing rapidly.

- 2.14** But the parts downstream from that change a little more slowly, and the parts downstream from that change a little more slowly , and so on. And as a result, these downstream parts still think that information here must represent the hand, even though the hand is gone , and the primary somatosensory cortex knows that. We'll come back to that point in a few lectures, so we'll understand that more deeply. But here's the point I want to make. There are many ways that the somatosensory cortex can readjust, and it doesn't even have to be an amputation. Anything that declines, anything that causes decline in the incoming signals, will cause this to readjust.
- 2.15** So for example, the same thing happens if the nerves from your arm are blocked . Let's say you put a really tight pressure cuff around your arm for a while, you can start measuring changes in the somatosensory cortex. Now, this is wild because this is pretty fast that this stuff can happen. We'll talk about the speed of this in a moment. The same thing happens if you anesthetize the arm. So if you're not getting the signals from the arm, the somatosensory cortex says, "Oh, OK, that arm must be gone. I'm not getting data from there, so I'm going to start readjusting my map here."
- 2.16** Same thing happens, by the way, if you tie two of your fingers together. Your brain in the somatosensory cortex will start representing it like one finger because the way that it understands multiple fingers is because you're constantly doing things and getting different data on these fingers. But if it's all tied together for a long time, it says, "OK, I got it. You just have one finger there , one big thing." Again, this is all the brain sees. It's dark in there. So how does it figure out what the body looks like? The answer has everything to do with how synapses adjust themselves.
- 2.17** So I've mentioned neurons before. You've got these cell types with these specialized processes coming out of them. So neurons are just like other cells in your body, except they've got these big extensions coming out of them. They have a lot of these processes. Think of them like tree branches coming off . These are called the dendrites. And then there's one branch coming off that's called the axon , which is what carries the electrical signal. When the body of the neuron has an electric spike, it gets carried along the axon.
- 2.18** And the key is, where one neuron touches another neuron, that's called a synapse. And chemicals called neurotransmitters are released from one neuron, and they bind to receptors on the other neuron, and that's how a signal is carried from one to the next. The key is that the strength of the connection can change, and there are many ways you can change it. You can release more neurotransmitters, you can express more receptors for the neurotransmitter here. There are lots of ways that you can make a stronger and stronger connection. And that's what neurons are constantly doing, is adjusting these things.
- 2.19** And by the way, for any of you who follow artificial neural networks, what's happening in artificial intelligence, it's exactly this idea, where you have units and you have connections between the units and you're just adjusting the strengths. That's

essentially the cartoon version of what happens in the brain is what got imported over to AI, and that works quite well. OK, but the question is, what is it that changes the strength? What is it that causes the strengths to change? How does the synapse know when to do that and when it doesn't? Well, the answer is that timing is everything.

2.20 So neurons spend a small fraction of their time hosting these electrical signals [noise], and the timing of when one neuron fires and the next neuron fires, that's critically important. So a typical neuron in your brain is reaching out and touching about 10,000 of its neighbors. And some of those neighbors are then firing in the next moment because of inputs that they're getting from everyone else, and some are not firing. And it turns out that, when this neuron fires and the next one fires, and they're synchronous or one right after the other, what happens is that the connection strengthens.

2.21 Now, this was originally proposed as a theoretical rule by a scientist named Donald Hebb, who said, "Look, maybe neurons that fire together, wire together," meaning, if these guys go off at the same time, then they strengthen the connection between them. In this example here, he said, "Look, maybe when you present a banana to the brain, all these cells are involved because this has got some yellow and that's got some curvature and that's got some whatever. And all these neurons happen to be firing at the same time, and they strengthen their connection with each other. And that's how you represent a banana in the network."

2.22 Whereas the apple assembly is different. This is representing red and the curve and the shape and the smell, whatever. And those neurons happen to be the ones that fire when you see an apple. And what happens is they strengthen their connection with each other. And so you get this network that represents apple. OK, now this was a very good idea. But when Hebb proposed this in 1948, there was no evidence that it was actually true. And it wasn't until 1973 that people were able to gather experimental evidence that there might be something to this.

2.23 So specifically, some scientists named Bliss and Lomo were able to show something very special, and they happened to be in a part of the brain called the hippocampus. I'll try to make this really simple. But what they were doing is, they were stimulating in one area and then measuring the response in the other area. And the way we talk about this is we say they were stimulating the presynaptic neurons and measuring the postsynaptic neurons. In other words, before the synapse and then after the synapse. And what they found is that, if you stimulate, zap, then you measure here a little activity, and they saw it was of a certain size and they could measure that.

2.24 But then what they did is they gave a whole bunch of spikes. So on the presynaptic, it caused a whole bunch of spikes. And what that does is it stimulates the postsynaptic ones very strongly. And what they found is that you get this, what's called the post-tetanic potential. So what you do is you give a whole bunch of spikes, and that causes the postsynaptic neurons to fire off. And what happens is, you see this big potentiation. You see this strengthening of the connection during that time. And then what happens

is that comes back down, but it stays at a higher level. And this is what's called long-term potentiation .

- 2.25** Now, the idea is, what they discovered is, if you zap a bunch so that the pre and the post-synaptic neurons are firing together, then what happens is you see this strengthening. You see that you have a stronger signal now on the postsynaptic side. Now, the reason I'm going into detail about this long-term potentiation or LTP is because this is absolutely critical to essentially everything we know about neural networks, real and artificial, is that they can change the strength. And this is how it was demonstrated experimentally, that this could happen.
- 2.26** And by the way, whenever you have something that can be strengthened, you can have it weakened also. So if you have the pre and the postsynaptic neurons always anti-correlated, so that when one is firing, the other one is not, then you get long-term depression, which means that gets weakened. And so between those things, you can change the strength of a synapse. Now, why does this matter for what we're talking about? Because in the young neighborhood of a new brain, what you've got are all these nerves coming into the brain that are carrying information about what's happening out there with the body. And what happens is, when they come into a young brain, their roots are pretty spread out. So everything is connected quite broadly.
- 2.27** So, how does this simple trick of neurons that fire together, wire together, how does this lead to a map of the body inside the brain? Well, what the key is, is that, as you go around the world and you bump and you kick things and you touch things, what happens is that neighboring areas of the body tend to be stimulated together. So these two parts might get stimulated together, but it's not that likely that that and that gets stimulated at the same time.
- 2.28** So what happens is, neighboring parts of the body that get stimulated together end up strengthening and wiring next to each other, and you get this all throughout the brain. So you end up getting a map of the body in the brain just based on a simple Hebbian rule . In other words, If you make sure that neurons that are firing end up wiring together, this ends up dropping out, which is to say, patches of the body that are near each other will tend to be coactive, and they'll tend to wire up next to each other. So that's how you get this body map out of it.
- 2.29** OK, now, here's the way to think about why it's dynamic. It's because, in order to maintain this body map, you have to constantly have activity coming in here. In other words, it is activity-dependent to keep this map of the body. So here's the way to think about it. This is a map of North America in 1750, and you had these different countries holding on to different territory. And what happened is that the waxing and waning of the French grip on the new world, the territory changed size over time. But what happened is the French ended up sending fewer ships than the British did over time. And they ended up losing their grip on the new world because they just didn't have as

much data coming across there. So in the face of fierce competition, they had just not shipped enough over to keep a hold on the territory .

- 2.30** And so the lesson from looking at North America is that holding on to territory is a full-time business , because now we don't have anything left of the French, except some leftover state names and stuff like that. This same story is exactly what plays out in the brain all the time, which is to say, when a part of the body is no longer sending information, it loses territory. So what happened with Lord Nelson's arm is that it was no longer sending ships up the nerves and spinal cord there. And so that territory gets taken over, just like the French got taken over. So Admiral Nelson's arm was France, and the cortex was the new world. And it started off with his arm having a healthy colonization, sending useful spikes of information up there. And in his youth, that ended up staking out a territory. But once he lost the arm, then his brain received no new information from there. And with time , the arm lost neural real estate.
- 2.31** Now, you see this not only with the body, but you see this with all the senses, too. This applies to any system that's sending information into the brain. So when you look at the brain of a person who is blind, what you see is that this area at the back of their head, what's called the occipital cortex, which is normally the visual cortex . Normally, we just call it, "Oh, yeah, I see that orange spot. That's the visual cortex." But it's not the visual cortex in a blind person. It becomes responsive to sound. It becomes responsive to touch . It becomes responsive to math problems, vocabulary words, all kinds of other stuff. Why? It's because the territory is not getting used and so it just gets taken over by other parts of the territory.
- 2.32** OK, so this lesson about holding on to territory, this applies to not only the body, but the senses as well, because for a person who is blind or who goes blind, there are no longer ships of data coming in through the optic nerves and colonizing the back of the head. And so that gets taken over. So now for a blind person, her visual cortex is involving things like how to read braille. And we will see this in a later lecture that if she now gets a stroke in the occipital cortex, she loses her ability to read braille. So this is what happens. It's a very fluid system. And I started off lecture one talking about the fluidity of the system and comparing it to rerunning a map of the world and how the countries could come out very differently if you reran history. This is the surprising thing when we look at what happens when somebody, let's say, loses an arm or goes blind, is that the whole system can rewrite itself.
- 2.33** And there's a side note that we'll return to, which is that age matters here. So in a person who is born blind, their entire occipital cortex gets taken over by touch and hearing and other things. And if you go blind later and later, then you have less and less of a takeover. The older the brain, the less flexible it is, just like the borders in North America shift very little now because everything is settled into place.
- 2.34** Now, I just want to mention that this story has grown even more interesting in recent years, because when new occupants move into the visual cortex like this, they often

retain some of the original architecture. Like, if you've ever been to Turkey and you see these mosques there, they were originally Roman cathedrals. And so it's the same architecture, but it's now different deities in there. As an example, an area that processes visual written language in the sighted is exactly the same area that's used when people are reading braille words with their fingertips. And similarly, the main area for processing visual motion in people with sight is the same area in a blind person that detects tactile motion, like something moving across your skin.

2.35 So these observations have led to the hypothesis that the brain is a task machine and it's doing jobs like detecting objects in the world or detecting motion, rather than thinking about it as a system that's organized by particular senses. In other words, brain regions care about solving particular types of tasks, irrespective of the sensory channel by which that information arrives. OK, so what I wanted to give a sense of so far is that when the input changes, what the brain is doing changes. And what's interesting is that it's not just rearranging things; having more is better. So if you have more real estate devoted to a task, things are better.

2.36 Look at somebody like Ray Charles, who came into the world blind and has an extraordinary sense of music and of instruments and so on. He can feel instruments and caress them, and he can become one with his instrument. People began to wonder about this a while ago. They thought, look, when we see people like Ray Charles, is that just a one-off ? He happens to be blind and he happens to be a good musician, or is there some relationship between them? As it turns out, many scientific studies now demonstrate that, if you are blind, let's say, you are better with your other senses. Why? Because you just have more real estate devoted to it. And having more real estate matters a lot because you have more resolution in everything you're doing, in his case, with touch and with hearing. And what we see, of course, are lots and lots of blind musicians like Stevie Wonder, like Andrea Bocelli, and more, because they can devote a lot of neural real estate towards hearing and touch.

2.37 Now, I'll give you an example of something that blind people can do also. This was a child named Ben Underwood, who, I can't remember what age he was, I think about eight years old, developed cancer in both of his retinas, the back of the eye, and he had to have his eyes removed . And so what happened is he ended up developing this useful and unexpected technique, where he would click with his mouth [clicking sound], like this , and he would listen for the echoes that were coming back. And that way he could tell, "Oh , I'm near a wall, near an open space. Oh, there's a space between cars here and so on." And he could do things like rollerblade on the street and avoid the garbage can and do all kinds of incredible stuff by bouncing sound waves off things in the environment.

2.38 Now, I saw a documentary about Ben, and the documentary starts off by saying he's the only person in the world who can echolocate. But as it turns out, that's not true. Thousands of people can do this. And in fact, the phenomenon of blind people

echolocating has been discussed since at least the 1940s, when the word echolocation was first coined in an article in Science called Echolocation by Blind Men, Bats , and Radar. And the author wrote, "Many blind persons develop in the course of time a considerable ability to avoid obstacles by means of auditory cues received from sources of their own making."

2.39 And what the author demonstrated actually is that this is things like their own footsteps or tapping a cane or finger-snapping or making noises with their mouth. All of these things work for them to get feedback about what are the signals from the world that are telling them what is the layout out there. And he demonstrated that this was because of the sounds they were making by using things like ear plugs or distracting noises, and demonstrating that they couldn't perform as well. So again, what this means is that you have a takeover of other senses, in this case of the occipital cortex, and that gives people more ability to do things like hear what's going on and localize sound.

2.40 Now, it turns out that you see this interesting thing about territory takeover everywhere. So I'll just give an example. This idea that more real estate is better sometimes leads to counterintuitive results. So it turns out that most people have three types of color photoreceptors, and that's how we see color, all the colors we see. But some people are born with only two types or one type or none, giving them diminished ability to discriminate colors. But it turns out that colorblind people don't have it all bad. It turns out that they are better at certain things, like distinguishing different shades of gray. And the military realized this a while ago, that people who are colorblind are actually better at certain jobs because they can spot the enemy camouflage better than people can with normal vision. And that's because they have the same amount of real estate devoted to vision, but they don't have to worry about all this other detail with color. And so they have better resolution in the gray scale.

2.41 So, the key is how much real estate do you have to accomplish any given task? And I'll tell you something that we've been seeing a lot of in recent decades, which is, there's this understanding that what happens with autism often is you get children or adults who are what we call autistic savants, who are extraordinary at some task, like hearing a piece of music and playing it, or this kid, Max Park, can do an extraordinarily fast Rubik's Cube. This other guy can fly over a city and look out the airplane, and then when he lands, he can draw everything in great detail, including the number of windows in the building and so on. Sometimes you have these things that seem like absolute magic. How can anybody do this?

2.42 Well, it turns out one hypothesis about what's happening in autism is that you have more real estate devoted to some particular tasks. So in Max's case, it's more real estate devoted to the Rubik's Cube. Now, unfortunately, if you devote a bunch of real estate towards something, the good news is you can be absolutely magical at it. But the bad news is that comes at the expense of other things. So what all of us have to do with

our brains is accomplish lots and lots of tasks . And if you want to be part of our deeply social world, you have to balance lots of things and devote your real estate in such a way that everything is balanced. And if you have such an imbalanced distribution, sometimes you have other deficits in terms of social things.

- 2.43** OK, so now let's look a little more closely at brain reorganization. So in the brains of blind people, when they are reading braille, they're touching the bumps on the page, what happens is, their occipital cortex at the back of their head becomes active. So this is a PET scan. It's an older way of doing brain imaging. What you can see is that their occipital cortex is active during braille. Now, again, this is the part of the brain that would be involved in vision if you were normally sighted, but now it's taken over by a different task in a blind person.
- 2.44** Now, I mentioned this earlier, but if somebody has damage at the back of their head, if a blind person gets a stroke and gets damage in the back of their head, they lose the ability to read the braille. They're less good at it now because they've lost some of the territory. So it's not just that this spread into the back of the head is not functional, it's decorative or something. It actually matters. And we know that because when it gets knocked out, they perform worse. People have also done this in ways where they say, OK, what happens if we blindfold someone carefully? Let's say you blindfold someone for five days and you teach them to read braille, compared to a group of people who are not blindfolded and you're teaching them how to read braille the same way.
- 2.45** OK. And what you find is that the people who are blindfolded will end up performing better . They will end up having a better ability to learn this new task because now they're using some of that real estate for learning the braille. But also you can determine if the spread into the occipital cortex matters because you can do something called transcranial magnetic stimulation. Has anyone heard of TMS? Transcranial magnetic stimulation. It's just a way of, essentially , you're zapping a current across the skull. So you can do this non-invasively from the outside. You're zapping, you're disrupting the neurons underneath it. You don't need to know anything more than that because the key is, what they do is they say, OK, now that you've learned how to read braille, we're going to zap your somatosensory cortex. So if I'm reading the braille and you zap my somatosensory cortex, then I can't tell what I'm touching there. And so not surprisingly , my performance goes down.
- 2.46** But then what they do is they zap the occipital cortex at the back of the head. It's a detailed graph, but the thing I just want you to notice is that when they zap the occipital cortex, which is what's shown in blue there, you get the same thing, which is a decrease in the performance in recognizing what is there because the people, the blindfolded people who have learned how to read braille, are using their occipital cortex for part of this. And if you disrupt the activity in the occipital cortex, they are worse at it. So all of this is just a way of saying that, when you don't have sight, the back of the head starts getting used. The brain doesn't like anything to lie fallow. You

know that term from farming? If you have a field that you're not planting anything on, that's called lying fallow. The brain doesn't want any territory to go unused. And so everything gets used and taken over. So if you go blind, or even if you're just blindfolded, that territory gets used, and it gets used functionally.

2.47 OK, now here's the key. The key is that this particular study was done over the course of five days. And then they did brain imaging, and they were able to demonstrate that, when these people were reading braille, their occipital cortex was active. And that was surprising to everybody about, "Wow, five days. That's so rapid that you would see a takeover already happening." But then some colleagues of mine at Harvard did a study where they blindfolded people and put them in the scanner, and they looked at the effect of sound or of touch. And what they found was something that came as a big surprise to everybody, which is that after—so when they were looking in the visual cortex and the occipital lobe, when they were looking for activity there, they found activity within an hour. At about 60 to 90 minutes, they were seeing activity going on for touching or for sounds in what we normally think of as the visual cortex. And these were normally sighted subjects who just had been blindfolded.

2.48 So what that tells us is that this is faster than anybody expected. This was the shock of the findings, was their sheer speed. The shapeshifting that brains do, this is not like continental plates moving on the Earth or something, but instead, this can be remarkably swift. What we're going to see in a later lecture is how the brain, at a neuron level, sets itself up to do these really fast changes, this mousetrap rapid change. But for right now, the important point is that the brain changes are faster than anyone would have thought even at the beginning of this century.

2.49 This led me, when I realized how rapidly the visual cortex can get encroached upon, this led me and a student of mine to a new hypothesis about why brains dream at nighttime. And what we realize is that this has to do with the rotation of the planet. So specifically, there's this chronic competition for real estate happening everywhere in the brain. And the visual brain in particular has to deal with a big problem, which is that the planet rotates into darkness. So it spends half its time cast into darkness. Obviously, I'm talking about our evolutionary history, not our electricity-blessed times, which are just in the last nanosecond here.

2.50 OK, so what I just talked about was the way that sensory deprivation causes a takeover from other senses. And so what we realized is, wait a minute, how in the world does the visual system deal with this unfair disadvantage? Because you can still hear and touch and taste and smell in the dark, but you can't see in the dark. And so what we hypothesized is that we dream in order to keep the occipital cortex active during the night. In other words, what you're doing is you're making sure that other areas don't take over the occipital lobe when it's dark out because, given the startling rapidity with which these other things can take over, the threat is really formidable, given that

you're spending almost 12 hours in the darkness. So the idea is that dreams are the brain's way of fighting takeover .

- 2.51** When you look at what's going on with dream circuitry in the brain, it's actually, first of all, ancient . Every animal dreams, as far as we can tell, everything, all the way down to houseflies, dream. And what they have is, at least in the human brain, you've got this very small nucleus in the midbrain that every 90 minutes during the night just starts generating activity. It goes straight to this little area called the lateral geniculate nucleus, which projects straight into the primary visual cortex. And that's it. Dreaming doesn't blast activity into other parts of the brain. It just blasted right in the first visual area. That's all that dreaming is. Because we are visual creatures, we interpret that as visual, and we see.
- 2.52** I mean, this is wild, right? Even with your eyes closed at nighttime, you're having full, rich visual experience. And that's because you're jamming a bunch of random activity into the primary visual cortex. Now, because the brain is a storyteller, we assign meaning to it. You don't just feel like you're seeing dots and whatever. You see, "Oh, it's this person, and this happens, and now I'm riding a camel, and now I'm seeing this thing." Whatever your dream is. But it's because you're imposing an interpretation onto the dream. But the point I want to make with this diagram is that it is just going to the visual cortex. And this is what you would expect if you had circuitry that's just trying to prevent takeover of the visual system.
- 2.53** So in the face of this chronic competition, you have the self-defense system for the occipital cortex, which evolved as a natural consequence of cyclically dropping into darkness. So dreams, we suggest, are the strange love child of the rotation of the planet and brain plasticity. And by the way, we call this the defensive activation theory, which is to say, your visual system is used to having all this activity coming in, and then it doesn't have it for a while, so it starts fighting back. It starts saying, "Hey, I better blast some activity in there."
- 2.54** And what we've noticed is that this general hypothesis explains several different things. For example, tinnitus, which is ringing in the ears, when people lose a particular frequency of sound, they're no longer hearing that, then they get this ringing in the ears [beep sound]. They get that ringing . And we think this is also an expression of the defensive activation theory, which is to say, your brain says, "Hey, I used to hear that, and now I'm not hearing it anymore." So it starts generating it. And by the way, I think this is why phantom limbs hurt, why you have phantom limb pain, because your brain says, "Wait, I used to be feeling stuff from there," and it starts generating activity to fight back against that absence.
- 2.55** Coming back to dreams and the visual cortex getting taken over, there are certain predictions that come out of this very wacky hypothesis. So one thing is that we know plasticity decreases with age. I'm not going to go into these graphs right now, except to say, we know that as brains get older, they become less and less plastic. Well, so does

REM sleep. REM sleep decreases a lot from the time when you're an infant to the time that you become an adult. OK, but that's just a correlation. That doesn't prove anything yet. So let's look at a different kind of prediction across animal species.

2.56 So what we have hypothesized is that, in a brain that's born essentially ready, like we talked about last lecture, where a lot of brains drop into the world, they already know exactly what they're doing, the idea would be, you wouldn't really need a lot of dream time. You wouldn't need as much of it because it's not that plastic. And so, for example, if you take an animal like the gray mouse lemur, which is a type of primate, within a month, it walks and weans from its mother, becomes a juvenile and able to reproduce within three months. So it's an example of a preprogrammed animal, let's just call it. But in contrast, this primate doesn't walk for a year, and it doesn't wean until three years, and it doesn't reach adolescence until 13 years. So this is a much more plastic thing.

2.57 So the hypothesis is that the more plastic a system, the more REM sleep it will need in order to keep its visual system defended. So what we did is we looked at primates. We looked at 25 different species of primates, and I won't go into the details of these graphs, except to say, we looked at these issues about how plastic the primates are and how much REM sleep they get at night. Sorry, REM sleep is rapid eye movement sleep. That's when you're dreaming and your eyes are doing that. It turns out this is easy enough to measure across species because you just set up these infrared cameras or whatever, and you can look at the ape and the gray mouse lemur and the rhesus monkey and whatever, and you can just track how much REM sleep they're doing every night.

2.58 So it turns out, these correlate very well, statistically significantly, which is to say, if you are a more plastic animal, you have more REM sleep. Now, no one would have expected this before, before we came up with this hypothesis, that these are related, that more plastic animals have more dream sleep, but there it is. So, the longer it takes an animal to mature, the more REM sleep they have. Now, by the way, sometimes people ask about, what about animals that are close relatives but have very different amounts of sleep? But what's interesting is if you look at the fraction of REM sleep, you find that it's the same fraction.

2.59 So again, the hypothesis would be that for a certain level of plasticity, the system needs to spend a certain amount of its time fighting back. Every 90 minutes or so, just blast random activity into the visual cortex to keep it defended. And in general, the less plastic brain is in less threat of takeover and less fraction of time needed to be in REM sleep. Now, I just want to mention, there are some question marks with the hypothesis, like, what about elephants?

2.60 Because elephants have really big brains as we do. Do they have REM? And the answer is not really. They don't, they don't have much in the way of REM sleep. They do have a little bit of it, but it's consistent with our theory, actually. And the reason is because

elephants only sleep an hour or two a night. And also, they have really excellent night vision. And so this is consistent with the theory that an animal that sleeps so little and has reasonably good nocturnal vision just doesn't need the REM sleep because they're not going to get that taken over by the other senses. And that's exactly what we find here.

- 2.61** Now, I just want to make a note about dream content because you might be wondering, well, wait, why if it's just about defending the visual system, why do we have this rich plot to everything? Well, the key is that everything is linked in associations in the brain. So if I'm walking over here and I smell coffee, that reminds my brain of, oh, OK, that's what coffee feels like, and that's what coffee tastes it looks like, and that's the name of my barista at Starbucks, and that's where I buy coffee and whatever. And so all these things are linked in the brain. That's how everything goes. This is what's called an associative neural network. And so you can think about everything in the brain is really like, if I ask you to think about a rabbit, that's linked with everything that you've ever associated with rabbits. That's all physically linked in the neural network.
- 2.62** OK. So I think what happens during dreams is that you're blasting random activity in there to keep the visual cortex defended. But what happens is that it's looser. You're still following all these paths, but it's looser what can go. So more random stuff gets activated. And that's why dreams have the characteristic of bizarreness. That's why for everybody's dream, there's bizarreness to it because you're walking down these paths but taking slightly unusual routes that you wouldn't have taken otherwise. So that's why dreams have a unifying thread, but they are more unusual.
- 2.63** One other interesting question is, how do blind people dream? So does anyone know the answer to that? How blind people dream? Do they dream? So blind people do dream every night, just like the rest of us. But their dreams don't involve anything about vision. Why? Because they don't have vision. They don't experience vision. They don't know what vision is. Let's say you're born blind. Instead, their dreams are about touch and hearing. So a typical dream of a blind person might be, OK, well, I was in my living room, but someone had rearranged all the furniture, and I was trying to understand what was going on with the furniture. And then I felt something in the corner, and I realized it was a bear, and I ran, and it was growling at me, and I was running from the bear and leaping over the furniture and so on. That's the dream of a blind person.
- 2.64** Why? Because the circuitry is still plugging right into the occipital cortex, but now that's not visual for them. Now it's about hearing and touch and so on. And so the plot of their dream is that, because this is very ancient circuitry. This is an ancient circuitry that doesn't have the expectation that you might be born blind. I can't even emphasize the degree to which this is very ancient stuff. As I said, you find it across the animal kingdom. You find this dreaming circuitry everywhere. And so even in situations where

you are born blind, you still have that activity blasting in there every 90 minutes during the night, and you have dreams .

- 2.65** I want to wrap up lecture two. What we saw is that the brain distributes its real estate to the body, to different parts of the body, based on the data coming up the pipelines. And if the data flow changes, let's say because you lose an arm, then the territory in the brain changes accordingly . And it's the same with the senses. If you go blind, your visual cortex will get taken over by neighboring territories and tasks. And if you go deaf, your auditory cortex will get taken over and so on. And as I said in the first lecture, what we're looking at is a dynamic fluid system that is, it doesn't care about borders that we draw in textbooks and so on. All it wants to do is make sure that no territory gets wasted . It wants to use all of it. So it distributes its real estate based on where the action is .
- 2.66** And finally, I suggested that because the system is so fluid, it runs into a special problem, which is it has to worry about and go through special efforts to make sure that the visual cortex does not get taken over at nighttime when the planet rotates into darkness. And that is why we dream. Thank you for your attention. I'll take any questions.
- 2.67** My question had to do with areas in the world that are not dark for long portions of time. Does that affect dreaming?
- 2.68** That's a really interesting question. You mean, if you are, if you're living in very northern or very southern climes? The issue is that because of the tilt of the planet, you're going to spend half the year where it's more dark, right? So, I would say, on average, to my knowledge , there's no difference in groups of people about that. But I suspect it's because, on average, over the course of a year, it's half the time is spent in the darkness. And again, because this is such ancient circuitry that has to cover essentially all animals, all it does is it says, "Hey, every 90 minutes during the night, let's pop activity in there."
- 2.69** And what's interesting is there's a whole separate literature on why we sleep, which is for many different reasons. And it also has to do with consolidating learning and memory and so on. Essentially, the brain goes into a different mode. At nighttime, it switches over and does this other thing. But the point is , whether or not you're sleeping, the point is that when it's dark out and you can't see, you have to make sure that it's defended, that territory.
- 2.70** Going back to adaptability of the brain to be flexible and learn new things and change its ways, as opposed to the windows that we get as infants to be able to gain certain certain behaviors or attachments, let's say, to a parent. In the case of Heidi, that was very extreme. She didn't see light. She was bound. There was a lot of different issues. But what about the typical depressed mother that just could not bond with the child? The child, let's say, had the survival needs met but didn't have that affection and that

touch that's extremely essential for the development. Can an adult somehow with brain plasticity therapies or however it's done in the next chapters , somehow get that fundamental need met in another way that doesn't depend on an exterior or an environmental factor?

- 2.71** It's a great question, and we're going to get into this in later lectures. But the sad truth about it is that all the major architecture is set down when you're young , and there are things that we can do with brain plasticity later in life, but it's tough. It's tough to take care of something that was done in childhood. The way I think about this is, for example, I live in San Francisco, and I found a picture of San Francisco from 1903, and you could see all the major roads and so on. And then a picture of it in modern day, you see all these tall buildings and so on. But the roads and the whatever is all the same. And that's essentially how I think about the development of the brain. You get these very fundamental things laid down. So it is tough to change it later. And we'll come back to this when we're talking about teaching an old dog new tricks.
- 2.72** And also it turns out that different parts of the brain have different plasticity through life. So there's not just one critical window, but there are multiple critical windows. But the point is, this is why childhood is so important. And by the way, this is a side note, this is where I give my charity money, is to things at the preschool level and before, because that's where you can really make change. I mean, in lots of neighborhoods, in impoverished neighborhoods, what you have is the mother has to go to work and, say, a single mother puts the babies in the daycare center where all the kids are just huddled in front of a television. They just watch TV all day. And the point is just to make sure they don't get into trouble or die, and that's it. And they're not being educated and loved on and stuff like that. Anyway , that's the important level where you want to make the change for exactly the reason that your question is.
- 2.73** So you mentioned a bit about psychedelics and how that affects the brain. I'm more curious about awe and the experience of awe or inspiration or something that pops out at you. I wonder, what does that do in terms of neuroplasticity in the relation that— what turns on and how does that change you psychophysiologically, I guess?
- 2.74** Great. So we'll talk about that later. But the short answer is, the brain makes changes based on surprise. So all your brain is ever trying to do is build an internal model of the outside world. And if the brain could do it perfectly, which is never possible because the world is too complicated, but if it could do it perfectly, it wouldn't have to burn any energy because it would say, "OK, I got it. I've got a perfect representation of what's going on in the world. Don't need to do anything." What happens is, we attend to things that are surprising, and that's where we say, "Oh, wow, I didn't realize that. I got to make a change there." And that's what all of brain plasticity is, is saying, "Oh, I got to improve my internal model."
- 2.75** And so in the case of something like awe, where we see something, you say, "Wow, I didn't even know that was possible," that's just one flavor of surprise of something that

your internal model didn't capture. You can probably think of examples of things that were awe-striking to you when you were a child , and yet now you're totally used to it. You don't even think about it. And so your brain isn't making changes.

- 2.76** OK. So could it also be how awe can also trigger a connection that clears up paths and makes it a through line through different paths? And it's like, and then everything ticks and it's like, da, da, da, da, kind of like that?
- 2.77** Yes. It turns out that what we're always trying to do is tighten our model , and say, OK, I understand this, I understand this, I understand this. And then at some point you realize, oh my God, those are the same thing. It's actually the same underlying physics principle. I remember when I was a kid and somebody told me that fog is just a cloud that's low. And I was like, oh, I just felt like this click of things coming together because now one can inherit the properties of the other. It just simplifies everything.
- 2.78** And we have this very strong drive to get a tighter and tighter model all the time. And so that's exactly right. When we realize something, that's the way that the brain says , "Ah, got it," and it makes a change. And now you've got a tighter way of doing it. And by the way, this is how all great physicists , Feynman, Einstein, whoever, what they're trying to do is get to the thing. It's like, OK, if I understand this, I can derive all the rest from there.
- 2.79** My question is on REM sleep. I think you were mentioning that every 90 minutes is when we have REM. I think I had read it was very short, less than a minute or so. My question is about sleeping and how I just track off my sleep. I had heard this before, and I was tracking my sleep in 90-minute intervals because I had heard that it was better to wake up during or right after a REM sleep, so putting an alarm after 6 hours or something like that. And then I had also read in the past that some cultures would sleep, would break up their sleep in two separate patterns, like 4 hours at night and then like 4 hours in the afternoon or something like that. What can you share about what are the best methods of sleep? How much REM we need, or depending on maybe male or female?
- 2.80** Yeah, great question. OK. So it turns out, yes, you have REM sleep every 90 minutes. And the key about the REM sleep part of your sleep cycle is that that is actually the lightest sleep. It's the closest to being awake. And so often we wake up straight from a dream. And you're exactly right in your question that that's when you want to wake up. If your alarm goes off when you're in deep sleep , that's really tough and you're disoriented and so on. If your alarm goes off and you happen to be in REM sleep, then it's easy. So generally, yes. And there are increasingly better ways of measuring with rings or devices when you're in REM sleep and setting your alarm in a range, saying, "I want to wake up somewhere in this range. Just make sure you wake me up during the REM state ." So that is correct.

- 2.81** As far as what's called polyphasic sleep, that's the idea of we , in modern life, have gotten extremely used to just having one phase of sleep. We're awake all day, then we hit sleep at night. But many cultures in the world do polyphasic sleep all over the world. They'll do this post-lunch nap, stuff like that. That's two phases of sleep. But people have tried all kinds of things where you sleep maybe just 90 minutes at four different times of the day. People have tried many different things. Da Vinci did this and others.
- 2.82** And there's always debate on the internet that you can find about which of these techniques is better. Is it better to sleep four hours and then take two naps? Or is it better to just sleep four times for 90 minutes? And so on. All of these are productivity hacks, and they're all plausible, and they all work. And it's just a matter of what works best for your body. But if you can make it work and get away with just a fewer number of hours of sleep a night, then you're better off.

Lecture 3: New Sensory Frontiers

At a glance — New sensory frontiers — creating new senses for humans.

- 3.1** Great. So now we're gonna talk about this issue of, can you create new senses for humans? Given that the brain is locked in silence and darkness and just gets information through whatever sensory pathways it has, could you do something to create a new sense? So it sounds like a wacky idea, and this is what we're going to talk about. So we're already quite good at marrying our technology to our biology. So you might know there are hundreds of thousands of people who are walking around with artificial hearing and artificial vision.
- 3.2** And the way this works is, you take something like a digital microphone and you have a little strip of electrodes that you're slipping into the inner ear, and people can come to hear this way. Or you take a digital camera and you plug an electrode grid into the back of the eye, and people can come to see this way. Now, here's the thing. As recently as sort of 25 or 30 years ago, there were a lot of scientists who thought this was never going to work. Why? It's because these devices speak the language of Silicon Valley, and that's slightly different than what your biological organs are speaking. And so people thought, no way the brain's ever going to figure this out, exactly how to interpret these signals from a microphone or a camera.
- 3.3** But as it turns out, the brain does it, and it does it just fine. Now, how do we understand that? Well, the big secret is, your brain is not directly hearing or seeing any of this. Remember that everything happening in your brain is in the darkness of your skull, and all it ever sees are electrochemical signals. This is your entire experience happening in here. You have signals that come in along different data cables, and that's all your brain has to work with and nothing more. Now, amazingly, the brain is really good at putting together a story of this, collecting these things, looking for patterns, building meaning on top of this. And eventually you have this, your entire subjective, colorful, noisy world that you experience.
- 3.4** But this is the really wacky reality about how we perceive. Now, the part I'm going to concentrate on is this, is that your brain doesn't actually know, and it doesn't care where it gets the data from. Whatever data is showing up in there, it just figures out how to best operate with it, how to understand what that data is, and what motor outputs it can do to get what it needs. And that's all it ever experiences. So this is a very efficient kind of machine. It's essentially a general purpose compute device. It just takes in all the data and it figures out what it's going to do with it.
- 3.5** And this, I think, frees up Mother Nature to tinker around with different sorts of input channels. And so I call this the PH model of evolution. And I don't want to get too technical, but PH stands for Potato Head. And the idea with this hypothesis is that all these sensors that we know and love, like our eyes and our ears and our fingertips and

our nose, these are all just plug-and-play peripheral detectors. They're sensors that you plug in, and the brain figures out what to do with them. You stick them in, and you're good to go. Now, viewing our peripheral detectors this way, this might seem kind of a crazy idea, because , after all, aren't there thousands of genes that are involved in building these different detectors ? And don't those genes overlap with other parts of the body and so on? Like, how could we really look at the eyes and the nose and the ears and so on as devices that stand alone?

3.6 So I actually dove deep into researching that question because I thought , look, if this Potato Head model that I'm proposing is actually correct, wouldn't that suggest that we should be able to find switches in the genetics where someone is born, like, without eyes or without a nose or without ears or something like that? First of all, as it turns out, not all genes are equal. So genes unpack in an exquisitely precise order, such that you have the expression of one gene triggering the expression of something else, triggering the expression of something else. And you get very complicated feedback loops . And as a result, there are critical nodes in the programs for, like, building a nose. And that program can be turned on or off . Now, that sounds crazy, but how do we know that?

3.7 Well, you can look at mutations that happen with genetic hiccups. So there's a condition called arhinia , in which a child is born without a nose. It's simply missing from the face. So this is baby Eli, who was born in Alabama, doesn't have a nose. He also doesn't have the nasal cavity or the system for smelling. And unless you knew about something like arhinia, which is very rare, this might seem completely surprising, startling, and difficult to fathom. But in our plug-and-play framework, this is actually totally predictable, that with a slight tweak of the genes, the peripheral device simply doesn't get built.

3.8 Now, there's something called anophthalmia, where a child's born without eyes. They just don't have eyes. And if you lift the eyelids, there's nothing but flesh under there. There's no eyes underneath there. Now, this sounds wild and hard to understand if you hadn't heard about this before. But again, tweak of the genes. You just don't get these particular peripheral receptors. There's something called anotia, where you get no ears. There's no ears on the child. This was a baby who was born who is missing a tongue, just doesn't have a tongue.

3.9 You might have heard of something called anaphia, which is where you don't have pain receptors in the skin. You just, you can't detect pain from the skin or inner organs. And so a child is totally insensitive to the sting of life's lesser moments. And so at first blush , by the way, you might think that it sounds great to be free from pain, but it turns out it's actually terrible and really dangerous because you need pain to tell you to withdraw your hand and so on. So these kids get burned and scarred and so on. But all of this falls under this idea.

- 3.10** So when we look at this constellation of disorders, it becomes clear that all these peripheral sensors unpack by dint of very particular genetic programs . And if you have, you know, these malfunctions in the genes, you just halt that program and you don't get an eye or ear or so on. And the brain does not receive that particular data stream. And so that's why this idea of the Potato Head model actually might make more sense than it first appears, which is to say, OK, Mother Nature goes through trouble to make some sensor to pick up on energy sources from the world. Like, we have photons bouncing off stuff, so why not build some eyeballs? We have air compression waves, so why not build some ears? You can pick up on those. You can pick up on pressure and temperature and stuff like that with your fingertips and so on. You can pick up on mixtures of molecules with your nose and your tongue and so on.
- 3.11** So the idea is, when you actually look across the animal kingdom, what you find is that Mother Nature has figured out how to build peripheral detectors for all kinds of weird things. So , for example, when you look at snakes, you find that they have heat pits to pick up on infrared light . So we can't see infrared light, but they can, because they have these special receptors for them. For us, we don't see, by the way, you guys might know this, but the electromagnetic spectrum, there's a part of it that we call visible light. You know, red, orange, yellow, all that stuff. That's less than a 10-trillionth of the spectrum. And all these other things like radio waves and X-rays and gamma rays and microwaves, all of that is light also. It's just not visible light.
- 3.12** Why? Because we don't have the proper biological receptors for it. So it's just invisible to us. We think it's not even there. But as I said, snakes can pick up on infrared light. The black ghost knifefish has electroreceptors all up and down its body. So it picks up on perturbations in the electrical field. So it navigates in the dark waters this way by feeling where the electrical field is changing. "Oh, here comes a rock, here comes a predator ," stuff like that. There's this animal called the star-nosed mole which has a nose with 22 fingers on it and it feels out its dark tunnels this way. And that's how it constructs a three-dimensional model of its world, with these fingers. We don't have a nose like that, but it's certainly possible to have a nose like that.
- 3.13** Birds, insects, cows, lots of other mammals can pick up on the magnetic orientation of the planet. They have magnetoreception and they can navigate this way and they know how to move around this way. What you find when you look across the animal kingdom are all kinds of other ways of picking up information that is out there. We happen not to pick it up, but they do. And my assertion is that Mother Nature, once she has figured out the basic principles of brain operation, does not have to reinvent the brain every time. All she has to do is say, OK, cool, here's a new peripheral device. I'm going to plug it in, and the brain can figure out what to do with the signals .
- 3.14** The lesson that surfaces here is that there's nothing really special or fundamental about the receptors that we happen to come to the table with. It's just what we have inherited from a long road of evolution and what we are used to. But it's not what we

have to stick with. And the best proof of principle for that comes from what's called sensory substitution. And this refers to feeding information into the brain via unusual sensory channels, and the brain just figures out what to do with the information. Now that might sound totally speculative, but the first paper demonstrating that was in the journal Nature in 1969.

- 3.15** So there was a scientist named Paul Bach-y-Rita, who was a physician, and his brother was a physician also. His father had a stroke and was, you know, really had bad deficits because of it. And Paul and his brother really believed in the idea of neuroplasticity. And so they forced their dad to do all kinds of things to recover. So, as an example, they would put some food and they'd say, "Come get it, Dad." And he'd say, "I can't. My muscles don't work. I can't move." And they'd say, "Well, you gotta go get the food." And the neighbors thought this was horrible and cruel, but what it did is it actually worked. And the dad ended up dying, like, 14 years later hiking a mountain, because he had essentially a full recovery.
- 3.16** And so this got Bach-y-Rita really interested and, you know, really interested in this idea of neuroplasticity. So what he did is something quite incredible, which is, he took people who were blind and put them in a dental chair, and he built this little solenoid grid behind them so it has these things that poke you in the back. And what he did was he set up a camera, and whatever the camera saw, you would feel that poked into your back. So you remember when you were kids, you ever do this game where someone draws something on your back and you try to figure out what's being drawn? OK, so it's like that, but you're feeling this in your back. Turns out, the blind people got so good at this that they were able to identify what was going on in front of the camera just predicated on what they were feeling in their back.
- 3.17** Now, remember, the signal's coming in through the skin of the back. That's a very weird way to get visual information, but they were able to do it. What I discovered later was that the seeds of this idea had started a long time ago. At the end of the 1800s, there was a gentleman who set something up where, for blind people, you had a helmet with a little buzzer on it. And when you saw light or dark, this thing would buzz. And he was trying to see if blind people could get good at saying, "OK, there's a light, there's darkness," and so on. There was another group who did this with a helmet, with a bunch of little vibratory motors, and whatever the camera saw, you would feel that on the head. And so it worked, except that the device was really heavy and got too hot. And so the whole idea had to wait for a while, but the proof of principle was there.
- 3.18** Now, why does this work? Why does this idea of sensory substitution work? Well, when you look at the cortex, these are three drawings of cortical cells by Santiago Ramon y Cajal, who was the great neuroscientist who won the Nobel Prize for figuring out sort of what the cells were like in the cortex, what things looked like down there. What you find is something quite surprising, which is that the cortex is essentially a one-trick pony, which is to say, everywhere you look in the cortex, it's doing the same

thing. It looks the same. And this is why, as we saw in the previous lecture, which is why territory can get taken over and swap where things are happening, because it's the same stuff everywhere.

- 3.19** What you find is essentially six layers, and you have the cells doing slightly different things in the different layers and some are talking to the layers next to them and some are just talking up and down and so on. But it looks the same everywhere. Which is to say, if I showed you, I opened a skull and I showed you a patch of brain, and let's say I had some magical microscope where you could see all the activity in there, and I said, "Hey, what are we looking at? Are we looking at visual cortex? Are we looking at auditory cortex? Are we looking at gustatory cortex? Or, you know, what are we looking at?" You wouldn't know, and I wouldn't know, because it all is the same stuff everywhere across the brain. It's the same game going on here.
- 3.20** OK. And that's why the visual cortex is able to get taken over by touch, by sound, things like that. OK, so the idea is that the cortex is actually just an all-purpose data processing engine. You feed data in, and it will crunch the data and figure out the statistical regularities going on in that. Now, by the way, you might say, you might say, "Hey, look, I'm a little confused. Why is this typically the visual cortex? Why is this the auditory cortex and so on?" It's because that's where the data cables plug in. So you've got these eyes and you've got these data cables going back, and they plug in right here. And so that's why it becomes the visual cortex. And typically you have the data cables from the ear plugging in to this area in the temporal lobe. And that's why it becomes the auditory cortex.
- 3.21** But it turns out, it doesn't have to be that way. You can actually, well, if in the right sort of circumstances in the lab, you can actually reroute those cables. So this was done in the year 2000. This is a picture of a ferret brain, which is why it looks funny. But what the scientists did is they took the, the data cables from the eyes that normally go to this part of the thalamus that we call the visual thalamus, and they just rerouted it so it goes to the auditory thalamus, which means that visual data goes into an area of the brain that's normally the auditory cortex. And what happened? It became visual. It became visual cortex.
- 3.22** And so, normally in visual cortex, for example, you find cells that respond to lines of different orientations. Now, you find that in what we would have thought was the auditory cortex is doing the same thing. And that's what this graph shows. I'm not going to go into detail on that, but what we would have thought of as the auditory cortex becomes a visual cortex. OK, so the brain analyzes whatever data is coming in along whatever data cable, regardless of where it comes from and why. And that's why we saw in the case of Paul Bach-y-Rita's invention, that visual data can be useful even if it's coming in through the back.
- 3.23** Now, as it turns out, there have been many modern incarnations of this. So one of them is called the BrainPort. And this is a small electro-tactile grid that sits on your

tongue. And what happens is, this is again for blind people, there's a camera. And the camera sees something. And whatever the camera sees, you're feeling that on your tongue. So it feels like little Pop Rocks on your tongue. And it's drawing out what the camera's seeing in terms of the edges. And blind people can get so good at this that they can do things like throw a ball into a basket at a distance or navigate a complex obstacle course, seeing through their tongue.

3.24 Now, if that sounds completely crazy, that you might be able to learn how to see through your tongue, remember that all seeing ever is these two spheres in your skull that are picking up on photons and sending spikes to the back of your head. You can send those spikes in through a different channel and you can figure out how to operate with that. Again, the brain doesn't know where the signals are coming from. It just figures out what it can do with them.

3.25 And there have been lots of versions of this. So this system uses an electro-tactile grid on the forehead . And whatever the camera is seeing for the blind person, they're feeling on the forehead. Why the forehead? Because they're not using it for anything else. And what these have in common, though, is that the brain can figure out what to make of information that we would normally think of as touch. And it turns out, touch is not the only strategy that works. You can do the same thing by turning vision into sound. So in this version of the sonic glasses, a blind person has the camera input, which is taking things in and turning it into sound. So things sound like [makes noise], using pitch, using volume, things like that, using localization to turn the visual world into a soundscape . And at first it sounds like a cacophony. And then blind people start getting better and better at it and understanding what the sound is telling them about how they're moving around in the world.

3.26 So this started off in the 1960s with these very bulky glasses, but nowadays, this is something you can just download onto your phone. So this is an app that's called The vOICe, and the three middle letters are capitalized as OIC. OK. But the idea is, as you're going around, it is turning the visual landscape in front of you into sound. So it sounds like this.

3.27 [audio]

3.28 So it's pretty extraordinary. That's what it looks like when you're turning vision into sound. And blind people can learn on this kind of thing. And so, one vOICe user who went blind after having 20 years of vision, this is what she had to say about it. She said, "You can develop a sense of the soundscapes within two to three weeks. Within three months or so, you should start seeing the flashes of your environment, where you'd be able to identify things just by looking at them. It is sight. I know what sight looks like. I remember it." Now, this is wild, right? Because she's getting the information through a totally different channel into the darkness of her brain. But her brain starts interpreting it like sight, which is a pretty wild thing .

- 3.29** If it seems surprising that a blind person can come to see through their tongue or through earbuds on their cell phone, just think about the way that the blind come to read braille. So at first the experience is touching these mysterious bumps with your fingertips. Soon it becomes more than that. Your brain just goes ahead and reads the details of the bumps for a direct experience of the meaning. What I mean is, when you read a book and you're looking at the letters, you don't think, "Oh, OK, that's a weird curve with a shape." You don't appreciate the details of the alphabet that you're looking at there. Instead, you just laugh and cry at the adventures of the characters you're reading about.
- 3.30** The details have—they're invisible to you, and now you're just directly having an experience of the meaning. That's exactly what it's like for a blind person with braille. They're laughing, they're crying, they're not thinking about the bumps anymore. And it's the same thing when you are getting vision through your back or through some electro-tactile grid or through your ears. With enough practice, the brain is learning how to translate that. And once it does, the understanding of the visual world becomes readily apparent. It becomes directly apparent.
- 3.31** OK. So in my laboratory, I got really interested in this question of, could we build sensory substitution for people who are deaf? Such that, you say something, and a person who is deaf, who can't hear you, is able to understand what you have said. So, in my lab, we built a vest with vibratory motors on it, and we mapped these from low to high. So they're picking up on sound in the world at different frequencies, which is, by the way, what your inner ear is doing. Your inner ear captures sound at the eardrum and breaks it up from low to high frequency, and then ships it off to the brain. So we built this vest, and then here's what this looks like.
- 3.32** So now that I'm speaking, the sound is getting translated into dynamic patterns of vibration. I'm feeling the sonic world around me.
- 3.33** So that's a talk I gave at TED about this, about this vibratory vest. And what we were able to demonstrate is that people who are deaf could come to hear the world this way. So here's a woman who's saying the word sound or saying the word touch out loud, and you can see where the frequencies are. So it's actually mapping just from low to high frequency. So you might be able to tell visually what's going on there a little bit. But the key thing is that when people who are deaf start wearing this, after a little bit of time, they can start to understand the language of the vest.
- 3.34** So this is Jonathan, who was our very first participant. He was born profoundly deaf, so there's a part of his world that's unavailable to him. And so we had him train with the vest for four days, a couple hours a day. And here he is on his fifth day. So this is my graduate student Scott, on the right. And Scott says a word. And Jonathan's job is just to write down what he thinks must have been said in the outside world. So Scott says the word "you." Jonathan feels this on the vest, and he writes on the board what he thinks he's heard.

- 3.35 So Jonathan is translating this complicated pattern of vibrations into some sort of understanding of what's being said. Now, interestingly, Jonathan's not doing this consciously in the same way that, when I'm speaking to you, you're not saying, "Oh, yeah, Eagleman's saying some high frequency, then some low and some medium and what." You're just hearing my voice. And by the way, you think you're hearing my voice out here, even though it's happening inside your head. But it's the same thing as Jonathan gets better and better at the vest. He's not having to translate, "Oh, let's see, there was some high and low." He's just saying, "OK, that equals that sort of word."
- 3.36 Now, the reason this is a big deal, by the way, I think, is because the only option for people who are deaf is a cochlear implant, which is \$100,000 and an invasive surgery, and we can make the vest for less than \$1,000. And that opens it up for the whole world. And I'll mention some other things. We had National Geographic over to our offices recently to film. Here was a deaf participant, but he's not the reason they were there. They were there because of his daughter, who was born blind and deaf. So we built a little miniature vest for her so that she can feel the sounds of the world for the first time by understanding this is vibrations on her skin. It's just a way of pushing the information into the brain via an unusual channel.
- 3.37 And so here, for example, is her grandmother taking her around and putting her feet on things, saying, "OK, that's soft, that's cold, that's hot," and so on. On the right, she's on one of these beds that's going up and down. And the grandmother's saying, "Up, up, up, down, down, down." And by the way, this is how you learn the world. It's all about correlations. It's all about making correlations with things. And so from there, we shrunk it down to a chest strap for children who are deaf. And the chest strap has vibratory motors on it. But from there, we were able to shrink it down to a wristband. And so the wristband has vibratory motors on it. And now this wristband is all over the world on the hands of, on the wrists of people who are deaf.
- 3.38 This happens to be our very first participant, a man named Philip. And I just want to give you one-second sense of what this is like for him to be able to experience the auditory world, which he was never able to experience before. OK. So we ran lots and lots of people on experiments. This is just a word cloud of things that we got back about people telling us what things they were picking up on, which things surprised them, what they were getting good at. I mean, one thing is people who are born deaf, they have no idea that microwaves make beeps or that the blinkers in your car makes a clicking sound or whatever. There's lots of discoveries that happen and so on.
- 3.39 But more importantly, they come to understand what's in the world, when someone's shouting their name, when someone's blaring a horn, when there's a dog barking, when someone needs help, stuff like that. And so I thought this was a very original idea, but as it turns out, hardly anything ever is in science. And I ended up finding a paper more recently from 1923, where something similar to this had been done. And this is the little picture you see at the bottom there, where a gentleman speaks in a tube, and

something like 30ft away, there's a deaf woman who's feeling the vibrations against her skin. And the question was, could she be able to understand what he's saying just by picking up on the vibrations on her skin? And she was able to. And that was from 1923 .

3.40 And then in the early 1930s, there was an educator in Massachusetts who developed a technique for deaf-blind students to be able to read the lips of speakers so they know what they're saying, but because they were blind, they couldn't just look at the lips. And so it's a particular technique called the Tadoma, where you put your fingers on different parts of the neck and the lips so you can understand what somebody is saying. And thousands of deaf-blind children have been taught this sort of method here, and they've obtained such proficiency at understanding language that way, it almost rivals the point of somebody who can hear. So the thing to note for our purposes here is that all the information is coming in through the sense of touch, and they're able to understand what's happening in the auditory world.

3.41 Now, that's what we did with, once we built the vest and then the wristband, that's what we did in terms of hearing. But then we thought, gosh, can we feed in other kinds of information? So , as an example, it turns out when somebody gets a prosthetic leg, it takes them a while to learn how to walk with it because they're not getting the normal feedback from their leg telling them, oh, this is where it's bent, and this is where it's touching and so on. So they have to stare at their leg while they're trying to learn how to walk. So we just put pressure and angle sensors into the prosthetic leg and fed that directly into the vest.

3.42 So the idea is, a person is getting the feedback that the leg would be giving them. It's just putting it in through a different place in the body, and that works very well. No problem for people to learn how to do that. So that is sensory substitution. The idea of taking information that would have gone into the body, through the ears, through the eyes, or through the leg, and just pushing it into a different place. But then I got really interested in this idea about, well, what about, could you go, not just substitute, but actually add something new? Could you enhance senses? So that's the idea of sensory enhancement.

3.43 So here, for example, is a colorblind artist named Neil Harbisson. And what he's doing is, he can't see colors because he's colorblind. So he set up this little camera that looks at color and puts different, turns that into different auditory tones so that he can experience color this way. And he has a great time understanding and hearing color this way by just enhancing what he normally has.

3.44 And with animals, researchers are taking this one step further. So mice are colorblind, but you can genetically engineer photoreceptors to give them color vision. And the question is, would their brains be able to understand that? And the answer is, it's trivial. You just put the color photoreceptors in , and their brains have no trouble understanding that and saying, "Oh, OK, cool, I can use that." And with these extra genes, then mice can understand and distinguish different colors, and they can do

these tasks where it depends on the color. And if you have a normal colorblind mouse, it can't do these tasks because it doesn't see any difference.

3.45 And this has been done in monkeys, too, which typically just have two types of color photoreceptors. People have genetically engineered a third type of photoreceptor, and now the monkey can enjoy a human-level color experience. And again, the way you test this is with doing tests where the monkey has to distinguish red and green and so on. And he has no problem doing this as long as he's got the proper receptors to be able to pick up all the information.

3.46 And it turns out that sometimes sensory enhancements happen accidentally. So many people go in for a cataract surgery and they have their lenses replaced, for a synthetic replacement. And as it turns out, the lens normally blocks ultraviolet light that protects your retina. But what happens is, you've got this artificial replacement lens that doesn't block ultraviolet. And so patients find themselves tapping into ranges of light that they couldn't ever see before. And they can start picking up on ultraviolet light and describing things.

3.47 So here was one person who was describing what things look like to him. And , you know, when he's looking at Colorado Rockies shorts or whatever, everyone else sees it as black. He sees it as a faint blue violet sheen. And so, you know, with his, you know, with his new superpowers of this lens, he can see things that other people can't, past the normal spectrum of colors, which gives him new experiences when he looks at sunsets or gas stoves and all kinds of plants and flowers which have information in the ultraviolet range. You guys may know this, but flowers have all kinds of information in the ultraviolet range because bees can see ultraviolet. And so there are little runways for the bees, and they know exactly where to go on the flower. But we can't see. We say, oh, it looks like a nice blue flower or whatever. We don't see all that extra information.

3.48 A similar type of sensory enhancement, a visual enhancement in this case, has been given to animals . In 2015, some scientists at Duke took an infrared light detector and plugged it straight into the brain of a rat. And the rat became able to use it and solve tasks based on infrared light. And so if you plugged that into the somatosensory cortex, it took longer for the rat how to learn to use it. If you plugged it directly into the visual cortex , the rats were super fast at learning how to use it. Like, within a single day, they could conquer the task. So the point is, you're taking this totally foreign signal, infrared light , but if you get it into the brain, the brain has no trouble saying, "Oh, yeah, OK, cool, I can use this." And then it just figures out how to incorporate it. It's quite stunning.

3.49 One of the scientists involved in that said, "I'm still pretty amazed. Yes, the brain is always hungry for new sources of information , but the fact that it actually absorbed this new, completely foreign type so quickly is auspicious for the field of neuroprosthetics."

- 3.50** OK. People have done all kinds of interesting experiments now. There's something called the FlyVis helmet, which just, you know, you've got your visual display, and it's just capturing 360 degrees. So you get to see the world in 360 degrees. Normally, we have a small angle of about 180 degrees. The, you know, the housefly is seeing 360. And so you can get the, you know, the joy of housefly vision easily enough, and you can learn how to use this.
- 3.51** So I was a scientific advisor for the show Westworld, so I put my vest in that. I don't know if any of you saw season two of it, but there's this point where all the robots have gone bad, and these military contractors drop in, and they're wearing the vest. And the reason they're wearing the vest is so that they can feel the location of the hosts, the robots, and know where they are. Even if they're on the other side of a wall or it's dark or whatever, they know their location. This was very successful for these fictional military contractors who dropped in, to be able to just immediately feel the 3D-layout of what's going on there. So I call this Vestworld now, for what it's worth.
- 3.52** But the reason I did this is because this is an experiment we're actually running. And we've been doing this for people who are blind. So for example, we have LiDAR set up here, and this gentleman who is blind can feel the location of everyone around him. He knows, "Oh, this person's over to my right and coming closer. It's getting more intense. And now moving around behind me." And on top of that we add navigation. So we say, OK, we want you to go to this conference room. So we're going to buzz in the forward direction, now turn left, now turn 45 degrees, stuff like that.
- 3.53** And he's able to do this, even though he's never worn something like this before. There's no learning curve here. Somehow he immediately has a sense of the 3D structure of space, such that if we tell him, OK, look, correlate this with somebody's voice and so on, he immediately learns, "Oh, OK, I get it. That's where the person is and that's what's happening around me." And so what we're working on are things about sensory enhancement that, you know, let's say, so you don't need the drug dog. You can have a vest that's picking up on different shapes of molecules that humans cannot pick up on normally and can just alert you that that's happening here. So could you do molecular odor detection and just feel the different substances?
- 3.54** And what we got really interested in also was sensory addition, which is to say, not just enhancing a sense with better color vision or something, but actually adding something completely new to expand the human experience. So, for example, could you feel, you know, real-time data from the Internet? Could you feed that right in somebody's brain and could they develop a direct perceptual experience? So let me tell you about that, but I'll tell you about a couple of other things first about sensory addition. So what several people have done, biohackers, is they'll implant a magnet in their skin and then they're able to, you know, not only pick things up like this, but they're able to pick up on, if they're diagnosing electronics, they can feel the electromagnetic cloud around something.

- 3.55** So if, you know, if they're looking at a whole bunch of wires, they're trying to figure out which wire is burnt out or something, they can just feel it with their skin because of the magnetics. They're getting a feel from it. They can't carry hotel card keys in their pocket, though, because they demagnetize it. And what if you could detect not only the magnetic field around electronic objects, but also the one around the planet? Because this is what animals do. I mentioned before that a lot of animals have magnetoreception.
- 3.56** This was the gentleman who got really curious about the way that, why can turtles return to their hatching spot, you know, years later? Why do migrating birds go from Greenland to Antarctica and back? And, like, how in the world do they do this? And so he wondered how they could accomplish their magic. And he correctly guessed that they might be using an internal compass. In other words, using the magnetic field of the Earth. So, some colleagues of mine in Osnabrück built this belt called the feelSpace belt, where these are all vibratory motors around the belt. And what you're feeling is where magnetic north is. So, wherever north is, you're always feeling that on your body.
- 3.57** And it turns out that, you know, at first it just feels like a weird buzzing. But people get really good at understanding where north is, how to orient themselves to the planet that way. What's extraordinary is that all the subjects felt like they got much better at understanding things like the orientation of the cities or buildings or whatever. They just got much better at understanding what was going on there. And it went very quickly from being a tactile feeling to being a spatial feeling, as in, like, "Oh, I get where I am, and where I am, you know, oriented towards north." The experience isn't one of sensory substitution, where you're like, "Oh, I'm just feeding in vision or hearing through a different channel." This is a new sense, a new sense of the magnetic field of the planet.
- 3.58** What I think is an interesting side note is that when subjects—actually, two things . One of them is when subjects take the belt off, they tend to retain a sense of where north is much better than other people. And I think it's because humans probably have a little bit of magnetite. So we probably have a little whisper of a sense of where the magnetic orientation of the planet is. But we never get good at being able to learn up on that and correlate it. But if you wear something that gives you signals where you can clearly correlate it, then even after you take the belt off, you're better at it .
- 3.59** The other thing I was going to mention I think is interesting is, it's a totally new kind of sense to have. And so once, when subjects talked to each other, they were able to communicate better with each other than they were to the experimenters. The experimenters were like, "Hey, what was it like?" And blah, blah, blah. But you can only put things into words if someone else has experienced the same kind of thing. And so the subjects were better able to talk to each other because they were able to get what the experience was. And so I'll come back to this at the end, but I think it's really

interesting as we develop new senses, whether other people who don't have those senses will ever be able to understand it.

3.60 For example, if you have a friend who is born blind, if you ever try to describe vision to your friend, they can't ever understand it. I mean, you can try and they might even pretend to understand you, just to shut you up after a while, but they can't understand it. Like, what do you mean you're capturing photons from half a mile away and you know when someone's coming or whatever? But they'll never get it. Just like they can't understand if you're colorblind . You can't understand what somebody means by purple or something. It just doesn't make any sense to you. So that's the same thing that's going to happen more as we create new senses.

3.61 Now, coming back to this issue of feeding in a real-time data stream from the Internet, this is an experiment we did for the lab where we're feeding in a real-time stream of data into a participant who's wearing the vest. And then two buttons appear on the screen, unmarked buttons, one blue, one yellow. And he makes a choice. And then he gets feedback a second and a half later, either a smiley face or frowny face. He doesn't know what's going on or why we're doing this. What we're doing is, we're feeding in real-time data from the stock market. He's making buy-and-sell decisions, and we're seeing if, by giving him feedback about where the stock goes over the next second and a half, whether he can become tuned in to the economic movements of the planet.

3.62 So this is still work underway, but this is a bigger experience than a human can normally ever have, the idea of being plugged into the stock market like this. We're also doing this with social media. We can scrape the Internet for any hashtag, let's say, and then feed that into—we do an automated sentiment analysis and then feed that into the vest or the wristband, where all we're saying is, are people saying , you know, positive, negative, or neutral? And you're actually plugged into the consciousness of millions of people all at the same time. You're plugged into what a whole planet is doing.

3.63 Now who would want to be plugged into Twitter? I have no idea, but it's a really interesting, it's a bigger experience than a human can normally have in terms of sensory addition. We've been doing this with pilots , drone pilots, who are plugged into the pitch, yaw, roll, heading, and orientation of their drone. So as they're flying the drone, they're able to feel exactly what's going on. It's like they've stretched their skin up there onto the drone and become one with the device.

3.64 Now I just want to mention a couple other things about where we can go from here. So there's many, many things we're doing. We actually have 70 different projects going on where we're doing things with sensory substitution, expansion, or addition. And if I was interested, happy to talk to you about that later. But what I want to talk about is other ways you can do it. What we're doing is non-invasive . We're just putting vibrations on the skin and getting information that way in. But what a lot of people are interested in is directly plugging into the brain. What can you do with that?

- 3.65** So, the problem here is not a theoretical problem, it's just a practical problem, which is, you know, drilling a hole in the skull, having people go in for open-head neurosurgery. There's a lot of risk anytime you're putting somebody on the operating table, and so there's always this question of how badly do you want it? So, you know, the mythology of Neuralink is like, OK, well, maybe you could text faster your friends just by thinking about it. But I'm not going to go in for an open-head surgery just to, you know, text faster. So there's an interesting question there, but I'll tell you where we are with this.
- 3.66** The idea is, you know, in the laboratory, people can make electrodes, very fine electrodes, and stick them into the brain, and you can measure and you can stimulate cells that way. And so this is where things could go from here. Instead of having non-invasive ways of pushing information into the brain, you could push directly in. There's also this idea of optogenetics, which maybe people have heard of. This is where you can shine light into the brain, and you have these special proteins that will detect different light frequencies and can make a cell fire or, in other cases, inhibit itself from firing.
- 3.67** The problem is you can't do this in humans now. You can do this in animals because you can get them virally infected with these little proteins, but you can't do that with humans currently. But this is the kind of things that are being studied. And what is in patients and has been in patients is sticking in these arrays of electrodes. And so people have been doing this for well over a couple decades now, making these brain-machine interfaces, BMIs, where you stick electrodes in and you can get information recorded from the brain and you can write information to the brain.
- 3.68** And the most recent version of this is Neuralink, which uses, you know, which drills a hole and then uses a robot to stick electrodes in very finely. The only advantage what Neuralink is doing is it's extremely fine and precise, and it's also wireless. So then you put this little cap back on, and whatever's going on in there, you can read that data from the outside. But this idea of sticking electrodes in the brain, this is the same thing that's always been happening. I'll also mention that there are vascular stentodes, which is just a different way of eavesdropping on what brain cells are doing by going in through the blood vessels. You know, so your brain is shot through with blood vessels. So you sneak into the blood vessel and you have little electrodes on the inside of the blood vessel that's listening to the neurons just on the other side of the blood vessels.
- 3.69** So, anyway, this is all the stuff that's going on about how you can listen to what neurons are doing and/ or how you can change what the neurons are doing. So these are the, this is what the future holds for us. And I think in the more distant future, what we'll do is build nano-robots where we're actually making, you know, atomically precise little robots. You're essentially doing 3D-printing of atoms, one at a time, make these little robots, and then you might, you know, swallow a pill with 100 billion of these little robots in them. They swim through your bloodstream, they go up past your blood-

brain barrier, they impregnate the neurons, and then you can have them there doing things where they're sending and receiving signals.

- 3.70** Now, that's, I would say, science fiction right now, but it's not going to be in 20 years. So this is something we're all going to see in our lifetimes. That's what's going on with technology. These are all just ways of doing the same thing that the vest or the wristband does, which is just pushing information into the brain. And the question is, will the brain just figure it out? Now, I just want to touch on another point here, which is, what does this mean when we start having new experiences?
- 3.71** Let's say I asked you to imagine a new color, right? No matter how hard you try, you can't imagine a new color. You're just, you're stuck there. And what this illustrates for us is the fence line of our qualia. Qualia is the philosophical term we use for, you know, our internal, private, subjective experience. You just, you can't imagine something beyond what you have already experienced. And so, as we create new senses, the really interesting part is that we won't have a vocabulary for it because the vocabulary that we use is all about shared vocabulary. It's about things we both understand. If I've eaten an egg and you've eaten an egg, then we can talk about, oh, it tastes like an egg, and so on, but if I've eaten some, you know, shmegegge that you've never heard of, and then we can't communicate about it anymore.
- 3.72** So it's just a really interesting world that we're going into, which is, if you develop a new sense, either from a brain implant or the wristband or whatever it is you're doing, you won't actually be able to communicate it to someone in the same way that you can't communicate vision to your friend who's blind. When it comes to us thinking about sensory substitution and expansion and addition, there's really no end to the possibilities on the horizon here, which is to say, you can imagine anything.
- 3.73** Just imagine an astronaut on the International Space Station right now. They float around and they look at all these monitors all day to make sure everything's going well. Imagine if you could just feel how the Space Station is going, the way that you feel how your body's going. Like, "Oh, I should adjust because my knee hurts, or whatever, or I've got an itch over there." Like, imagine you're just feeling the Space Station instead of having to float around all day. Or for that matter, imagine just feeling the invisible states of your own health, like feeling your blood pressure and the state of your microbiome and your blood sugar levels and stuff like that. These are things that are easily measurable. They're just normally invisible to you. But imagine making them conscious by having that data. Imagine having 360-degree vision or seeing an infrared or ultraviolet.
- 3.74** So the key is this, I think. As we move into the future, we're going to increasingly be able to choose our own peripheral devices. Because what we've seen is that the brain can adjust. The brain doesn't care at all. It doesn't know about eyes and ears and nose and whatever. All it knows is, "Hey, I'm getting these spikes in. This is relevant information to me. Because when I do a motor output, you know, there's some, there's

something that I can match up here and I can find correlations across the other senses." That's all the brain cares about. And so what this means is we don't have to wait for Mother Nature's sensory gifts on her time scales. It means that, like any good parent, what she's given us is the cognitive ability to go out and define our own future. And so the question is, how do you want to experience your universe?

3.75 OK, so in this lecture we saw that the brain can interpret any information that comes into it. And although we're used to eyes and ears and nose and fingertips and so on, we can actually plug in any kind of information. And as long as the information is relevant to the brain, which we will see as the topic of later lectures, the brain will incorporate that into its interpretation of the world. Thank you for your attention. I'll take any questions.

3.76 So , like, first of all, that's just absolutely insane. And my question is kind of the inverse. So you talk about how we can interpret information or stimulus that we before couldn't. I guess I want to flip that on its head. So there are people that have, I guess you could say, cognitive disabilities. So like people with autism, they don't have, I don't know what it is, but they don't, because I don't know enough about it. But it's like they don't have enough understanding of, like, facial expressions and things like that. Could it be flipped around to where you could use those? You could benefit from their more sensitive parts of what they sense and use that so that they're better able to communicate or vice versa? We're better able to communicate with them? Has that been explored at all?

3.77 Actually, yes. So this is a spin-off company from the company that I spun off from my lab, which is, what it does is, it listens in real time. So for a child with autism , the microphone is listening to the conversation partner and it's just doing machine learning to say whether the person that you're talking to is angry or happy or sad or whatever. And it just buzzes in different ways to let you know that. And so for a kid with autism who is unable to read the emotional content of the conversation, they're just totally, "Oh, I see. The person's sad. OK, I guess I should act different as a result." So it's a super simple solution. Yeah. So what we can do is leverage all the kind of sensors we have, the machine learning we have and whatever, to just tell somebody on their skin what the answer is.

3.78 In terms of the future and having a bunch of different new senses, do you ever wonder or worry about what that could mean in terms of, is it possible to have too many senses? Like, what would that do to the human brain if we're, we're tuned into, like, too many things? Maybe there's an evolutionary reason as to why we have certain limits, and what that might look like in society if certain people have economically decided to choose certain abilities where other people cannot.

3.79 As far as adding new senses, what we know is that we can add a sixth sense , probably seventh and eighth and ninth. What we don't know is how many we can add until you start seeing blurring of other senses. So in other words, what the brain is really good at

doing—we'll see more of this in the coming lectures—but the brain is really good at sharing real estate and figuring out how to move things around in there. And so I don't know what the limit is for senses . Probably not a hard limit, but instead, if you add 12 new things, maybe your vision gets kind of blurrier or something.

3.80 What's interesting is that I think people will become different from each other. Right? If you're picking up on the magnetic field and I'm picking up on social media or whatever, and we try to tell each other what it's like, we might not have much of an ability to do that, but I will say that's a different degree of the same thing that happens now. If one of us is Democrat and one's Republican, or you're from somewhere in the world, I'm from somewhere else in the world, like, it's already the case that people have pretty different experiences . And even though they think they're speaking the same language, they're actually speaking pretty different languages.

3.81 So if we're from different political parties, justice or liberty or what freedom means or something like that, we both think, oh yeah, we're saying the same thing, but we're not. We might have completely different things going on inside our heads. And so I think as people become more and more different, it's, we'll just see more and more of that. You know, if I'm a hang glider and you're a stamp collector or whatever, we're already pretty different on the inside. So in my opinion, as long as the technology is inexpensive enough, then it's not really an issue. It's not like, hey , who can afford the Rolls-Royce? It's, hey, who can afford something that's, you know, several hundred dollars?

3.82 Cell phones, for example, are so inexpensive that they spread around the world. And even in the poorest countries around the world, you find everybody with cell phones. And in fact, many countries leap-frogged America because, for example, we went through the personal computer revolution, and a lot of countries couldn't afford that. But once cell phones came out, everyone got those, and they were doing all the great stuff with it.

3.83 My question is kind of similar. Since the human has evolved through the survival of the fittest and, through its own striving, like Rocky, boxing and that whole thing, how is that going to change us as humans if we could just press a button, become more intelligent or become more attractive or become without our own human will ? Because, for instance, if I fall in love with somebody, it's because of the way he is, the way he strove , the way he learned, the way he did his, you know. But if he just pressed a button to become this excellent creature , what is that? You know, how is that going to change our dynamic as humans?

3.84 Let me say two things. So one is that you can't, there's no such thing as pressing the button to become smarter or something, and here's why. For example, you remember the scene in the Matrix where Trinity calls the operator and says, "I need to know how to fly a B212 helicopter," and he, da, da, da, and then suddenly shows up. OK , that's total fantasy and always will be because my brain is totally different from your brain. So

there's no program to teach how to fly a B212 helicopter because you can't know the forest, my forest of 86 billion neurons and your forest of 86 billion neurons. The way to fly the helicopter is totally different, the way that you'd have to push it in there and the way you'd have to push in there.

3.85 So the way it happens now, and probably always will be, is that I have to put the 10,000 hours into learning it, into figuring out how to make my system adjust so that I learn how to fly. It would be the same with yours. So there's no such thing as putting a button to make yourself smarter or more intelligent or something like that, or better looking. I mean, except for things like, I don't know, all the cosmetic things that people do and that exists. But I just want to say, on a deeper level, what's interesting is every time a new technology comes along, people always worry about this. Justifiably so.

3.86 But when the printing press came out in 1440, people said, "Oh my God, the answer's just sitting there on the shelf. How are kids going to learn? They're going to be so stupid because they can just go and look up the answer anytime they want." Of course, obviously with the Internet, people said the same thing and so on. In my opinion, kids are going to be much smarter with each generation because of the access to the information. In other words, with the Internet, we all grew up—I'm older than you guys are—I grew up with a lot of just-in-case information, as in, OK, just in case you ever need to know that the Battle of Hastings happened in 1066, here we are.

3.87 But what kids get now is lots of just-in-time information. As soon as they're curious about something, they just look it up and they get the answer. And the reason that matters is because when you are curious about something, you have the right cocktail of neurotransmitters present so that that sticks. And so that's why kids, this next generation is going to be so much smarter than, let's say, my generation. I run into 13-year-olds all the time who say, you know, they say something really smart and I say, "Wow, how did you know that?" They say, "Oh, I saw a TED talk on it."

3.88 Well, that's cool, because when I grew up, I had my homeroom teacher and hopefully the person knew a few things more than I did, but that's all I had. But now you can go to the smartest person in the world, giving the best talk of their lives in 15 minutes, and get the information. And you can keep surfing around doing that. So I just, I think it's extraordinary the world that we're going into. I think people will be smarter, but not because of not striving, just having better access to information.

3.89 OK, so my question was, if we create new senses, will that create new genes that can be passed on through, like, evolution to, like, your children?

3.90 Great question. The answer is no. So it turns out that anything you do, you know, any kind of information you're pushing in doesn't have any effect on the genome. Just like if I go and I get a nose job and I, you know, shave all my hair and I, what, like, that doesn't change anything about the genes that I pass on to my kid. In other words, let's say I had a nose that stuck out this far. I could get a nose job and look really good, but

my kid would still have a nose that stuck out that far. That's, yeah, yeah. Doesn't change genes.

3.91 So you mentioned that our brain is really good at balancing real estate with senses. But what about our sense of belief or what we believe? Like, religion or spiritual? Do we have a real estate for that, or can that expand and contract or?

3.92 Sure, yeah, Great question. It turns out there's not a region of the brain where we say, "Ah, that's the religion area." But instead, it has to do with everything that you absorb from your parents and your neighborhood and your culture. That's what it has to do with. And you know, there are 2,000 religions on this little planet that we're on, right? Because you absorb whatever's poured into your nervous system . In other words, if there were one true religion going on, you'd expect that that might spread everywhere equally.

3.93 But that's not what we see. Right? You don't find a blossoming of Islam in Springfield, Ohio, or a blossoming of Protestantism in Mecca. You just don't find that because we all absorb whatever our cultures around us are doing. So religion is a really good example of brain plasticity , how you're born. And as I mentioned, if you were born 30,000 years ago, you'd believe in completely different deities and you'd have a whole different thing that you're willing to fight and die over with the next tribe and so on. So it is something that we get entirely from brain plasticity.

Lecture 4: Motor Mastery

At a glance — Motor mastery — the output side of plasticity.

- 4.1 OK, so last time we talked about how to create new senses, and today we're going to talk about the opposite, how your brain drives your body. So not the input, but the output. OK, so I'm going to start in 1963 when the Spider-Man comic came out with this character named Dr. Octavius, who was a, who was working with radiological samples. And so what he did is he built these robotic arms that came out of his body and were mind-controlled, so that he could actually do the, you know, work with the radiology samples and not get poisoned that way. So this is what Dr. Octavius was doing.
- 4.2 But then there was a chemical explosion in the lab and he got brain damage and turned to a life of villainy. And so he became known as Doc Ock. And he used these eight arms to, you know, run around and break into banks and have multiple hand combat and stuff like that. And so that was 1963 that this fictional thing was proposed. And what I find really surprising is how quickly this sort of idea has gone from fiction to fact. And that's what I'm going to tell you about today.
- 4.3 I want to return to this thing that we talked about in the last lecture about the maps of the body in the brain. So you remember right around where you wear headphones, you've got these maps of the body. One of them is all this sensory input, and one of them is the map that drives the output of the body. And so today we're just going to talk about the output. And so you remember Lord Nelson, who I mentioned, who had lost an arm? So what we talked about is that when you lose an arm, the map shifts. Both the sensory and the motor maps shift to represent your new body. And the key is, how do you measure something like this? So back in the 1960s, the way you do it is by sticking an electrode in the brain. And you move that electrode in different places and you stimulate and you see where the muscles are twitching. And that's how you figure out the map.
- 4.4 But nowadays there's a different way you can do it. You can use what's called transcranial magnetic stimulation, where you're just zapping and causing activity underneath the skull. And you can go and figure out somebody's map this way, and you can see very easily what their map is by zapping and seeing which muscle is twitching. And then you can see if there are changes in that map. All of this is going to serve as a big clue for any of us who are interested in when you look around the animal kingdom and you see this massive diversity in animal bodies and shapes. I've always been fascinated by this, ever since I was a kid, that you can have animals that are shaped so differently and they have different things like prehensile tails or they have claws, or they have, you know, tentacles or whiskers or trunks or things like this. These are all things that they can control with their body.

- 4.5 But when I got older and took biology classes and so on in high school, and I realized this is weird because they all have super similar genomes. So it's not like these animals are built out of extremely different instruction manuals. It's just that they have different things that they can control with their body. And so it took me years to figure that out. You know, this question of, like, OK, and when you look at animals, how do mountain goats get so good at leaping up rocks? Have you ever looked at a mountain goat doing its thing on a cliffside? It's unbelievable. Or owls with their wings plunging down on a mouse. They're really good at what they do. Or, you know, a frog will sit there and a fly will come by and the frog will hit it with its tongue. It's quite extraordinary.
- 4.6 And so the question is, these things have different motor capabilities and they're really good at it. And so how does all that work? To understand this, let's return to the Potato Head model of the brain, which I introduced last time, in which various sorts of things can get attached, and the brain just figures out what to do with it and how to operate it. So I'm going to suggest that this exact same principle applies to outputs. So whatever you've got, you know, you got wings, weird things, trunks, whatever, the brain can figure this out. Any kind of device that you have that you can operate, Mother Nature can treat this as a plug-and-play motor device.
- 4.7 And so, you know, this is true whether it's fingers or flappers or wings or fins, whether you have two legs or four legs or eight legs, the brain just says, "OK, cool, that's what's plugged in. I'm going to figure out how to operate that." Talons, wings, all that stuff. So the idea, just like last time we talked about the senses, is that Mother Nature doesn't have to redesign the brain every time once she's figured out the principles of brain operation. All you need to do is plug and play.
- 4.8 Now you might say, wait a minute, last lecture you talked about this with senses, and you made this genetics argument that things like eyes and ears and noses can— you know, there's genetic programs that build those peripherals. And I showed last time that you can get born without a nose or without eyeballs or without an ear or something. So the question is, is the same kind of thing true for the motor system? Some years ago, people discovered these genes called homeobox genes, which are like master controllers that change what happens. So one of the first homeobox gene discoveries was a mutation in fruit flies where their legs grow where their antenna normally would be. And there's a reverse mutation where you can have the antenna grow where the legs normally would be.
- 4.9 And so when I saw this, I thought, my gosh, it's so strange with, genetically, you can control this stuff so that, you know, you say, hey, unpack legs, or unpack antenna, or, you know, you can just put things in different places. When I say you can put things, I mean that's what the genome is doing. It just says, "OK, I'm going to put this thing here." OK, and so you might say, well, look, if there are these master controller genes that build things, maybe we'd expect to find weird things in humans. Like, why aren't

humans born with tails sometimes? And the answer is, they are. So there are a number of babies born every year with tails. And you might not have known this because this always gets surgically removed right at the beginning.

4.10 The tail, of course, is just an extension of the vertebral column that keeps going. And in our species, it just so happens that we don't have those. In great apes, we don't have those. But it's an easy enough genetic tweak that you get a tail, and so it just gets removed. But the point I want to make is all it takes is just a tweak of the genome for that to happen. And children are born sometimes—it's not that common—but children are born sometimes with an extra limb. So this is a baby named Jie-Jie who was born in China recently and has three arms. And sometimes what happens is that one of the arms does worse than the other in utero. And so it's an easy choice to make about which one to remove. But in the case of Jie-Jie, both his arms were good and strong, and they were connected all the way to the shoulder blades. And so it was a difficult surgery to figure out what to do there.

4.11 But the point that I want to make here is that tails and extra arms, what this illustrates is the way that body plans can change unmistakably with very small genetic mutations. And it goes without saying, of course, that when you look around any crowd, you find lots of genetic differences between people. Some people are taller and shorter, some people are wider and thinner. People have, you know, longer arms or stubbier fingers or a big toe that's longer than the second toe, or, you know, wider hips, broader shoulders, all these things. There's a lot of genetic diversity going on all around us.

4.12 When you look at chimps who are our nearest neighbors in the animal kingdom, they are genetically nearly identical to us. But they've got these really obvious differences in the body plan. For starters, they have, you know, the insertion point of their muscles into the bones is different, so they have much more leverage. That's why they can swing in trees easily and so on. It's much harder for us. You know, we try and say, "Oh, my arm hurts," but yeah, they can swing around because they just have different, slightly different things. Their toes are longer, their hips are turned more outward, stuff like that.

4.13 Now, the point is, they're almost genetically identical to us. The chimpanzee brain perched inside the dark throne room of the chimpanzee skull, it doesn't have to be completely rewritten to figure out how to use this and swing around and walk on these, with these hips and so on. It doesn't need to be rewritten. All it needs to do is figure out, "OK, how do I drive this chimpanzee body?" Just the same way that human brains don't have to be rewritten to figure out how to compete in a ping-pong tournament or do a pogo stick or dance salsa or anything like that. You just figure it out. You figure out how to control your body appropriately.

4.14 OK, so the point that we're going to see in today's lecture is that the brain figures out how to best drive the machinery in which it's embedded, OK? And so to understand the power of this sort of thing, consider Matt Stutzman, who was born without arms, and

he was very attracted to archery. And so he learned how to manipulate a bow and arrow with his feet. And he notches the arrow into the string with his toes, and then he lifts the bow with his right foot and a strap holds the apparatus against his right shoulder. So he positions the thing, he finds the target, and then he lets it fly. And Matt is not simply talented at archery. He is actually the best archer in the world. He has the world record for the longest accurate shot.

4.15 Now, this is probably not what the doctors would have predicted for him when he was born and they saw him there. But if there were naysayers in his life, it's probably because they didn't realize how readily his brain would adapt its resources to solve problems in the outside world. So the brain readily adapts whatever it's got, whatever machinery it's got. And it turns out, there's this beautiful dog named Faith who I have videos of, but I'm just going to show a picture today. But this dog was born without four limbs.

4.16 And so what does she do? She just walks bipedally, and Faith, you know, just moves around. She looks kind of like a human. She just walks around on her back legs and she's very mobile and obviously gets a lot of attention, and everyone watches her when she's moving around. Although you might have guessed that dog brains are sort of preprogrammed to drive dog bodies, they're not. They can just as easily do this sort of thing. And we'll come back to this question of, why don't all dogs do that? We'll come back to that in a few minutes. But the point is, her brain can figure that out.

4.17 OK, so what archers born without arms or dogs born without forelegs, what they shine light on is this fact that brains are not predefined for particular bodies, but instead they adapt themselves to move and interact and succeed in the world. And this is not just about the body that you're born into, but whatever opportunities come along. So, for example, there's a bulldog in California named Tyson who has mastered skateboarding. He's excellent at this. So he goes and he jumps on the skateboard, and then with his front paw he scrapes the ground and gets a bunch of speed. And then, you know, he puts all his paws on the thing, and he leans his body, and he's pretty much just as good as a human skateboarder that way. And when he's done, he lets the skateboard slow down and then he hops off and he grabs the skateboard in his mouth and comes back.

4.18 Given the conspicuous absence of wheels in the evolutionary history of dogs, what this demonstrates is the adaptability of brains to steer new possibilities. There are actually lots of dogs who do stuff. There's this dog named Sugar who took up surfboarding and has been inducted into the International Surf Dog Walk of Fame. And if you watch Sugar and all her competitors doing this stuff, it's quite extraordinary. They're quite good at it. They surf the wave and they figure out how to do it. What strikes me is, forget Sugar, just be amazed that there is an International Surf Dog Walk of Fame because it's so crazy.

4.19 So canine brains aren't typically studied in the context of whether they can hang ten on a surfboard, but they certainly can be. All it requires is the opportunity, and then the

brain will figure that out. Now, the thing to note about Tyson the skateboarding dog and Sugar the surfing dog is that they're really good at what they do, at riding the streets or the waves. The question is, how did the dogs get so good? So that's what we're going to explore. And to get there, we're going to look at a related thing here, which is, how do young Homo sapiens get so good at speaking language?

- 4.20 OK, so a baby learns how to produce language not by genetics and not by surfing Wikipedia to look things up, but instead by babbling. So what a child does is babble. Sounds come out of her mouth, and her ears pick up on the sound. So, you know, coming out that way and going in this way. And her brain can compare how close those sounds were to what she's hearing from her parents and other people around her. And what happens is she gets positive feedback for some kind of utterances, and most, you know, not others that don't make sense, but the things that make sense. Everyone says, "Oh!" and pays lots of attention to her, and she says, "Oh, I guess I did something right." And so in this way, the constant feedback allows her to refine her speech until she speaks English or Chinese or Hindi or Farsi or whatever, any of the 7,000 languages on the globe that people speak.
- 4.21 Now, that's babbling. But what's interesting is, we learn how to control our bodies with something similar. And this is what I call motoric babbling, or motor babbling. And this is how babies learn how to control this giant meat suit that they're in. So observe that same baby in her crib. You know, she bites her toes and slaps her forehead and, you know, tugs on her hair and bends her fingers and hits the bars of the, you know, the crib and so on. Because she's learning how her motor output corresponds to the sensory feedback she gets. It's the same thing as doing language, but now it's just a broader sort of thing. So in this way, she or he learns to understand the language of the body. In other words, how the outputs map onto the next inputs. And this is what we do with motor babbling.
- 4.22 And this is how we learn, how to eventually walk and how to navigate food to our mouths and how to stay afloat in a pool or dangle on monkey bars or master jumping jacks. Or anything like that, is by trying and getting feedback. And even better, what we're figuring out is not just how to master whatever bodies we have, but how to attach extensions to our bodies and then master those. So think about a bicycle or any machine that our genome didn't see coming, but here it is. So our brains originally shaped themselves for climbing trees and carrying food and fashioning tools and walking long distances and so on.
- 4.23 But riding a bicycle is a whole different kind of challenge. You have to carefully balance your torso and move your legs in a certain way. And when you want to change directions, you do so by changing your arms this way, which is not what you do when you walk. And when you want to stop, you do it by squeezing your hand, which is a totally weird thing to do. It's not like anything we've ever had in our evolutionary history. But despite the complexities, any five-year-old can master that, can

demonstrate this extended body plan into the resume of the motor cortex. So remember, it's her motor cortex that's figuring out exactly how to run this new body that she's on. And that's what she's doing here. This is what we're going to understand how.

- 4.24 OK, and by the way, I want to say this isn't just about typical bicycles. I don't know if any of you guys ever saw this TED Talk, but this guy named Destin Sandlin was given a bicycle that his engineer friend had made where it was a reversed steering thing. So when you turn the handlebars to the right, the front wheel turns to the left and vice versa. So Destin thought, "Oh, I can do this. This is no problem." And so he ended up filming himself over the course of weeks. I think it was maybe a couple months. And, you know, it's so hard to ride. And you know, because every time you turn, it goes the other way. And it's very counterintuitive. So the thing is, it was not as easy as having a cognitive understanding.
- 4.25 So Destin is an engineer, but even though he knew full well that when you turn this way, the wheel turns that way, it was not sufficient to be able to ride it. He was super terrible at it for a long time. And this actually relates to a question that came up yesterday. Someone asked during Q and A about whether, you know, you can just press a button and know how to do something instantly, but the answer is, you can't. It's impossible to do that because what you need is actually burning the practice into your circuitry. This is what we'll talk about in just a moment. Anyway, so Destin knew what was going on, but couldn't do it until he put lots and lots and lots of work into it.
- 4.26 And once he did that, he finally started getting the hang of it. Why did he get the hang of it? It's because each time he made a move, he'd get feedback. OK, you're falling to the left, you're crashing into a mailbox, you're swerving in front of the pickup truck, whatever. He was getting feedback from the world. So his brain would put out an output and get feedback and say, whoops. And so he used that feedback to modify the next moves. And after several weeks of practice, he got really good at it. He mastered this strange bike in the same way that you would master, that a child masters a normal bike, which is by motor babbling.
- 4.27 And if any of you have ever been in a country where the steering wheel's on the other side of the car, you know, it takes some real practice to get used to that. And it's scary at first, but if you keep doing it for a while, you're getting feedback, and that's how you learn how to do that. And this is how Tyson and Sugar learned as well. Now, I just want to mention something which is, it seems strange that you can learn how to do the steering wheel on either side, or Destin could learn how to do a bike this way and that way. The key is that when he got back onto his normal bike, he crashed on his normal bike. But pretty quickly he got the hang of both of them. And now when he jumps on one bike or the other bike, he's able to do the right thing. Exactly like when you're in the other country, if you go over to Ireland or something, you drive on the other side. No problem.

- 4.28 Now, this is what in the field is called schema. So your brain has a schema where it says, "Oh, OK, got it. I'm in this context. So this is how I run my motor system. Now, I'm in this context. This is how I run it." OK? So, for example, when you're on a bicycle, you're, you know, when you want to come to a stop, you squeeze your hand. But obviously, if I'm walking, I don't do that. I don't think, "Oh, I want to stop, so I'm going to squeeze my hand." Why? Because I have a schema. I know the context that I'm in.
- 4.29 Here's an example by which I recently got a deeper view of my schema, which is, I was throwing a football around with some friends. And the football ended up hitting my truck and the rearview mirror fell off. So, no problem. I figured I'd glue it back on later. And what I noticed is, when I was driving, I kept looking up towards the top of the trees, and I kept thinking, "Wait, why am I looking over there?" And I realized it's because, obviously, my mind was looking to the rearview mirror so I could see behind me, right? But now the rearview mirror was missing, so I was just seeing the trees.
- 4.30 But the thing is, I would never do that outside of my truck. I would never look up here to see behind me. But totally unconsciously, my brain knows when I'm in the truck that that's how I can have this other visual experience. So that's what finally got me to glue it back on, because it was so crazy and annoying that I kept looking over there. But that's the key. So when I'm in my truck, I've got a particular schema. When Destin's on his reverse bike, he has a particular schema. In other words, your motor system says, "OK, here's how I control my body in this context." And so it's quite extraordinary.
- 4.31 Let's return to motor babbling. So there's some kind of extraordinary, which is that, it's not only the way that babies learn and bicyclists learn, it's also become a new approach in robotics. So this is a robot done by a colleague of mine named Hod Lipson. And the way this robot learns how to walk is by making a model of itself and constantly improving that model with feedback from the world. It does motor babbling, and it figures out what's working and what's not working. And that's how it figures out its own capabilities. In other words, you don't preprogram this robot. You just let it figure out. So all that the Starfish robot has is this instruction of "go from the left to the right across the table."
- 4.32 So what it does, it looks a little creepy, sorry. But it's pretty interesting because what it does is it figures out, "OK, I'm just going to try all these things." And all it has is a gyroscope in the middle that's telling it, "Am I falling over?" And so on. And it just figures out how to use its limbs this way. This is how it figures out how to move. And then what happens is he'll, like, break a leg off of it. And it figures out how to walk with a broken leg. Now, the key is, what it tries to do at each moment is, "OK, what is the next move I'm going to make that best distinguishes the available remaining hypotheses about how I should move my body?" But the point is, it's using motor babbling to figure out how to operate its actuators.
- 4.33 OK, which is a very straightforward. I mean, I know this looks creepy, but it's a very straightforward way of doing robotics, which is exactly how animals learn in the world.

Here's the idea. Building a babbling self-exploring robot is way better than preprogramming the robot. And this is what Mother Nature figured out a long time ago. In the animal kingdom, every animal comes to the table with only, you know, like 19,000 genes. You don't have that much room to tell it how to do everything. So what do you do? You build a system that just figures it out. That's the key. And this is the trick that allows the surfing dogs or the skateboarding dogs to master their craft. They're babbling with their bodies. They figure out, "OK, I'm going to try various moves and positions and postures and balances, and I'm going to assess the result."

- 4.34 So, you know, Tyson the bulldog figures, "OK, I'm going to lean to the left. Does that work?" You know, Sugar the surfing dog videos are, "I'm going to, you know, move my body this way. Does that dump me into the cold water or does it accomplish what I want?" The key is that the feedback allows the motor system to fine-tune millions or billions of parameters on the fly. "OK, I'm going to try this. I'm going to get this feedback. OK, great. And this is going to do it." And this is how a body learns its environment. It figures out its interaction with its environment. It comes to grasp its capabilities and its, and the consequences of its movement.
- 4.35 And so there's this feedback between the internal and the external world. And that's the key to motor babbling. But I also want to mention, it's the key to everything we do, the same with what I'm calling social babbling. Consider how you learn and how you continue to learn interaction with other people. You constantly are putting social actions out into the world. Like, "OK, I'm going to say this funny line at the party," and then you get feedback about whether that worked or not. That modifies how you behave next. You rove the space of possibilities. Your whole life you've been doing this. You try out multiple personas when you're young. You act differently with your friends and your parents and this and that.
- 4.36 And, you know, you see, you know, should I cry in this situation? Is that going to get me what I want? Should I be defiant in this situation? And so on. And even when you're an adult, you figure, "OK, should I tell this strange joke at the party or should I tell that at work, or should I not do it?" Or whatever, you figure this out. OK, so just like the human who at different times is operating a mountain bike or a hang glider, ice skates or whatever, you adopt different schemas for different social situations as well. But the key is, just like with the motor feedback, you are relying on social feedback. It turns out this is how you think.
- 4.37 So thinking is just like motor movements. So I don't know if you guys ever thought about this way, but, you know, when you make an action, like, I'm gonna lift this up or whatever, when you're thinking about, "OK, what if I do this and this and this?", it's exactly like you're manipulating things in the outside world, except that you're not actually moving and manipulating something. You're manipulating things on the inside world. That's what thinking is. And so the key is, you know, thinking is remarkably similar to motor movements this way. And the neural storm of activity that represents

a thought, like, "Hey, what if I do this?" That is the same as, you know, doing that. It just doesn't put activity in the motor cortex that shoots down your body.

- 4.38** The key is that you do this with the same sort of babbling. You think, "OK, what if I did this? No, that's not gonna work. OK, what if I did this?" You simulate it. "Oh, that's dumb. What if I—oh, hey, that's pretty good. I'm gonna do that." And so, some thoughts map well onto the world. Like, "Hey, I wonder if I pulled that cord, if that would start the lawnmower. Oh, that actually works." But other things don't. Thoughts might get tried, might get babbled by your brain, like, "Oh, what if I throw my pancake across the room? Would that be interesting?" And then you simulate that and you reject that thought and so on. But just like movements and speech, your thoughts need to learn how to operate in the world.
- 4.39** Back to just the body and motor system, the principle that we're talking about is that the brain figures out how to control whatever body it's in, whether dog genetics give you four legs or two legs or wheels instead of paws or a surfboard instead of a surfboard-like skeleton. The dog brain doesn't have to be redesigned. It just says, "OK, I'm going to recalibrate." And so think about how effectively this strategy creates biodiversity. Because the idea is that a live-wired brain does not need to be swapped out every time you make a genetic change. Like, OK, now we're going to make a mutation that causes a trunk here. You don't need to redo the entire brain. You say, "That's cool. We'll figure it out." And you can make animals fit any habitat this way, whether hooves or toes are appropriate, or wings or fins or forearms or whatever, says, "Cool, I got it."
- 4.40** So the fact that Mother Nature doesn't need to do anything extra, I assert this is the only way that this could have worked. The only way that evolution could have possibly worked is by the strategy of just recalibrating the brain, because it would take too long otherwise if you had to re-, you know, re-figure out the principles of the brain every time. So evolution wouldn't work unless body changes were really easy to deploy and brain changes could follow without difficulty.
- 4.41** This massive flexibility, this is why we can so easily install ourselves into new bodies. Anyone here see Aliens? Anyone see the Aliens movie? OK, so there's the scene at the, at the end where Ellen Ripley, the protagonist—you know, there are these big factory suits where you can get in this thing and control this giant robotic body, and she ends up having a fight with the giant alien at the end. She climbs into the suit and fights the alien. And at first she's trying to figure it out. You know, she's controlling this whole body. But eventually she lands a punch on the alien's mucousy face and then, you know, lands another punch. She figures out how to do this.
- 4.42** But presumably what Ellen Ripley is doing is the same thing. Like, "Oh, am I leaning too far to the left? Am I, you know, whatever?" She's getting feedback with her new body, and as a result, you know, is able to avoid being alien snack food that way because she's able to learn a new body plan rapidly. And the key is that it's easy to see these

new associations if you look around with, like, forklift drivers or crane operators or laparoscopic surgeons who are doing this with, you know, robotic surgery and so on. All of these people get out of bed every morning to pilot strange new bodies. I mean, it's just not that hard.

- 4.43 Although Ellen Ripley is fictional, this is the principle that applies to rollerblades and unicycles and wheelchairs and surfboards and whatever. Anything that we can lash onto our bodies, we do. And the specifics of the device's weight and joints and movements, this all works its way into your brain circuitry. And in fact, in the earliest days of plane aviation, like really when planes were first coming out, what people would do is tie ropes to their body and they would change the wing shape and so on to land, take off things like this. They were really, literally, becoming one with their airplane. Now, it's a sort of different thing where you just sit back, most of it's on autopilot, but they were actually one with their plane.
- 4.44 And I think about, as a neuroscientist, I watch this all the time. Like when I watch a concert pianist and I'm watching the concert, I'm like, "Wow, she is one with her machine." Like, she and the piano are kind of one mega-organism. And that piano has worked its way into her brain such that she knows exactly what is going to happen there. It's the same with chainsaw lumberjacks or drone pilots or anything. Their brains have incorporated their tools as an extension of their body. And by the way, this is the same with blind people who are using a cane. The cane not only extends away from their body, but it also extends into their neural circuitry.
- 4.45 Consider what this means for our near-term future as humans. So imagine that you could control a robot at a distance, you know, just with your brain activity. So you got your robot across the room and you say, "OK, can you get me a cup of coffee?" But, sorry, you don't say it, you just think it. Like, "Get me that cup of coffee over there." You know, and you want it to lift its arm. It does that. You want it to squat or pirouette or whatever you want it to do. It would heed your mental command without delay or mistakes. Now, this sounds like science fiction, right? But this is already underway. So let me start by going back to 1995.
- 4.46 So does anyone know who Jean-Dominique Bauby was? Does anyone know that name? OK, so he was the editor of Elle magazine in Paris. He was on top of all the social circles. And one afternoon in 1995, he had a terrible stroke that plunged him instantly into a deep coma. And he awoke 20 days later. And he had what's called locked-in syndrome, where he was perfectly conscious and aware, but he couldn't move any of his body, with one exception, which is he could move his left eyelid. So he could wink his left eyelid. It was the single thing that he was able to do. He couldn't speak, he couldn't cry out, he couldn't move his toes, he couldn't do anything.
- 4.47 So what happened is, he had these wonderful persevering therapists who, coming in, and what one of them figured out is that she could get him to communicate by saying every letter of the alphabet in order of frequency. So you'd say like E, T, R, S. You say

every letter. And then he would wink his eye when it was the letter that he wanted. And so one letter at a time, he could communicate like this. And what happened is, with the help of these angels, he ended up writing an entire book this way. Does anyone know what the book is? Does anyone? Can you guess what the book is? It's called *The Diving Bell and the Butterfly*. Anyone heard of that book?

- 4.48 So it was a big bestseller. And what happened is, unfortunately, he passed away two days after his book was published. But this became a giant bestseller all across Europe and around the world. And everyone who read this cried their eyes out because it was about what it is to be trapped. And his metaphor was about being trapped in a diving bell. But it really makes you appreciate what it is to be able to have communication between your brain and this giant robotic meat suit so that you can move around and do things, right. It really gave everyone a deep sense of what's going on there.
- 4.49 Now why couldn't Bauby move? It was because his spinal cord was damaged from the, from the stroke. Now here's the key. You know, your brain says, "OK, here's what I want to do." And then it sends these electrical signals down your spinal cord. From the spinal cord, it goes to all these muscles, and that's how you move your body. You know, the best thing would be if you could just fix a broken spinal cord. But unfortunately we're not able to do that even today. You can't just fix a spinal cord. It's too complicated. You've got all these fibers going on.
- 4.50 So the next best thing, though, really the solution that we have is, could you eavesdrop on the brain? Instead of using eye blinks, could you actually eavesdrop on what the brain is trying to do? So this really began in earnest, I would say, about a year after Bauby's death, just by coincidence. There were researchers at Emory University who implanted a brain-computer interface into a locked-in patient named Johnny Ray. His motor cortex was able to get signals through this way because they were eavesdropping with these electrodes, and they could—so he could pass that on to the computer. So he could draw a circle or open some things with the computer or whatever, because you were listening to the motor cortex. And even though he was completely locked-in, there was a way of communicating in a way that was better than eye blinks.
- 4.51 OK. And then by 2006, there was a football player who was paralyzed, and he could crudely open and close an artificial hand using signals from his motor cortex. Control lights, open email, play the video game Pong, draw a circle on the screen, that kind of thing. And this was about 100 electrodes that were implanted in his motor cortex. That was sort of unpolished technology, but it proved the possibility. And by 2011, a neuroscientist named Andrew Schwartz and his colleagues at University of Pittsburgh built a very beautiful prosthetic arm.
- 4.52 And there's this woman named Jan who is paralyzed, completely paralyzed, from the neck down. And she was able to control this robotic arm with great suppleness and subtlety, just by her thoughts, as in, "I want to move that arm over there," and it moves

according to her thoughts. Now, I'll come back to this in a minute, but it takes a lot of practice. Why? Because the researchers don't know precisely what the motor cortex is trying to do. And so they're just pulling the signals and making that make the robotic arm do something. And so it's up to Jan to figure out, "OK, wait, I want it to actually reach over here and grab this." So she has to learn it in the same way. I don't know, I guess it would be something like if I tied a bunch of weights to your arm, and at first you're doing the wrong thing, but you'd get a lot of feedback and do a lot of motor babbling, and then you'd figure out how to use it. That's what it's like for Jan to be able to do that.

4.53 So, as Jan said, you know, "I'd much rather have my brain than my legs." And what she meant by that is, if you have your brain intact, you can learn how to control a robotic body like this. She's able to use this technology because her brain is rewiring to understand how to use this new limb, just like you would understand a reverse bicycle or a surfboard or Ellen Ripley's mech suit. Now, some of you might have seen recently with Neuralink, which is another company that's injecting electrodes into the brain. The only difference here is that it's wireless and smaller.

4.54 And so this was the first patient, and the patient is able, again it's the same idea. You're eavesdropping on the motor cortex and saying, "What is the cortex trying to say?" And what they will be doing soon enough with this is allowing him to control things this way, including—there are lots of ways of doing this. One way that scientists are thinking about is, what if we actually hook things directly up to the muscles so that when you think about, you know, I want to move my forearm, you're actually stimulating these electronic devices that move your forearm. Or you can do something where you build an exoskeleton, where you have the exoskeleton that is, you know, around a paralyzed body, and you think, "OK, I want to move the, the legs. I want to move the arm," and so on.

4.55 The idea, as this develops, which it's moving fast, is you put in higher and higher density electrode arrays, and then you can say, "All right, I'm going to control the body that way." Now, what's fascinating is, it's not only about controlling your own body, but could you do something like add an additional limb to a body that already exists? So this has been done. This is in 2008. There was a monkey with two normal arms, a normal body, and they used his motor cortex to control a third arm, a metal arm. And he's able to use this arm to put a marshmallow into his mouth, to grab the marshmallow and stick it in his mouth and so on. As he figured this out and did a lot of motor babbling on it, he was able to control the robotic arm from the marshmallow-gathering, all without any—at first, he was using his real arms to try to do it, but then now he keeps his arms still and just uses the robotic arm. It became a new limb this way.

4.56 And so the way that this monkey learns how to control this robotic arm recalls Doc Ock, right? It's the same thing where Doc Ock gets to use these extra arms, even while doing

other tasks with his fleshy arms. What the monkeys have begun to do is devote neural real estate to these robotic arms, distinct from the natural limbs. They divide their resources and assign them to different appendages, real or metal. Now, for both Jan and for the monkeys, the robotic arms weren't directly connected to their torso, but they were instead connected by a bundle of wires. But what if you could do this where the arm wasn't even in the same room? Could you control a robot across the world? So it turns out, yes. This was also done in 2008.

- 4.57 A colleague of mine named Miguel Nicolelis and his team at Duke University hooked up electrodes into a monkey's motor cortex, and the monkey walked on a treadmill. And when the monkey would walk, that controlled the walking pattern of a robot in Japan, halfway across the globe. As the monkey walked, his motor cortex signals were recorded and translated to zeros and ones and sent across the Internet to the laboratory in Japan. And then like a big, you know, five-foot-tall metal doppelganger, the robot would walk too.
- 4.58 So, in the not-too-distant future, it seems inevitable that we will mind-control robots in our factories, underwater, on the surface of the moon, all from the comfort of our couches. Because, why? It's because our cortical maps, after extensive training, can incorporate the robots' actuators in the same way that we can control a bicycle or anything else. They become like tele-limbs to us, as in, you know, limbs at a distance.
- 4.59 So the truth is that our meaty bodies have evolved very specifically for the atmosphere of this planet. And so that means for space exploration, it's really tough. You know, as it stands now, we have to get in these big suits, these big space suits, and they have to be super careful, make sure nothing leaks, and you got oxygen and all this stuff. The reason all this has to happen is because our bodies are really terrible for going onto the moon or into outer space. And so this, I assert, is probably our best strategy for space exploration, is controlling robots with our minds that are essentially part of us. They're tele-limbs.
- 4.60 One thing I've been wondering about is, you know, what would an expansion of your body, what would that mean for your conscious experience? And so I think the answer is that the robot that you're controlling would become a part of you. You would feel like it's part of you in exactly the same way that you feel about your limb. Like, I have no doubt that, you know, my leg is a part of me. I don't even wonder about it. And by the way, we have experience with tele-limbs. So when you look in a mirror and you move, that thing over there is moving exactly as your brain predicts. So your brain makes a prediction. "OK, I want to lift this arm." Oh yeah, that thing did exactly the same thing. It's always perfectly synchronized with you. And I think this is why we come to understand our mirror image as ourselves.
- 4.61 And I suspect that if you had a robot and you said, OK, you know, you have the brain coming, OK, lift an arm, and it lifts an arm, then it becomes part of you. And so, you know, Mother Nature is very talented obviously at making things that have, you know,

sinew and bone and whatever, but she wasn't, you know, she never invented Bluetooth, but we have. And so we can make that happen. And you can have these tele-limbs become a part of ourselves.

4.62 And so my assertion is that whatever you can control becomes a part of the self. And, you know, I don't know if any of you saw Star Trek, but there's a thing called the Borg which just tries and goes to incorporate everything. And everything that it can incorporate becomes a part of itself. Except, and the one thing it can't control is Captain Picard. And so he retains his identity. But I think that this hypothesis, that what we can control becomes itself, to my mind, this allows us to understand certain disorders. There's something called asomatognosia. What I'm going to propose here is that the way to understand this is all about predictability. Like, I want to make this move, and then the move happens.

4.63 So asomatognosia translates to not knowing one's body. Gnosia means knowledge. Somato means body. So, somatognosia means knowing one's body, and asomatognosia means not knowing one's body. And so what happens is, someone gets damage to a particular part of their brain, usually from a stroke or something like that, and they are no longer able to control a limb. They just, they, you know, they're not able to control, let's say, their arm. And as a completely wild result of this, they will deny that the limb belongs to them.

4.64 So here are some different patients, you know, just saying, you know, oh, this is the, this is the neuropsychologist's hand. It's not mine. Or, you know, this is my sister's hand. It's not my hand. My hand's on my belly, but I'm just too fat that I can't see it, or I don't know, it's not mine. It's just the two of us. So I guess it's your hand. You know, things like that. What happens is the patient will deny that this thing is hers or his. This is one of the wildest things that I've seen in neurology. But I think it's totally understandable if we take it from this point of view, that what's going on is that you no longer have the prediction. Your brain says, "OK, now I want to move this hand," and it doesn't move. So you say, "Oh, OK, I guess it's not mine, then." It's not part of the self. It has fallen from the brotherhood of the self.

4.65 Now coming back to the issue of controlling other bodies, you know, you don't necessarily need prosthetics or brain surgeries to control their bodies. There's a whole developing field of robotic, of avatar robotics, where you control a robot at a distance just based on Bluetoothy things where you move your hand and the robot moves and so on. There's one called Shadow Hand. There are all kinds of interesting companies coming up. One is, for example, using an avatar robot in a disaster situation. So there's a big, you know, fire and there's chemicals and whatever. So, you, the operator, puts on the suit and you're moving this robot. You know, could be eight miles away, but you see what it's seeing and you're moving its body by making your own motions. And you get to experience being in another body that way.

- 4.66 And what's interesting, by the way, is everyone's making these bodies look sort of like humanoid bodies. But you certainly could make it any kind of strange-bodied avatar that you want, in the same way that we learn how to use skis or trampolines or pogo sticks or whatever. Presumably an avatar operator could learn how to operate any sort of weird and wonderful avatar body.
- 4.67 OK, now what's cool is that you don't even, nowadays, you don't even have to build external stuff like that to really test this, to really understand what you can do with new bodies, because you can do this in VR. So one of my colleagues at Stanford, Jeremy Bailenson, he made a game where you have to pick up boxes of certain colors as fast as you can. But in this game you have three arms. So you've got your two normal arms that you're controlling with your controllers, and the third arm is controlled by the, by the rotation of the two controllers. And it only takes a few minutes for people to figure out, to get really good at it and use all three arms to do the picking of the boxes.
- 4.68 And that's, it's quite extraordinary to see how rapidly we can change this. This is what's called homuncular flexibility. Remember, the homunculus is the little, the little man, the map of your body inside your brain. Homuncular flexibility just means this issue that you can learn how to control a new body quite rapidly. And there's something I've been thinking about philosophically, which is there are certain hypotheses. For example, one is about language, that the language you speak determines what you think. And this is called the Sapir-Whorf Hypothesis, which is that the way that you are able to control language and so on, and across different languages it's different, changes the way that you think, the way your thoughts go.
- 4.69 I've been wondering a version of this, which is, does the way you can control your body change the way that you think? So this is a gentleman that I met who lost his arm in a smelting accident. And so these prosthetic engineers made this very beautiful prosthetic hand for him. But what they realized, sort of accident, I don't know if this is on purpose or not, but they realized, oh, there's no reason to actually, like, put tendons in there that stop the wrist. So he can just sit there and rotate his wrist around and around and around like this. And so, you know, what does that mean, if your body can do something totally new like that, that other people's can't do? Like, so when he thinks about, "I'm going to change this light bulb," is it a different kind of thought that he's having? Is it a different kind of consciousness that he's having?
- 4.70 OK, so let's summarize today's lecture. So, as we saw with Matt the archer or Faith the dog, brains learn how to, you know, they adjust to drive whatever body they are, they find themselves in. And like Jan's robotic arm or this monkey with the marshmallow, neurons can add new sorts of things to this. They can operate new hardware additions, and networks in your skull accomplish this by motor babbling, by putting out commands like lean to the left and then assessing the feedback like, oh did that work? You know, the skateboard tilted and wobbled or whatever, and then adjusting the parameters, and that's how it climbs the mountain of expertise. And all these stories

that I chose to tell, they all point to the same fundamental principle, which is whatever kind of skeleton you come to the table with, with whatever kind of muscles, whatever kind of possibilities for locomotion you discover, the brain on the inside does not need to be redesigned. It simply recalibrates.

- 4.71 And so the, you know, just in closing, I'll say the thing that's so interesting to me is that our progeny, I think, in the very near future, won't have to limit themselves to the boundaries of their bodies the way that we do. Instead, they can extend across the universe according to whatever is under their control. OK, so I'm going to wrap this lecture there and I'll take any questions. Yes.
- 4.72 Assuming that all of these technologies will eventually help people with ADHD, bipolar, or even psychopathy, until we get to that point, I wanted to know, it has said that we use only 10% of our brain at this point. And you've been talking about how the brain has tons of real estate, 800 billion neurons. I wanted to understand—or like a forest—I wanted to understand then, why is it that we sort of stay in a hamster wheel of our thoughts and our lifestyle and limited potential, even though many people would like to advance, such as, you know, addictions or becoming, you know, better at a sport or becoming more successful or world peace or ending domestic violence, or, you know? It's like we are stuck. And so I wanted to know how that can help, you know, mental illness and psychopathy. And is it related to receptors like dopamine and serotonin, like the reward system that has to be in place?
- 4.73 Let me say a couple quick things, which is it's not true that we only use 10% of our brains. You actually use 100% of your brain all the time, even when you're sleeping. So that's a myth. And I think the reason that myth is so sticky is because people like to think, "Oh, if I could just tap into that, I can be so much smarter or whatever." But what I'm talking about here is how your brain learns how to drive your body and how you could add extra sorts of limbs. I will say that addressing things like psychopathy or ADHD, that's probably not going to happen from brain electrodes. Because the only reason this works well is because you're eavesdropping on the motor cortex, which is the area that is trying to tell your arm to do something.
- 4.74 And so you're just saying, "Hey, what are you trying to tell the arm to do?" And it's all right there. But something like, you know, take psychopathy, it's, there's a million brain changes. It's not like there's one area that causes someone to be a psychopath or not. It's that the whole brain is wired differently in terms of empathy and future prediction and blah blah, blah. So there's no way. You'd have to stick electrodes everywhere in the entire brain. So that's not likely to happen soon. But driving the body is actually a much easier problem in a sense.
- 4.75 Kind of going off the theory of whatever you control becomes a part of you, what do you imagine the emotional extent of that being? Like, if emotions will also extend out to whatever you are controlling or in pain, how do you imagine that will work?

- 4.76 It's a lovely question. I can tell you, when there are army operators that control robots that are essentially bomb-sniffing robots and they control these things and it becomes, I interpret it as a part of them. It's like a tele-limb. And what will happen sometimes is the robot will get blown up in an explosion, and the soldier will end up crying because it's like they've lost a part of themselves. And I think that is absolutely right, that we probably attach emotions to everything we can control. I mean, for example, if you damaged your hand or something, you'd feel so, you'd care about that so deeply because that is part of you. And I think it's the same thing as we extend ourselves across bigger spaces.
- 4.77 So we've spoken about dreams, and then we spoke about how thoughts, we can think through situations. And so, in a way, it's like you can think so that your thoughts can die, so that you don't have to, in a sense. I'm curious to, and then you spoke about VR. Can you—I know lucid dreaming, people that lucid dream, they can kind of train in their dream, and that will actually incorporate an improvement in waking life. I wonder, is that possible, too, with what the technology that's being created? So can you use things to project and practice motor skills, like playing guitar or wrestling or something, without actually moving your body, but still, like, because your brain is working on it, can it be incorporated in, like, your actual physical being?
- 4.78 It's a really interesting question. There is a lot, I know a lot of athletic coaches who really believe in this idea of visualization, which is, you know, really picture the golf move or the swimming move or whatever. It probably has a little bit of an influence. But I have my doubts that you could actually do some great gymnastics thing or some dive by only visualizing it and not trying it. Because what you get from the world is the feedback that your brain wasn't smart enough to predict, which is, "Oh, I didn't realize that if I do that, the wind or my body weight or whatever." Because when your brain's making a prediction, it's always doing it imperfectly, just predicated on the little bit of knowledge that you have. So that's why you got to be out in the world trying things.
- 4.79 So you have been talking about the ability of the brain to use resources such as external limbs, robots, and stuff. So is it possible for the brain to use other brains to control other bodies?
- 4.80 Really interesting question. The answer is no, only because, in order for my brain to control your brain, I'd have to sort of somehow hook up all my neurons to all of your neurons, which would be a very thick cable, like 86 billion wires thick. So there's no clear way for me to do that. But it's easier, of course, for me to control a robot because I can send signals saying, I could actually send very small signals saying, "OK, now do this thing, like, get me the coffee." And that's just one bit for me to say, "Get the coffee." And then it goes and has some autonomy on its own.
- 4.81 But I think the idea of one brain controlling another brain is not possible in any way that I can foresee. Just because your brain is a vast inner cosmos and so is mine. And we'd have to figure out how to interface those. But of course, this is what language is,

is this very low-bandwidth, super low-bandwidth way of interfacing cosmoses. So you ask me a question from your universe, I answer back in some words. And yeah, we've got some linkage between our brains that way. It's just, it's not as high-bandwidth as we would want.

- 4.82 So you explain about how the motor system uses feedback, let's say, and the language too. And famously, like famously, the Stephen Hawking can communicate this way. Is it possible to, like, do the same thing with the reception part, like mind communication, in some way? There is some development in the area without using like our phonetic system to speak?
- 4.83 Let me make sure I understand. You mean to get information in or to get information out?
- 4.84 Out, like you and me. Can we communicate without the use of our phonetics?
- 4.85 Oh, oh, oh, oh. The answer is, I think that will be possible someday. OK, so in the same way that we use language right now, which is pretty low bandwidth, in theory, I could hook a part of my brain called Broca's area, which is, you know, essentially doing language output, to your auditory cortex. So we could do this thing where, like, my output is connecting to your input. But it would be a terrible idea, and here's why. Because the reason that we have all the, we've got all these filters with language that make it so that I might be thinking 100 thoughts like, "Oh, my toe itches, and I'm hungry or whatever," but like, you know, in this moment, I'm talking to you. And so all these unconscious thoughts get squelched while I'm doing this.
- 4.86 But if you could eavesdrop deeper into my brain, it would be a pretty messed-up conversation. So I don't think there's any purpose, only because we have evolved for literally millions of years to converse this way. So thank you. In fact, this is part of the mythology that people talk about with Neuralink or other brain implant devices is, "Hey, maybe I wouldn't have to type on my computer. I could just think. And everything I think shows up on the computer screen." Terrible idea.
- 4.87 I was wondering, with the story of Jie-Jie, the Chinese baby with the third arm that was fully functional. Other than the obvious reason of maybe the parents fearing that she would be judged or bullied in the future, what was the reason of operating and removing the arm?
- 4.88 Oh, it's an interesting question. I think it's probably exactly that. We live in a world where everyone's got two arms and so it's straightforward enough to do a surgical removal of the arm. It's a really interesting question whether if you re-ran that and had other parents, whether they would make a different choice. But I don't know. I mean, I suspect I would make that choice if I had a child with a third arm. I'd probably remove it because there's no harm in removing it. Jie-Jie's fine with two arms, but yeah, that's just an ethical question that would have to be asked of the parent themselves.

Lecture 5: The Science of Skill

At a glance — The science of skill and the technology of attention.

- 5.1** OK, so now we're going to talk about the technology of attention. So I'm going to start with a story about the Polgar sisters. How many people have heard of the Polgar sisters? Do you know them? OK, so their father, a guy named Laszlo Polgar, had these three girls and he loved chess and he loved his daughters. And so he launched a small experiment, which is that he and his wife homeschooled the daughters and they trained them rigorously on chess. So every day these girls would hop and skip these chess pieces across the 64 squares. And that's how they grew up.
- 5.2** So by the time the oldest daughter, Susan, turned 15 years old, she was one of the top-ranked chess players in the world. And in '86, she qualified for the men's world championship, which was the first time a female had ever done that. And within five years, she had earned the men's Grandmaster title. Then in 1989, her 14-year-old sister Sophia achieved fame for a stunning victory that she did in Rome. It was called the Sack of Rome, what she did, and she qualified for the men's world championship, and she went on to become an international master. And then there's the youngest sister, Judit, who is by all accounts considered the best female chess player in the world. And she achieved Grandmaster status at the age of 15 and remains the only woman on the World Chess Federation top 100 list.
- 5.3** So, what accounts for their success? Their parents lived by this philosophy that geniuses are made, not born. So they trained the girls every day. They not only exposed them to chess, but they gave them, you know, they fed them on chess, they gave them feedback, all the girls' feedback about, like, oh, approval and whatever the opposite of approval is. You know, all this stuff was predicated on how they were doing in the game.
- 5.4** Now, the reason I'm telling you about this is because what we've seen in previous lectures is that the brain reorganizes itself according to its inputs. But not all the information streaming into the pipelines is equally important. Brains readjust themselves based on what you are spending your time with. So if you decide to make a career change and become an ornithologist and watch birds, more of your neural resources will be devoted to learning the subtle differences between the wing shapes and the colors and the breast coloration and beak size and stuff like that. Whereas, previously, your neural real estate devoted to birds might be very crude, like, is that a bird or an airplane?
- 5.5** And when you look at champions in something, it's quite extraordinary. I saw this video of Magnus Carlsen, who's a world chess champ. And this other guy would set up the chessboard in different ways and say, "OK, what game is this from?" And Magnus Carlsen, it was beyond belief. He'd say, "Oh, that's the Anand and Ivanchuk game from

1994. Oh, that's Howell and Carlson, 2002. That's the 18th move. Or that's Kasparov versus Karpov." Like, he could recognize, just by looking at the board, these games. And so it's so striking to watch something like this, because presumably none of us in this room could do that sort of thing, because if you're not a chess player, it all looks the same to you. It just looks like chess pieces.

- 5.6 But what this shows is that, to have real expertise in something, it means you've actually devoted meaningful real estate to that. And there's an interesting story about the violinist Itzhak Perlman. After one of his concerts, there was this admiring concert-goer who came up to him and said, "I would give my life to play like that." And Perlman said, "I did." And what he meant by that is, every morning he drags himself out of bed at 5:15, and he takes a shower and eats breakfast, and then he practices for four and a half hours, and then he eats lunch and does some exercise, and then he practices for four and a half more hours. And he does this every single day of his life, except for concert days, where he only does the morning practice.
- 5.7 And so the point is that brain circuitry comes to represent where you're putting your time, what you're doing. And so, to the degree that the cortex of a highly trained musician like Perlman morphs into something meaningfully different, such that we can see it with the naked eye. You can see, even to an untrained eye, what the brain of a violinist looks like. If you pay careful attention to the motor cortex, so this is looking down on the brain from the top, that little part that's emphasized in pink there, that's essentially an extra fold that you don't find in violinists. This is what is called the omega sign because it looks like the Greek letter omega.
- 5.8 But the point is, for the violinist, you know, they're having to use their fingers in great detail here on the left side. And so the body and the brain are linked up on opposite sides. So the right motor cortex has to deal with all this very fine motor movement. And so you have to essentially build more stuff there, so you get more stuff packed in and this extra wrinkle. And so the brain of a violinist, the right hand is just mostly bowing. It's not as detailed. So you don't see that on the right-hand side.
- 5.9 By the way, so even though I just told you the story about Perlman, this actually isn't Perlman's brain. This belongs to someone else famous. Whose brain is this? This is Albert Einstein's brain. And you might ask, why did he have the omega sign? Does anyone know? Because he was a violinist. He would play violin. OK, so brain circuitry comes to reflect what you do, what you're spending your time with physically. And it turns out, you can distinguish the brain of, let's say, a violinist and a piano player. OK, so if you look at, let's say, Perlman the violinist and Vladimir Ashkenazy the pianist, they both share the same dedication to their craft and countless hours of practice and the grueling schedule and all that.
- 5.10 But their brains look so different that you can easily tell which brain belongs to whom, because a string player has the omega sign just on the right hemisphere because his left fingers are doing all this work, whereas a keyboard player has the omega sign on both

sides because both fingers have to do extremely detailed work. And so simply by eyeballing the brain at autopsy or in MRI or something like that, you could immediately know who is a keyboard player and who is a string player. You can tell what kind of musician is in your scanner.

5.11 Now you can read even more from the brain's organization. Just, you know, just as an example, imagine that you're working a guy who puts the lid on the jar. So in the first case, that's using fine motor movement of your fingers. And the second case, it's mostly wrist and forearm. And it turns out that whichever job you do, that's what will expand in your brain. So in the first case, the representation for the fingers will expand. In the second case, it'll be the opposite. The wrist and forearm representation in the cortex expands. So in this way, what you do over and over changes, shapes your brain.

5.12 And we see that with cortical remodeling in many situations. So not only the violinist, but if you take up juggling, let's say as an adult, you take up juggling and figure out how to do it, it's not just the motor system, but also visual areas of your brain increase because you're having to really watch and pay attention. And so that increases in a way that we can just see with the naked eye. If you spend months learning how to read braille, the bit of your cortex that represents your index finger will grow. That representation of that will grow in your brain because that becomes more important. There was a recent study I saw that if you grow up with Pokemon, you actually have Pokemon represented in your brain. And, you know, so my 12-year-old boy has a lot of time spent looking at Pokemon, and I worry about the fact that that's going to stay in his brain his entire life.

5.13 Anyway, the point is that brains reflect not simply the outside world, but your outside world, what you are doing with your time. And this is what underlies getting good at something. So professional tennis players, like Venus and Serena, they spend years training. They spend years doing this over and over. So the right moves will come in the heat of the battle. So, you know, step, pivot, backhand, smash, all this stuff. They train for thousands of hours to burn these moves down into unconscious circuitry of the brain. So in the first lecture I was talking about, you know, like the cup stacker and the soccer players and things like that. That's what Venus and Serena are doing. They're putting these hours in, so it gets burned down into the unconscious circuitry.

5.14 If they tried to run a tennis game just based on cognitive skills, as in, oh, probably somebody should do this in this situation, probably a backhand would be useful here, there's no way they could win. You have to actually burn that into the circuitry to do that. So you might have heard of this 10,000-hour rule which suggests that you need to practice a skill for that many hours to get really expert at it, whether that's, you know, spelunking or saxophone playing or surfboarding or whatever. Now, although the precise number of hours is impossible to quantify, the idea is, the general idea is correct, which is that you just need massive amounts of repetition to carve those subway maps into your brain and make it fast and efficient.

- 5.15** So you remember the guy with the reversed steering on the bicycle? I mentioned, although he cognitively knew how that worked, that proved totally insufficient to drive the bicycle. He needed to invest the weeks or months of practice in it. There was an experiment with a monkey where they trained him to use a new tool. So he had to get his food by using this rake. And what they were looking at was neurons in the brain and what the neurons were encoding for. And this is related to the last lecture. What happens is that the neurons essentially started coding for the rake itself as part of the monkey's body. So it became part of the monkey's body.
- 5.16** But the key thing that I want to emphasize here is that the brain needs to practice with the tool for this to happen, because as the control, what they did is they just had a monkey holding onto a rake. And you don't get any changes in the brain that way. The brain actually needs to see that there's practice on something happening with that. So exactly equivalent to this would be if Venus and Serena just held a tennis racket all the time, but they weren't using it and they weren't in the heat of the game. They would gain no expertise with it. So this is why there's this 10,000-hour rule, of 10,000 hours of actually working hard and doing something with it.
- 5.17** Now, the neural effects of intensive practice, this isn't just motor systems and whatever and, you know, swinging a tennis racket and manipulating a rake for the monkey, but it can also apply to inputs. So there was an interesting experiment that was done with medical students who have to practice for their final exams. They have to study for their final exams for three months. And the gray matter volume in their brain changes so much that it can be seen with the naked eye. So, that's what this paper showed, is that you could just look at it and see the change.
- 5.18** Now, just as a quick side note, it's interesting because if you ever want to study something, one of the good ideas is to go find a group of people who's already doing the kind of thing that you want and then just ride on top of that. Anyway, you can see these changes. The same with, like, mirror writing. You can find similar physical changes in the brain that you can see if you train people how to read mirror writing. So you can take something totally made up like this and teach it that way. By the way, this mirror writing belongs to who? Do you guys know? This is Leonardo da Vinci. He used to write that way.
- 5.19** OK, so here's an example that's really quite striking, which is, there was an extensive study done on black cab drivers in London. So I don't know if you guys know about the way the black cabs work in London, but traditionally, at least before Waze and Google Maps and all those things came out, so for the last whatever number of decades, they had to memorize every street in London. London is a very big, complicated city. And these guys would have to take this test called the Knowledge, where they would have to recite routes from memory. OK, I take this and go that bridge and I go over there. And they'd have to learn the entire thing, which is an almost impossibly difficult task of understanding spatial layout and navigation.

- 5.20** So again, one of these situations where people are already doing something. Researchers came in and studied what happens in their brains. And what they found is that the black cab drivers were able to, you could see in their heads, changes in an area called the hippocampus, which is involved in spatial navigation and layout. You could actually see an enlarged hippocampus in these guys versus control participants. What you spend your time on changes your brain. You are more than what you eat. You are everything you digest. Like, cognitively, what you're digesting.
- 5.21** And this is how the Polgar sisters were able to become world champion chess players. It's not that they had some gene that codes for skill in chess. It's that they practiced over and over, and they chiseled pathways in their brain that encoded the powers of rooks and knights and bishops and pawns and kings and queens. Brains come to reflect their world. But how? How do they do that? Well, let me tell you. Here's how we're going to get to our answer.
- 5.22** So, I recently saw an online meme. It said, it was a picture of a human brain that said, "Hey, I think your phone just buzzed in your pocket." And then across the bottom, it reads, "Just kidding. Your phone isn't even in your pocket, you moron." And so this is referring to the phantom phone vibration, which is a scourge of the 21st century. Obviously, it never existed before. And what's going on is that, you know, you guys probably have all had this before, where if there's any sort of momentary spasm or quivering or shaking or touch on your leg, your brain interprets it as, oh, my phone's ringing. Because as long as what's happening is even vaguely similar to the vibration in your phone, your brain will decide that somebody interesting is trying to reach you just now.
- 5.23** 30 years ago, had you noticed that same twitch, you would have interpreted it differently. Like, oh, there's a bug, or somebody's brushing past me or some movement of my clothing, something like that. So why do you interpret it differently now? Why does your brain, given the exact same stimulus, think that it's your phone now? The answer has to do with how your brain optimally interprets things that exist in a range. There's a range of twitchy, brushy feelings that all get pulled into this interpretation of the cell phone.
- 5.24** And here's the way I think about it. I think about it like a hilly landscape. Whenever you see something like this, where there's a lake in the middle, the key is that raindrops can fall anywhere around here, and they'll all come to the lake. Right? This is what happens when water flows down a valley into the middle. And this is what happens when your neural networks fashion themselves from experience in the world. You have a big range of things that all have a single interpretation. So in the case of the phantom cell vibration, anything that's even vaguely similar will roll down the hillside into the lake, and you'll say, "Oh, great, it's a buzzing phone." And so, as long as the signals are even vaguely close, they'll get pulled into that.

- 5.25 Now, this idea of having a landscape, this is, I think, a very powerful way to understand a lot of what happens in the brain. For example, this is what happens with language. So it feels totally natural that you can understand the sounds of your native tongue. Whereas, if you listen to foreign languages, like a language that you have no exposure to, you can't even hear the difference. The person says, maybe you try to impersonate. And they say, "No, no, no, it's this and this." You say, "Wait, I think I'm saying the same thing you are." And they're like, "No, those are totally different sounds." But you can't even hear those. OK, why? It's because of the landscapes. It's the way that your brain has fashioned itself versus the way their brain has.
- 5.26 So if you look at all the ways that humans can make sounds with their tongues and their mouth, it's a relatively smooth continuum. But we, you know, in terms of the way we understand it, we divide it up into things like, oh, there's an E, there's an I, there's an A, there's an O, and so on. We divide them up into things, and the way we divide them has to do with this landscape. So what's interesting is that, you know, you learn from experience that, you know, when your father says E, that's, that's E, and your babysitter says, like, E, says it a little louder, you know, a little longer. It's slightly different or whatever. But you realize, oh, they're saying the same word there.
- 5.27 And maybe you have a, you know, Texas friend who says, I can't impersonate, but whatever. Yeah, a Texas friend says E in some Texan accent. Or you have some Australian friend who says, oi, and he means an E and whatever. Anyway, you learn from experience that all these things represent E because of context. And so you've got this valley where all these things are interpreted the same way so you can understand what someone is saying, irrespective of the accent that they have. And in neighboring valleys, you've got the A and the O and the I and so on. And this is the way that you take this smooth continuum of sounds and you have things with what we call attractor states.
- 5.28 OK, and what that means is that somebody who grows up differently from you do, their whole valley and landscape looks very different from yours. So let me give a very specific example. So take two friends. A kid born in Japan named Hayato and a kid born in Palo Alto, California, named William. And they're friends. And from their brains' point of view, there's nothing different about them. But in Osaka, Hayato hears Japanese all around him from day one. And in Palo Alto, William hears the tones of English. And you have different sounds carrying different meaning in these two things.
- 5.29 An example of what these two kids hear differently is the distinction between R and L sounds. So in English, that carries meaning, that distinction. But in Japanese, there's no difference there. There's no meaning carried in that distinction. And so, as a result, William's internal landscape makes two different valleys. There's the R sound and there's the L sound, but Hayato's does not. That develops a valley where R and L flow to the same lake. William's brain, having passively listened to tens of thousands of hours of English words, has learned that there's information carried in that distinction,

whereas Hayato's brain doesn't. So it sounds indistinguishable. The auditory system begins universally and then wires itself to maximize the distinctions of your language, depending on where on the planet you happen to stick your head out of the womb.

- 5.30** So I'll give you an example of this R versus L distinction. You can measure this in babies, and it turns out that, by 10 to 12 months, you see this difference. So if you measure babies at 6 to 8 months, let's say American and Japanese infants, they can't tell the difference when you go "rrrlll," when you make that change. And the way you measure this is by the speed at which they're sucking the nipples. So they, they'll suck a nipple, and when they hear a change in a sound, they'll suck faster. That's how you can tell with an infant when they've heard a change in a sound.
- 5.31** So what happens is, by 10 to 12 months, the American infant, when you switch from R to L, will start sucking faster, and the Japanese infant will not. He just doesn't hear that sound. So by 10 to 12 months, you're already getting these landscapes that are differing. And obviously, if William had been born in Osaka and Hayato in Palo Alto, they'd have exactly the opposite results because it's nothing about their brains or their genes. It's just about what they are exposed to in terms of the thousands of hours of language that they are listening to.
- 5.32** So in the same way this cell phone buzzing isn't about something that you were born to detect. Its high relevance has carved out this valley in your neural landscape, such that you've got this broad catch-all valley for anything that's even vaguely like a twitch on your leg. So just like with Hayato, with his R and an L, you're combining twitches and vibrations and whatever and quiverings into a single interpretation of what happened, which is, "My phone rang." So, from what we've talked about so far, we might think, OK, if you have repetitive practice at something, repetitive exposure to something, that's what's carving your landscape. But it's actually a deeper principle that's going on here.
- 5.33** And as one inroad to that, I'll take another study by the same woman who did that R and L study, a woman named Patricia Kuhl, which demonstrates that there's a big social component to learning. So what she did is she took American infants and Chinese-speaking infants, and she was looking at things like Mandarin and whether you can distinguish sounds in Mandarin. What she did is she took these American infants and either they were exposed to Mandarin sounds with a person who spoke to them, or from the exact same sounds, but now just played on audio, just an audio file, but same exposure to the sounds. And what she found is that, if you're actually there with a human, they got much better at distinguishing the sounds. But if they're just exposed to the audio, that actually was not sufficient.
- 5.34** So the point is, it's not simply about being exposed to things that carves your landscape. You need something deeper. And what it is is relevance. Relevance. And so what do all these people have in common? The Polgar sisters and Ashkenazy and Perlman and the Williams sisters? What they all had in common is they wanted to make

something happen. So the brain's goals are critical for how and why it changes. So for, you know, all these people on the screen here, they had a desire to achieve their expertise.

- 5.35** So, you know, imagine, imagine that Venus and Serena had a brother named Fred, and Fred was forced to do the same number of hours of tennis practice, but he hated it. And he, you know, his parents forced the racket in his hand and made him do all the same stuff, but he found tennis repugnant. And he never got good feedback from people, from classmates about his performances. And he didn't win any contests, and his elders didn't lavish him with praise. So what would be the result of all that practice that Fred did? His brain wouldn't change in the same way. In fact, it probably wouldn't change at all. We'll come to that. We'll see some examples of that.
- 5.36** His brain would not show the same reorganization. Why? Because it's not simply about doing the backhand and all that stuff enough times. It's about getting the right sort of praise and feedback. The motions would be misaligned with his internal incentives, and he would not become a champion. So how did Matt Stutzman get so good at archery? What's the reward for him? Go ahead and shout something out. What's the reward for him? Yeah, gold medal. That's right. Yeah. Feeling like he's good at something. Sense of purpose. [inaudible audience]
- 5.37** Yeah, that's right. That's right. Fame, sense of purpose, gold medals, all these things. That's what drives the changes in his brain to make his motor system able to do this sort of archery. Same with Faith, the dog. So when I told her story in the last lecture, I just sort of said it, like, magically, Faith could walk. But in fact, why did Faith learn how to walk bipedally? Why don't other dogs learn how to walk bipedally? It's because Faith needed that to get to her mother, to her food bowl, to her water. Faith needed that. There was reward in doing that. She would have died otherwise.
- 5.38** So the question is, could all dogs walk bipedally? And the answer is yes. Other dogs could do that, too, but they've got no motivation to do that. There's no particular reason for them. But Faith had a reason, and that makes all the difference in the world. Her brains, you know, she couldn't call DoorDash or something, right? So her brains had to figure out how to drive this body to get to the reward.
- 5.39** And how about Jan Scheuermann? She had to learn how to control her body. And here's a woman who had been an athlete, and she'd had this great life, and now she ended up becoming paralyzed and was in a wheelchair, paralyzed from the neck down. And so this idea of having, you know, the ability to have some control in the world, and by the way, she was able to get chocolate to her mouth this way, too. That probably helps to learn how to do this again. But, you know, finding control of the body and being a participant in some cutting-edge neuroscience experiment, there's a hundred reasons like that why Jan is motivated to do this. And that's what's absolutely required for making changes in the brain.

- 5.40 And you can demonstrate this sort of stuff in the laboratory. So just as an example, you know, imagine there's an experiment where somebody is tapping out Morse code on your foot. Someone else is playing musical notes, and in order for you to win money, you have to ignore the music and just pay attention to the Morse code on your foot and decode the signals that way. Well, if you do that, you'll find an expansion of your somatosensory cortex related to your foot. And, you know, you're getting paid to do this thing, and your brain will actually make that bigger.
- 5.41 But now you switch the task. They say, OK, now ignore the Morse code and just pay attention to the notes and tell me when you hear these subtle differences in the notes. Now it's the opposite. Now you're getting rewarded for listening to the music. And your auditory cortex will expand. Why? Because that's what's getting rewarded. That's what is relevant to you in this moment. And that's why Fred Williams gets no better on the court, because he derives no reward from doing the tennis thing. And so his brain, like yours, the maps of the neural territory, reflect the strategies that have won feedback, won positive feedback. So your reward systems mold your circuitry.
- 5.42 OK, and what's really interesting here is that this opens up new clinical avenues. So imagine that a friend of yours suffers a stroke, and that hits part of her motor cortex, and as a result, one of her arms becomes mostly paralyzed. You guys probably see, I don't know if you've noticed this, but sometimes you'll see someone whose arm is just sort of hanging there, and they'll have, you know, they'll be doing stuff with their other arm, but they'll have one arm that's just sort of hanging there. That's because they've had a stroke, and now their brain can't control that arm very well.
- 5.43 So what typically happens is that, because that arm is not getting used, it becomes weaker and weaker, and the person is depending on their good arm to do things. What the lessons of live-wiring offer is a totally different strategy, which is called constraint therapy. And what you do in constraint therapy is you take the good arm and you strap that down so they can't use that, and they're forced to use this arm that's super weak. And they're motivated, right, because they want to get the food to their mouth. They want to get their pants off when they need to go to the bathroom. They want to reach the coffee cup and get it to their mouth. They have all these motivations that they need to do.
- 5.44 And it turns out, constraint therapy is the single best approach when someone has what's called hemiparesis, where one of their arms is weak. Their motivation is all the things we can think about, wanting to live a self-sufficient, dignified life. And so it starts off as really frustrating because their good arm is strapped down. Turns out it's the best thing that you can do. And by the way, you also do this with kids in casts. You make them use the other arm, the weak hand, to get better at that.
- 5.45 And by the way, you remember the Silver Springs monkeys that I mentioned at the beginning, where their arm had been cut? The reason that the researcher had done that is because he's the guy who discovered this issue about constraint therapy, which

is, when you have damage to the nerve, this is where you, this is where it was realized that the way to do it is to strap down the good arm and then you get better at using the bad arm. So it seems like a paradoxical solution to make things worse for somebody, but it's actually the best medicine.

5.46 And finally, recall what we saw about blind people with echolocation. It turns out that people with perfectly good, normal vision could do this too. Any of you could learn how to echolocate. It's just that you probably don't have the motivation to do it. But if you go blind, you are really motivated to do it. And then your brain rewires and starts really paying attention to the echoes off of things because it matters. So what it takes is effort and motivation to rewrite your neural maps. That's what it takes. And what's interesting is that reward, like I mentioned, if you're paid cash for the Morse code or music thing, whatever, you know, cash or cookies or gifts, these are all good ways to rewire the brain.

5.47 But more generally, change is just tied to anything that is relevant to your goals. So if you're in the far north and you need to learn about, you know, ice fishing and different types of snow, that's what your brain's going to encode because it is relevant to your life. And in contrast, if you are equatorial and you need to learn how to identify different types of mushrooms to eat and whatever, your brain will devote its resources accordingly. Not because you're getting, you know, cash or cookies in the moment, but because it's relevant to your longer-term goals of staying alive.

5.48 So, using relevance as its north star, this is how your brain picks up on important details in the world. What you have is these billions of neurons, and think of it like a blank canvas, when you drop into the world. And what it's doing is figuring out, OK, what world did you happen to drop into? And what is the thing that is relevant for you? And that's how you develop deep expertise at whatever has relevance to you, whether that's chess or basketball or theater or badminton or Greek classics or cliff jumping or whatever it is. That's how your brain figures out what it's going to do. What are you getting reward from? What are you getting positive feedback from? What is relevant to your long-term goals? When a task is aligned with your goals, your brain circuitry comes to reflect that.

5.49 And I think you can make an interesting analogy here with the United States government or any government of any country. But I happened to stumble on this poster a while ago that shows all the different departments of the US, and what's interesting is that this stuff changes over time. And so because I love analogies, I was thinking about how it's like a brain in this sense. So as an example, after September 11, 2001, the organizational maps did a lot of shifting because suddenly the world changed. You know, the World Trade Centers were destroyed. And so, the United States' brain responded to this injury by establishing the Department of Homeland Security, which sucked up a bunch of other areas. And it changed things and you know, changed

different parts of the brain based on what was relevant. And that's the thing that was relevant right then.

- 5.50** And you know, it wasn't even, that's not the first big shift. It did this in 1947. It's spawned the CIA and the National Security Council, whatever. The point is that, just like brains, governments constantly redraw their maps based on what is relevant, what's going on in the world. Budgets swell and shrink to reflect priorities. When external threats loom, you have the military pocketbook expand. When peacetime follows, you get these social initiatives, gain territory, and so on. But it's just like that, where the territory is changing based on what is important.
- 5.51** Here's the question. How does the brain know? How does the brain figure out what is important? How does it know what is relevant and when it should change? OK, well, there are various things that have been proposed, many of which are all, you know, simultaneously true. So for example, one way that the brain changes things is when things co-occur. Like, if this and this always occur together, then I change the real estate to say, "Oh, OK, I got it. These are now bound together in the tissue. I get that these happen."
- 5.52** But I just want to say, you know, this thing of changing the brain just by things being correlated, you have to do that slowly because often in life there are lots of things that are accidentally correlated. Like, you know, I happen to close my refrigerator door at the same time that a dog barks outside. I don't want my brain to learn, "Oh, the fridge somehow is causing a dog to bark." So what you want to do is, there's very slow change that happens there. So people have studied that for a long time, about how do brains sort of generally absorb the patterns of the world. But that sort of slow change is not the whole story, because often brains do one-trial learning, where you learn something just like that.
- 5.53** So, for example, if you touch a hot stove, you know, you learn immediately not to do it again. You don't have to learn several times about this thing. And so you've got these emergency mechanisms. Like, you have a part of the brain called the amygdala that's involved in saying, "Hey, emergency!" To exist, to make sure that these life- and limb-threatening events get permanently retained. And so you can learn things in one trial. But the story of one-trial learning actually goes deeper than that. Because imagine you're very young and your favorite aunt teaches you a new word and says, "This is a pomegranate." Well, you didn't need to learn this in an emergency situation. And you don't need her to tell you this a thousand times.
- 5.54** Like, artificial neural networks, you know, OpenAI and all these things. They have to see things thousands or hundreds of thousands of times, but you don't, right? She says, this is a pomegranate. You're like, OK, got it. It's a pomegranate. Why? It's because your aunt was salient to you. You loved your aunt, and you wanted to identify that piece of fruit because you wanted to eat that piece of fruit. And so you derived social benefit from being able to know a new vocabulary word and being able to ask for it and so on.

This is not one-trial learning because of threat, but because of relevance, because it is relevant to you to now know the name of this thing.

- 5.55** Now the question is, how does that happen? Well, it happens because of a neuromodulator called acetylcholine. So you have these two small nuclei in the brain that broadcast acetylcholine broadly all over the place. And so, these signals turn plasticity up and down. And what they're doing is, when you need to make a change in the brain because something is relevant, you spit out some acetylcholine. And then the brain says, "Oh, OK, cool, I'm gonna make changes here." If you didn't have any acetylcholine, then the brain is locked in place and is not making those changes.
- 5.56** So there's a couple interesting things to note here about acetylcholine. That induces plasticity, both for reward and punishment. So these nuclei don't care whether something's rewarding or punishing. As long as it's emotionally valenced like this, you're going to cause plasticity to happen. It's only active when you are learning a task. Once you know a task, the acetylcholine, there's no more release of that happening. Finally, it tells the brain area to change, but it doesn't tell the specifics of how to change. It just says, essentially, "OK, I'm loosening up this area. Go make your changes." But it doesn't tell it exactly how.
- 5.57** So let me give you an example of what this looks like. So imagine that I take a piano and I play a particular note. Let's say, you know, what is that? The D. I play the D a number of times. That note triggers activity in your auditory cortex. So every time I hit it, you know, your auditory cortex is active. But over time, you know, that doesn't mean anything to you. It doesn't mean anything to the maps of your cortex. But why not? It's because it doesn't have any particular meaning to you.
- 5.58** But now, let's say every time I play the D, I give you a cookie. And so the note accrues a meaning. And so now the territory devoted to D expands. It actually expands in your brain, in your auditory cortex, because suddenly that's a thing to look for in the world because you're getting a warm chocolate chip cookie every time. So now let's say I don't have any more cookies available. So what I do instead of training you with cookies is I just stimulate your acetylcholine neurons. You get the exact same effect. The territory expands in your brain, because that's what these acetylcholine neurons are about, is saying, hey, something important is happening. The D is associated with reward. The presence of acetylcholine is the brain's way of saying, "Pay attention to this. Get better at detecting this."
- 5.59** So again, as we've seen in various things, the brain marks relevance by increasing territory. The more real estate, the better. And we've seen this before. You know, we talked about in lecture three, like with blind musicians who are actually better at detecting things. You know, typically blind people can tell a lot about your, you know, your accent and where you're from. They can localize things better. It's not that they're getting any different signals in their ears than you're getting in your ears. It's that

they're paying more attention. They've got more real estate to devote toward it. So this is the key. If you have more real estate, you'll be better at it.

- 5.60** And this thing I mentioned about, if you don't have acetylcholine, you won't get more real estate? I'll give you a very specific example of this. So this was a study that was done where you had two groups of rats that were trained to grab pellets through a small, high slot. And so in one group, you had normal release of acetylcholine, and in the other group, that was blocked. And so what happens is, for the rats with normal acetylcholine release, they learned how to do this. They grew the area of their brain that was associated with this very difficult motor task, and they got better at it. And for those rats for whom the acetylcholine release was blocked, their brain real estate didn't change at all, and they got no better at the task.
- 5.61** So we've seen this in various ways. I just want to repeat what this means here, is that the brain changes that make you better at something, it's not just based on the repeated performance of the task. Because they did this the same number of times. It requires these neuro-modulatory systems to encode the relevance. And without acetylcholine, the 10,000 hours is wasted time because you're not making the changes to your brain.
- 5.62** So recall Fred Williams, who hates tennis. Why didn't his brain change, even though he had the same number of hours of practice? Because the neuro-modulatory system, this acetylcholine system, is not engaged. He's drilling the backhands over and over, but he's like the rats who are grabbing the pellets with no acetylcholine release. Because Fred doesn't love tennis, he's not getting the appropriate feedback. He hates it. And so just like those rats, there's no changes in his brain. There's no behavioral improvement.
- 5.63** I mentioned earlier that the taxi drivers in London, they're famous for having to memorize the entire map of London. This is known as the Knowledge. And they would train for months, actually years, on this task. And there were these physical changes in the structure of their brain as a result. But they were doing this because the maps were relevant to them. This was their desired employment. This is what was going to pay for their home mortgage or their child's tuition or their upcoming marriage. This mattered to them. It was relevant.
- 5.64** But what's interesting is, since this study of the cabbies came out, the need for this kind of memorization has diminished, because now it's just as easy to have Google servers memorize the maps of London and, more generally, all of the streets interlacing the whole planet. So this leads us to the notion of AI and some deep questions about it. Because AI, you know, we're in this explosive moment of discovery with AI. It's not simply that AI, you know, most technologies we build and we know exactly what it's going to do, but AI has what's called technological overhang, where there's all kinds of stuff we didn't even expect it to be able to do. It's going great.

- 5.65 But there's one thing that's still missing, which is, at least with current AI algorithms, they don't care about relevance. They will encode whatever we ask them to encode. And this is a very useful feature of AI. We love AI for this. But this is why it's not exactly humanlike, because it just doesn't care which problems are interesting or germane. It's just going to take whatever we feed it, whether it's distinguishing zebras in a million photographs or tracking flight data from every airline on the planet. It has no sense of importance except for statistical importance.
- 5.66 And so, contemporary AI by itself could never decide that it finds irresistible a particular musical play, or that it doesn't like certain types of music, or that it's heartbroken by King Lear's speech over his dead daughter. AI can dispatch 10,000 hours of practice in 10,000 nanoseconds, but it doesn't care about any zeros and ones over any other zeros and ones. And so, as a result, AI can do very impressive things, but it's not like a human.
- 5.67 I want to turn now to the issue of the brain of the digital native, because this comes up a lot. And as a neuroscientist, I get asked this, you know, essentially every talk where people have kids, they all want to know, like, what is, what is going on with my kid's brain in this new digital world? And I'm very invested in this issue of how what we know about the modifiability of the brain bears on how we teach our young. So here's the thing. The traditional classroom consists of, you know, a teacher with PowerPoint slides with bulleted lists on it, droning on. But the fact is, that's actually quite suboptimal for learning because the students often in classrooms are not particularly engaged. And so there's no, there's little to no brain plasticity. The information isn't sticking.
- 5.68 OK, now, we're not the first generation to make that observation. The ancient Greeks had noted that there are seven levels of learning, and they said the highest level is when you are curious about something, when you care about something. That's when your brain is making changes. They didn't say it that way. They didn't say your brain is making changes. But that's, that's the issue. It turns out that you need the right combination of neurotransmitters present to get plasticity, and that happens to correlate with curiosity and engagement. So when you care about something, that's when changes happen.
- 5.69 And this is why, by the way, you know, when you join, like, a study group in college and, you know, you meet to study or whatever, those always are useful because then you've got this whole social mechanism engaged and you want to impress each other and have the right answer and stuff like that. You know, it activates the brain's social mechanisms that involve engagement. Now, back in the 1980s, the author Isaac Asimov gave an interview with Bill Moyers. And it was a brilliant interview, because this really struck me when I saw it, because what had happened in 1980 is that Asimov had noted a couple things that were spot on.
- 5.70 One of them is he said, look, in any classroom, it's going too fast for half the kids and too slow for the other half of the kids. And so his point was, what you really need is

individualized education, something that's adaptive to that student. And what he sort of squinted into the future and foresaw was, he said, "I foresee a time when there's one central computer that has all of mankind's knowledge on it, and everyone has a cable to their house from this central computer, and everyone can access this data."

- 5.71** Now, obviously, it was the Internet that he sort of foresaw, but the details were wrong, were hard to see, but what an incredible thing. Because what he envisioned is everyone gets access to the world's knowledge, but a kid can go into this sphere of humankind's knowledge via any door that they want to. So if you're a kid who loves baseball, you might say, "OK, well, wait, what's this RBI statistic? I don't even get what this is." And then you come in there and, oh, statistics. Oh, math. Oh, that's cool. And you're learning that. And if you're someone who, if you're a kid who loves ballet dancing or whatever, and you start studying things from that context, from that point of view.
- 5.72** Oh, and then the leap. Oh, that's, oh, and you spin. You spin faster when you do that. And whatever interests you, you can come into the world that way and get the education that's maximally useful for you because you're curious about it, it is relevant to you, you care about it, it is meaningful to you. And everything we've seen in this lecture is, comes down to that, which is, to make brain changes, it has to be relevant to you. So he foresaw this idea about the Internet. And I think it's the greatest thing that's happened to education in the history of humankind.
- 5.73** I'm a real cyber-optimist on this point. And when you look at a kid, let's say, drilling down through the blind molehills of Wikipedia, I think we've all had this experience. You start, you're looking up something on Wikipedia, and then eight clicks later, you're way over here, like, wait, how did I get here? But that's great. That's actually optimal for education because you care. At every moment, every click, you're like, oh, I'm really curious. What is that? What is that? What is that? That's the way to learn stuff. The Internet allows students to answer things right when it pops into their head.
- 5.74** I mentioned this the other day during Q and A, but this is really meaningful because we all had this, you know, just-in-case education, or you guys are all younger than I think, but I had just lots of facts dumped. But it's very different than just-in-time education, where as soon as you're curious about something, you go to Wikipedia, you look it up, you go to YouTube, you watch a video on it, you get what's going on, you put on VR, you shrink down to the size of an atom, you see the nucleus, you expand to the size of the cosmos and understand something going on there. It's a very different sort of thing to get education in the context of your curiosity, because then it is relevant and your brain changes. The changes stick.
- 5.75** The Chinese have an expression which is that "an hour with a wise person is worth more than 1,000 books." And the reason is because with the wise person, you can ask questions and you can, as soon as something is, you're curious about it, you get that feedback and you're able to, you know, that way you have the molecules of relevance and reward present, and that allows the brain to reconfigure. Because when you toss

random facts at an unengaged student, it's like throwing pebbles to dent a stone wall. It's like trying to get Fred Williams to absorb tennis.

- 5.76** And in this slide, I just want to mention, I think that there are a lot of great opportunities with the gamification of education. So you've got adaptive software that keeps students right at their point of struggle, where finding the right answer is frustrating but achievable. And so they're right in that range. And each student goes home with different homework. OK, you're on level two, you're on level four, you're on level six. Go home and do half hour, you know, answer six questions. And every student gets to be at the right level, and the questions get harder for them.
- 5.77** And the role for the teacher is to teach foundational concepts and to guide the path of learning. But fundamentally, given how brains operate in terms of how they rewire and how they assign territory, what we need is a neuroscience-compatible classroom in which students drill into this sphere of human knowledge by following the paths of their individual passions. You know, I can just tell you the root of my optimism, which is that, you know, we've never before had this opportunity to have all of humankind's knowledge in our pockets. And the great advantage of that comes from a simple fact, which is that everything that humans create, everything we put out, is a mashup of things that we've been exposed to before.
- 5.78** So you see all these things, and that's in your warehouse of knowledge. And we're remixing this. We're bending, breaking, and blending it. And as a result, children now live in a time that is unparalleled in its richness. So if you look at the history of humankind, things went very slowly because there weren't that many discoveries. And what happened is it's just blown up in a way that's hyper-exponential, and it's only getting faster and faster. And in large part, this is because of the Internet, because as soon as someone discovers something, it goes worldwide, instantly, and everyone can work on that. And things are just going faster and faster.
- 5.79** But the key is that young minds have this opportunity to cross-link things, to say, "Oh, I saw this fact over here, and I saw this fact over here and this one over here. What if I put these together? I got the next thing." They can take things from totally different domains and put these together at a speed that previous generations could have not ever imagined. And this is partially what explains this exponential increase in human scholarship. We have more, you know, mashups than ever. We have faster communication than ever. And, you know, while it's not clear what all the social and political consequences of the Internet will be, from a neuroscience perspective, it unlocks a completely richer level of education and brain changes.
- 5.80** OK, so let's wrap up this lecture. We saw earlier in the course that the brain real estates, sorry, the brain distributes its real estate according to the information that's coming in. But what we see now is that the information has to be relevant. It has to be linked to reward or linked to the long-term goals of the organism. And the larger principle that ties all that together is relevance. Your brain adjusts itself according to

what you spend your time on, as long as those tasks have alignment with your rewards and goals.

5.81 So for a person who goes blind, you know, expanding the other senses, which we talked about before, that happens if it's relevant, which it almost always is, for a blind person to pay more attention to what they're hearing, what they're touching. If a blind person happened to pass her fingers over braille, but there was no relevance for it and she didn't care, then the brain wouldn't change. There would be no rewiring there because the right neuro-modulators wouldn't be present. And similarly, if adding a new tele-limb to your body has high relevance to you, then your brain will learn how to do it, just as Faith the dog in the last lecture mastered a new body plan. So that's relevance. And in the next lecture we're going to talk about the role of surprise. So thank you for your attention.

5.82 So in terms of natural inclination and curiosity, do you believe that people are born or have certain natural inclinations as they're born with, or do you simply believe that it's a reward system based on social interactions and things like that? If so, how do you explain the study where kids who were naturally inclined to drawing were given more rewards for drawing, and then they were actually less likely to draw for fun on their own? And how does that work with the brain real estate, because you would think that would expand their brain real estate?

5.83 Yeah. So a couple things there. As far as nature versus nurture, it's always both. So everything that we have seen, everything that you'll ever see in neuroscience, it always involves both. So kids come to the table with certain predilections, certain skills, and then it's a matter of what they get rewarded for. But, you know, take, like, Michael Phelps. I wish I could be as good a swimmer as he could, but he has a longer wingspan. There's no way that I could ever beat him swimming because he genetically drops into the world more predisposed for that.

5.84 As far as this issue about rewarding a kid too much, I'm a little skeptical about that study because what it depends on is the kid's motivation for drawing. And so that can be predicated on lots of things, not just feedback from the adult, right. Some kids just like drawing more. They'll do that all day long because they like it. And they say, wow, that really looks like a horse. And they're so rewarded internally from the fact that they've been able to create something and so on. So I just, I think it's a little bit more complicated than what the parent does in terms of reward. There are ways to give good reward, to give good feedback, so that you're encouraging the kid to go to the next level. But I think those studies generally are maybe a little exaggerated.

5.85 I work with a lot of athletes, and so the 10,000 hours in sports area is very interesting to me. What role does motivation play when the athlete actually does not like the sport? So I think a really relevant example is Agassi. In his autobiography, he speaks, he says he hates tennis, like 20 times, yet he became one of the best ever, right? And just seeing other examples of athletes, I know it to be the case. I'm just not sure whether

it's a matter of, they don't really hate the sport or they find the reward in different ways. What's your thought on that?

- 5.86** I'll tell you my intuition. It's that they don't really hate the sport. So it's easy for Agassi to be a champion and say, "Oh, I hate it." But the truth is, there are millions of people who actually hate tennis, and they actually never became Agassi or went onto the court very much. I think it's just a cheap, easy thing. The other thing, look, every doctor I know says they hate being an MD. Why? Because it's hard. Any job you do is hard. But to answer your question, I don't believe him that he actually hates tennis because it just takes so much work to get there.
- 5.87** I knew a guy who was, he had a twin brother, identical twin brother. They were both 6 foot 8. And so they got recruited to play basketball their whole lives. And his brother actually liked basketball, but he hated basketball. And as a result, he wasn't good at it. And he was doing everything he could to actually get out of that and to go, you know, get a scholarship to college, but then quit the team as fast as he could. So, but, you know, fundamentally, I think the root of your question is that motivation has to be something that is inherent and internal. So there was something that Agassi liked about it—you know, the attention, the girlfriends, the money, the trophies, whatever it was that was driving him. There was something about that constellation of rewards that kept him going.
- 5.88** You're a leader in this. I just wonder how you want to present it. And I think you're doing a great job, by the way, because I love the technology and the representation of it all. I just was curious about how you would— what is it?
- 5.89** The answer to it is, what I spend my time worrying about and working on is how to make education better and better, given the resources of the Internet. There are a million ways that kids waste a lot of time on the Internet, a million stupid things that people do. But the question is, can we actually figure out how to use the tools of relevance to let people educate themselves? I mean, I'm totally in favor of this notion called the flipped classroom, where you have the, you know, the teacher, when you— what you do is, you have the kids go off and do for their homework, all their learning.
- 5.90** And when they come together in class, the teacher's just there to, like, present a few foundational comments to make sure everyone's doing OK and so on. But all the real work happens outside the classroom when everyone's using adaptive software that's, that's really relevant to them. And at last year's TED conference, Saul Khan, who runs the Khan Academy, presented this marriage of AI and Khan Academy, so that you have your own tutor who's actually tutoring you in a way that's super specific.
- 5.91** So if you say so, if you get the wrong answer, it's smart enough to say, "OK, let's take this a different way. Here's another way to think about. Oh, you didn't get that? OK, try it this way over here." It's not that hard to get AI to do that, but so valuable for the student. That's like spending the hour with the wise man. But now it's AI.

Lecture 6: Prediction Potentials

At a glance — Prediction potentials and the role of surprise.

- 6.1** OK, welcome back, everyone. We're going to start with an element of surprise today, which is, in the 1980s, something very strange started to happen, which is that a lot of people started to report that pages in the books looked red. They looked tinted red. Which it wasn't. It was just black and white. But people reported that they were seeing the text and the space in between the text as being a little bit red. And this was very weird because this only happened in the 1980s, and it didn't happen before or after. So what was going on there?
- 6.2** Well, to understand this, we have to step back 2,400 years to a horse that was stuck in a river. Now, many people noticed this horse. This was in Greece. But one person who saw this horse happened to be an extraordinarily observant and curious person named Aristotle. And so he watched the rescue operation as people went into the river and they managed to pull the horse out, and he was staring at it for a while. But the thing that he really noticed was what happened afterwards. When he looked over to the banks, to the trees and the rocks on the side, it looked like everything else was flowing in the opposite direction from the river. It looked like there was movement going on there.
- 6.3** Maybe other Greeks noticed this, but Aristotle, being a particularly curious person, wrote this down and really thought about this. This was the first recorded illusion. And this is an illusion known as the motion aftereffect. If you watch a bunch of motion in one direction, like, he was fixed on the horse in the river, so the water was going past, then it looks like everything else is going the other way. Now, the way that you might have noticed this, or you can notice if you haven't before, is stare at a waterfall. Next time you have an opportunity to get to a waterfall, stare at the waterfall and watch it for a while. Keep your eyes fixed there. And then after a while, look over to the rocks on the side and you'll see everything appearing to move up. Everything is appearing to move.
- 6.4** And so this is the motion aftereffect, and we're going to understand today why that happens and what it reveals to us about the brain. But first, we're going to experience it ourselves for about 30 seconds. So, watch this animation while we're talking. I just want you to keep your eyes fixed. There's a blue dot there, so just keep your eyes fixed on it. And what you're seeing is animation of movement that's going mostly up and stuff like that. And then what's going to happen is, we'll do this for about 30 seconds, and then you'll see a picture of a waterfall, and just keep your eyes fixed there. And even though it's just a still photograph of a waterfall, it'll look to you like it's actually moving.

- 6.5 So, keep your eyes fixed on the blue point. You might see that it looks like the waterfall is flowing down, if you stare at the thing. OK, great. What you're experiencing is the motion aftereffect here. Because you watched motion in one direction, now you're seeing it going in the opposite direction. So the question is, why do these motion aftereffects happen? So the first hypothesis on this is that the reason it looks like the world is neither going up nor down is because you have competing populations of neurons, some of which code for downward motion, some of which code for upward motion. And these are always communicating with each other and essentially inhibiting each other.
- 6.6 And under normal circumstances, there's equal inhibition. So both of them are holding each other in place, and neither of them can do much. And what happens is the idea is, if you put a lot of, let's say, downward input, let's say you're staring at the waterfall, if you are putting in lots and lots of downward input, then that population wins. But the idea is, maybe that population gets fatigued. So because it just gets tired after a while, it uses up a lot of energy. Now, when the downward motion stops, there's more energy in the upward neurons, and so you have this impression that things are moving up as a result.
- 6.7 And obviously, whatever your neurons are telling you is what you will perceive in the world. So if your neurons tell you there's upward motion, then that's what you will see. The idea here is fatigue. You burn up a bunch of energy in your downward-coding neurons, and so that temporarily is exhausted. You've got this unbalanced activity. But why doesn't this explanation work, this fatigue explanation? The answer is, next time you're in front of a waterfall for a while, watch the waterfall for a while, and then close your eyes tightly, and keep your eyes closed, and do it for three hours, and then open your eyes, and you'll see the motion aftereffect.
- 6.8 So it somehow isn't about passive fatigue because it'll last for hours and hours. What this tells us is it's not a temporary depletion of energy resources and neurons, but instead there's something deeper going on. The illusion comes about, not because of passive fatigue, but because of an active recalibration. In other words, the brain is actively saying, "Oh, I see. The world is like this, and I'm going to recalibrate it." So, the idea is that you're watching a waterfall, your system gets exposed to downward motion for a while, and after a little bit, it comes to assume, "OK, I got it. This is the new normal. The world has a bunch of downward motion in it." And so if you keep staring at the waterfall, you're not getting new information about that.
- 6.9 And so, as far as your brain is concerned, this is the new reality. It's a world where there's more downward motion than up. And so your visual system carefully recalibrates to mirror the expectations of the world that you're in to expect more downward than upward activity. And now when you look away from the waterfall, then you see this recalibrated set point. Now it becomes obvious, because now the brain is exposed to something that's not moving and it says, "Oh, well, that must be upward."

- 6.10** OK, and so what's happened is the set point has shifted. The balance between up and down has shifted. Why? It's because the brain always wants to set up a ground truth. It always wants to say, "OK, got it. I understand what's happening in the world," so that it can be better at detecting what? What's that? No. Close. What did you say? Say again. Abnormal. Yeah, that's right. Detect change. What it's trying to do is say, "Hey, is anything changing out there? Because I've got an internal model. I think I know what's going on in the world, and I want to maximally detect if anything changes from what I think is going on out there." So surprise, abnormal, those are all right. That's the same. That's the right idea.
- 6.11** If the descending water flow, because you're staring at it, if it's no longer informative, your circuitry readjusts itself so that it can be more sensitive to new information. OK, so let me give you some examples of this kind of active recalibration. We experience it all the time. So, if you've ever been on a small boat that's rocking around, and then you get off and you step onto the land, it feels as though you're still on the water. What you're actually experiencing there is a negative aftereffect of the water's motion. In the same way that when you watch the downward motion, then you're seeing the negative aftereffect of upward, you're actually feeling the negative images of the waves, but you wouldn't know that. It feels like waves to you.
- 6.12** But I'm going to show you a really stunning one here. This is something called the McCollough effect, but this one requires a little bit of time to get going. So what we're going to do is first take a look at these black and white lines. There's nothing special here. But what I'm going to expose you to is red vertical lines and green horizontal lines. And I just want you to stare at these while we talk because it takes about two or three minutes of staring at this. So go ahead and stare at that while we talk about the next thing.
- 6.13** So the next thing I want to tell you about is an illusion that you might have noticed if you are a runner at the gym. So if you run on the treadmill, you may have noticed that when you get off the treadmill and you're walking towards the locker room, it seems like everything is flowing past you faster. You guys notice? That's called the treadmill illusion. And the reason it happens is because, normally, when you are using your legs and running in the world, as you're doing that, the world is flowing past you. And so you're used to, OK, I send motor commands to my legs, and I get this visual feedback, this flow that's happening, called optic flow.
- 6.14** But what happens when you get on a treadmill is that your legs are moving, but nothing is moving past you. And so your system actually recalibrates the connection between, oh, OK, I get it, moving my legs. It doesn't lead to visual flow. So now you get off and you're walking in the real world and everything seems to move faster. So why does it happen? It's the same as Aristotle's horse or the waterfall or the post-boating wobbles. It's that your brain is readjusting its expectations about the world. In this case, the issue

of how you move your legs and how that changes the visual scene flowing past your eyes.

- 6.15** OK, I'm going to tell you about another example of recalibration. This is an experiment we did in my lab that involved time perception. So you hit a button and that causes a flash of light. So every time I hit the button, I get a flash of light. OK, now I inject a small delay in there, let's say a 10th of a second, so that when you hit the button, there's a little delay and then the light flashes. OK, but because your brain sees that every time it hits the button, it causes the flash of light, it quickly begins to recalibrate when it believes the flash is occurring, and it pulls that closer to simultaneity. So it thinks, oh, yeah, the flash is happening right when I press the button because you're the one causing it every time. And so your brain actually adjusts to the timing of those.
- 6.16** So what we now do is we suddenly drop the delay down to, let's say, no delay. And now when you hit the button, you believe that it happened before you press the button. You say, I didn't do that. The thing flashed just before I hit the button because it's gotten used to the delay and it's recalibrated to that. And so your impression of cause-and-effect can be reversed. So this was a quite stunning thing that we were able to show in my lab a while ago.
- 6.17** OK, so now, hopefully you've been staring at these lines long enough. So this is known as the McCollough effect. And what the idea is, is if you stare at red vertical lines and green horizontal lines for long enough, your brain is going to start associating the red and the vertical, and the green and the horizontal. And because your brain feels like, "Hey, those actually shouldn't be associated, there must be something wrong here," what it starts doing is a negative aftereffect to that. It starts separating those out. It says, "OK, it must be that the vertical should not go with red, so I'll make that a little greener, and green shouldn't go with horizontal, so I'll make that a little bit redder."
- 6.18** So now, if you look back at these black and white lines, what you might be able to see is that this space in between the horizontal lines look reddish, and the space between the vertical lines looks greenish. Now, generally, you have to stare at this for several minutes. So I hope it was long enough. But the key is that, what happens is, that your brain has detected, "Hey, these things are correlated, this particular orientation line, a particular color, and it shouldn't be." And so it's trying to separate those out. And what you get is a negative aftereffect from that that is dependent on the orientation.
- 6.19** And again, this has nothing to do with fatigue, because some researchers a while ago, they showed that if you stare at this for about 15 minutes, the effect lasts three and a half months. So this is not about fatigue. It's about actually resetting something. OK, so let's come back to this thing I said at the beginning about people thinking in the 1980s that books looked red in color. Why is that? It's because, in the 1980s, personal computers just came out, and it was green horizontal lines of code. And so people would stare at these all day. And then when they looked at black and white horizontal

lines, they thought it looked reddish. And so they thought that something was wrong with the publisher.

- 6.20** And by the way, the same thing happened with the IBM logo. So the IBM logo was horizontal lines like that. And people would always say to the IBM company, "Hey, I think there is some reddish in the way you printed that." And they'd say, "No, it's just a black and white logo." In this case, it's gray and white or something. But their logo is normally black and white, and people have this illusion about it. OK, so what does this tell us? This tells us that, how much motion you think is in the world, how stable you think the ground is, whether vision flows past us when we're walking, whether lines have color to them. None of this is decided by our genes. This is all calibrated by our experience. And so what we're going to see today is how and why.
- 6.21** Take this as an example. I don't know if any of you have ever tried this, but if you look at a scene that's completely all red, like in this room here, if we made the entire thing red and you just stared up close at the wall here, what would happen is that the color would quickly drain away and it would look neutral to you. So if this was red, and I stood right up against it, it would pretty quickly just look gray or white to me.
- 6.22** So one way you can try this is, take a colored ping-pong ball and cut it in half and put it over your eyes. And what it will look like to you is you'll just see what is called a Ganzfeld, which means the entire field just looks red. But in a few moments, you'll see that you stop perceiving the color and it just seems neutral, like a gray color. And this is weird. Why does it happen? Well, it's because your visual system assumes, oh, the whole world has somehow gotten redder, and that doesn't make sense. And so it adapts, it cancels that out, so that you can be sensitive to change, so that you can see other things that might happen.
- 6.23** By the way, I recommend you do this. If you just find a colored wall somewhere, just walk straight up to it and just stare at it for a while, and you'll be able to demonstrate this to yourself. OK, you've probably seen this if you put on, let's say, yellow sunglasses, right? You put it on, like, whoa, everything's so yellow. And then you walk around for a while and you totally forget that you're wearing them. And then you take it off. You're like, whoa, everything's blue. Everything's different again because you've calibrated to the thing, and now you have an aftereffect that happens.
- 6.24** Now, the scene, by the way, doesn't even need to be featureless to fade away, like with the Ganzfeld. So in 1804, there was a Swiss physician named Ignaz Troxler, who noticed something really stunning, which is if you have these blobby colors and you stare at a central point, so stare at the nose of the Cheshire cat here. Stare right at that point for about 10 seconds. What you'll see is that all the colors fade away into the background. The whole thing goes away, and eventually you're not seeing any of that at all. So you're just looking at a blank, gray square.

- 6.25** Now, by the way, what you might notice is you still see the teeth in this particular one. So what's the reason that you still see the teeth? No. Good guess. What's that? Say again. No, not because of danger. No. Generally, you can't make them go away. The teeth will remain. It's because they have sharp edges. So it's only the blobby colors that go away in the Troxler illusion. Anything with a sharp edge stays there.
- 6.26** Let's come back to this in a second. I just want to show you this is a demonstration that combines things in a pretty stunning way. So stare right at the middle. And what you'll see is, do you see a green dot moving around? If you still keep your eyes right in the middle. See a green dot that's moving around? OK. The interesting thing about this illusion is there is no green dot. All that's happening is the purple dots are disappearing. But if you keep your eyes really steady there, what happens is the purple dots go away, and all you're seeing is the aftereffect of a purple dot, used to be there, and now it is gone. And so your brain sees a green aftereffect, a negative image of the aftereffect. But there's actually no green dot there.
- 6.27** So that's just quite a stunning way to combine the Troxler illusion and see that, even though you're not seeing it, some part of your brain is still seeing it. And when it goes away, you know that it's gone away. So here's the question. Now you know what the Troxler effect is, which is that these colors can disappear. So why, when you're walking around, what prevents the scene from becoming Troxler-esque? What prevents everything from going away in the scene? Why aren't all the motionless features disappearing?
- 6.28** The edges, exactly right. They've got sharp edges. The world is full of really clear distinctions between things, whereas the Troxler effect is blobby color. There's a deeper reason, too, which you generally are not aware of. Generally, most humans in the world are not aware of this, but your eyes are constantly jumping around. Your eyes are moving all the time, and they actually make really large jumps three times a second. This is called a saccade. So your eyes are constantly doing this. And in between the big jumps, big saccades, you've got microsaccades. So it jumps to a new spot, and then it does little jiggles, and it jumps to a new spot, and does little jiggles. Observe a friend's eyes, and you'll see that her eyes are jumping around three times a second doing this. And you can see the micro-saccades, too.
- 6.29** But it's weird and interesting that we're not aware of our own saccades. It's because we make an internal model of the world. We say, oh, there's the world out there. And your eye is going and doing active investigation of the whole thing. But in between, when it's looking somewhere else, looking somewhere else, you just have an internal model of what's going on out there. It's just collecting new data to add to the model. So you're not even aware that your eyes are doing this.
- 6.30** OK, but why do your eyes do this? Why do your eyes jump around all the time? Why are they—even between the big jumps, why are they making little jumps? It's because any image that stays perfectly fixed on the retina becomes invisible to you. And one

way you can prove this yourself is, if you have contact lenses, just take out your contact and draw something on it, a little shape or triangle, and then put your contact back on. And for the first moment, you'll see this shape, and then it'll, in a few seconds, fade away and it will become invisible to you. Because even though you're moving your eye to constantly refresh the scene, that is staying in the same place on your eye, and so it's not getting refreshed.

6.31 And anything that is not refreshed becomes invisible to you right away. In other words, just like the Troxler effect, features that don't change yield very little information about the world. And what we see is that all the important information in the world, this is what will become clear, it all is about the flux, the change. That's where the important information in the world is. So I'll give you an example of this, in case you don't have contact lenses. It turns out that there is something that's constantly in the way of your eye all the time, and it's totally invisible to you. Can anyone guess what it is? What's that? Oh, also true. I guess that's true, yeah.

6.32 But you have something that's actually in the way of your whole vision, which is, you've got this very rich vascular system. You've got blood vessels that are in front of your retina. And this is in front of your eye all the time, and you don't know it. But the way you can see it is when you're at the eye doctor and he shines a light in your eye at an angle. The key is that it's very unusual. Your brain is not used to having light come in at an angle. And what you actually see—have you guys ever seen it? You see this flash of something, like this, and you think, oh, I just saw something weird. What you're seeing is actually the shadow of your blood vessels. You're seeing the shadow of it cast by the light. It's an unusual angle. So suddenly your brain notices that, when the light gets shined in there.

6.33 But the point is, the reason you never see them is because your blood vessels are sitting on your retina like that. And no matter how much you move your eye, they never change position. So even though that's in your way, your brain has predicted it away, and it doesn't see it. It doesn't see it at all. Just like the triangle on your contact lens, it doesn't see it because there's no change there. So it disappears entirely. What's interesting is that the strategy of ignoring the unchanging, what this does is this keeps the system poised to just detect when something transforms, when something shifts or moves.

6.34 And at the extreme, this is how reptile visual systems work. They can't see you if you're standing still. Did you know that? If you stand still, a reptile can't see you because it only registers change. And maybe you guys remember the scene from Jurassic Park where the little boy screams and the guy, I forgot his name. Anyway, he covers his mouth. He says, "Don't move. He can't see us if we don't move." And so they sit there perfectly still, and the Tyrannosaurus rex comes and breathes on him and his hat flies off and so on. But they don't move. And so what does that tell us? It tells us, reptiles don't bother with position. And yet reptiles have been with us for hundreds of millions

of years. So there's no problem with that. They do perfectly fine. It's perfectly sufficient to not detect fixed position. All they detect is motion.

6.35 And that's generally what our systems do, too. We can detect things that are fixed, but in part, that's because we're making all these saccades. So now with this in hand, let's return to the waterfall illusion. So why doesn't your visual system decide that the waterfall is just standing still? Why doesn't it just say, oh, OK, I'm going to cancel that entirely? Well, first, it might be that there are limits to the recalibration, which is that you can't recalibrate that much. There's another possibility, which is you just haven't watched it for long enough. If you actually stare at the waterfall for a year, then you would recalibrate. We don't know. But if you put enough time into staring at this thing, it might go away entirely. And ever-present background motion would become totally invisible to us.

6.36 Now, that leads to a very wacky but logically sound speculation that I've published before, which is that, are there parts of the world that are actually invisible to us that should be obvious? Imagine there were something like a cosmic rain that existed your entire life. It's never not doing that motion or something. If that were the case, it would be completely invisible to you. It would be totally invisible to you because your brain has never seen otherwise, and so it cancels that out. It doesn't see that at all. And the interesting part is, if the cosmic rain suddenly stopped, then you'd see the cosmic rain going up, and everyone's like, "Oh, my God, something has happened. Look at all this stuff going up in the world." But in fact, the whole interesting thing has ended, and all you're seeing is a motion aftereffect.

6.37 This situation could happen with any sensory channel. Imagine there was a cosmic beep, beep, beep, like some cosmic alarm clock that's been in existence ever since you were born. You wouldn't hear it. It'd be totally invisible to you. And only if the cosmic alarm clock broke would everyone on the planet suddenly say, "Oh, my God, there's a sound." But it's the absence of the sound that you're actually experiencing. So, successful adaptation makes regularities invisible. We have been talking about this, about these illusions, as the results of adaptation. But there's another way to look at this, which is as prediction.

6.38 Now, what's the difference? It's just that adaptation is passive and prediction is active. So if you subtract away the motion of the waterfall or the rocking of the boat or the triangle on your contact lens, that's equivalent to predicting their continued experience. It's the same thing as saying, OK, I'm predicting that these things are supposed to be there. So by doing this, you get rid of data that's expecting to continue. So think about your retinal blood vessels. The reason you get rid of them is because your system predicts that they're going to be there, so it ignores them. It actively says, "OK, I know that's going to be there, so I don't need to worry about that." And so only when those expectations are violated, like when a light comes in and shines a shadow you haven't seen before, does your brain burn any data on representing the data.

- 6.39** This is a general theme, which is that brains have to be extremely extraordinarily energy-efficient. I mean, ChatGPT, in many senses, is smarter than we are, but it uses an enormous amount of energy. I mean, essentially, like the energy of a small city. Whereas a brain exists and does all its stuff on just a handful of watts of energy, like a dim light bulb is all the energy that the brain spends because it has been crafted by millions of years of evolution to be so ruthlessly efficient. Part of that efficiency comes from saying, all right, what can I predict? And if I can predict something, then I don't have to burn any energy representing that all the time. So I'm not going to burn energy representing my retinal blood vessels because I know they're going to be there. So it's free for me to say, "OK, as soon as I've got that prediction in place, I don't need to spend any energy."
- 6.40** That is the key, is that the brain is massively energy-conscious. Anything that's predictable or even partially guessable, the system saves energy by structuring itself, by changing its circuitry, to say, "Look, I want to waste as little energy as possible." And so, a quieter nervous system means that you've got better expectations or fewer violations of your expectations. And what it means is that things in the outside world are going approximately as predicted.
- 6.41** What's really wild is that, in neuroscience, most people think when they're measuring in the brain and they measure spikes, they think, oh, that represents this thing in the outside world. But I've been suspecting for a while that when we see those spikes, it just means, oh, my prediction in the outside world wasn't accurate. I didn't know you were going to suddenly present a face, and that's why I'm spiking, not because I'm a face neuron who cares about faces. It's that I feel like my predictions were just violated.
- 6.42** For better or worse, we could never perfectly predict the world. The world is just way too big and complicated for us to ever get that right. But if you could get it right, what that would be represented by is a totally quiet brain. You'd say, OK, got it. I got everything totally predicted. So the key is this. When you make changes to the circuitry, you make adjustments after what? What's that? What do you say? OK. Put a noun in here. Neural adjustment. When you make a neural adjustment, close. No, close. Closer. Yeah, even closer. Yeah, that's right. I mean, that's right. It's surprise. That's what surprise is about.
- 6.43** So, what happens is I'm trying to predict the world. I'm trying to see what's going on. If I'm getting everything pretty much right, then I don't have to change anything in my brain. There's no brain plasticity required because I've got a good model of the world. It's only when I'm surprised, "Whoa, what was that?" Oh, I didn't know this happened or that, or the person was going to say that, or there's this button on the coffee machine or whatever. Then I say, "Oh, whoops, I wasn't expecting that." And then I make changes to my neural circuitry. That's what's going on there.
- 6.44** It's for these reasons that you first notice, when I put up the Troxler effect, you see the thing, you see the blobs of color on the screen there. You notice it. Why? It's surprising.

You've never seen that thing before. Then you stare at it, and it goes away because your brain says, "OK, I got it. There's no surprise there." Or when you put the triangle on your contact lens, you put it on, your brain says, "Wow, it's a big triangle there." But then it goes away. Why? Because your brain says, "OK, I got it. No more surprise there."

- 6.45** And interestingly, when you walk around your home, your brain is actually paying very little attention to anything. Why? It's because you have good predictions about it. But when you're traveling and you're in a new place, you've never been there before, your brain is on fire with activity because it's looking around, trying to understand everything, the street signs, the building names, the maps, the whatever, because you don't already have a good model for what to expect. And as a result, you're in there, you're really in there and you're paying attention.
- 6.46** The key is, what you're trying to do here, is build up a good internal model of what's going on. So the question is, how does the internal model update? What's the neural technology that allows you to zoom in to the data points that are unclear, that don't match what you already think is going on? In other words, I think I've got a model of what's going on, but then I look over here and in one of these chairs, there's a big teddy bear sitting there, and I say, "Whoa, what is going on there?" Because that doesn't match my internal models for what I thought was going on. So what's the neural technology that then allows me to update my internal model?
- 6.47** It's attention. This is what attention is. All attention is, is I say, "Oh, there's a teddy bear," and then my eyes go there, and I pay attention to it at the expense of other things. But this is the point. You pay attention to the unforeseen bang or brush on your skin or a surprising movement in your periphery or just the thing that you didn't think was actually there. And this is what allows you to put your high resolution sensors on the thing to understand what is the problem.
- 6.48** And then you usually answer. You say, "Oh, OK. Oh, I see. That was just the sound of a lawnmower. That was just the kitten, or that was just a housefly, or whatever." And now your internal model is updated. You say, "OK, good. Now I know what's going on." But things that we can predict, we don't pay any attention to. For example, what is the feeling of your shoe on your left foot right now. You can pay attention to it, and if you do, you realize, "Oh, I'm getting all kinds of sensation from my shoe on my left foot." But the interesting part is that, until I asked the question, there was no reason to pay any attention to it. So you don't do it. You don't pay attention to your shoe until you get a pebble in it. And then suddenly, there's a change that calls for attention. So I put my attention on that, and then I'm able to take care of it.
- 6.49** The difference between predictions and outcomes is key to understanding a strange property of learning called blocking. Maybe you guys saw this in a psychology class or something. But here's the interesting thing. If your brain is predicting something perfectly, it doesn't need to make any further changes. So the example is, let's say I train you that a bell leads to cheese. So every time I ring the bell, I then give you some

cheese. OK, and so now I add to that that a light comes on and predicts the cheese, too. But the bell is still there. And now the question is, if I just show the light, will that predict cheese? As in, will the dog salivate, or will you salivate? Things like that.

- 6.50** And it turns out the answer is no. The light comes to not predict it all. Even though it should, even though your brain should learn, "oh, yeah, the light predicts the cheese," your brain completely doesn't get that. Why? It's because your brain has learned, OK, the bell predicts the cheese, and that's consistent every time. So now adding other stuff, your brain doesn't care. It actually blocks that because it says, "I already know a prediction in the world, and I don't need to learn other things." The existence of this blocking thing, it only makes sense that we understand that changes in the brain only happen when there's a difference between what was expected and what actually happens. So the only time the brain makes a change is when there's a violation of expectation, and otherwise, it says, "No, I'm not making a change at all."
- 6.51** So, with this background in hand, it's easy to understand some other things, like how drugs modify nervous systems. What happens is that, when you consume a drug, whether that's whatever drug, marijuana, cocaine, whatever, what that does is it changes the number of receptors for that drug in the brain, because your brain essentially comes to expect the existence of that drug. It says, "Oh, I'm getting cocaine in there all these times. And so I'm going to express more and more receptors that bind this because I'm expecting it's going to exist out there." And the reason we know this is because you can look at the brain of a person after they've died and determine what the person's addictions were, just by looking at the molecular changes. If somebody has been doing cocaine, they've got a lot more receptors for cocaine.
- 6.52** Now, the key is this. This is why people become desensitized to drugs or tolerant to the drugs, because the brain comes to predict the presence of the drug, and it adapts its receptor presence around that, so that it can maintain a stable equilibrium when the next hit comes. So, in a physical, literal way, the brain expects the drug to be there, and the biological details have adjusted themselves accordingly. And because it's expecting it to be there, that's why people need more and more to achieve the original high. And so this recalibration is the basis of the ugly symptoms of drug withdrawal, because what happens is, the more the brain is adapted to the drug, the harder the fall when the drug is taken away.
- 6.53** And so, withdrawal symptoms, they vary from shakes to depression, whatever. But what they all have in common is the powerful absence of something in the world that was anticipated and not delivered. And this is just like the motion aftereffect. And this, I have proposed in one of my books, is exactly the same with heartbreak, which is that, in this way, everyone you've ever come in contact with leaves an impression in your brain. Essentially, you've got the negative image of everyone you've ever come in contact with, your lovers, your friends, your parents. They fill in the expected shapes, everyone in your life.

- 6.54** And just like the feeling of the waves after you've departed the boat or the craving of the drug when it's not there, so your brain needs those people in your life because you've adjusted your brain to be around those people. And so when someone moves away or rejects you or dies, the brain struggles with these thwarted expectations and has to adjust, has to readjust to a world without that person. It's exactly like drug withdrawal, where it expects the person to be there and the person is no longer there and it has to readjust.
- 6.55** So now I want to bring this back up a little bit, and I want to understand the big picture of what the brain is doing with all these changes that it's making. So what I'm going to propose is a theory called infotropism that I've developed, about when the brain makes a change, how it's always trying to maximize the information it gets from the world. But let's start by looking at something like phototropism, which you're familiar with, which is, this is the active maximizing light capture in plants.
- 6.56** So what the plant does is it changes its position. It adopts new positions. It changes its circuitry, in a sense, so that it can capture the maximum amount of sunlight. So you switch the position of the light, it'll all follow it. And what's interesting is, if you watch a plant in fast motion—I'll just show that again—if you watch it in fast motion, you'll see it doesn't grow straight towards the light source. Instead, it overshoots its trajectory a little bit and then undershoots a little bit and so on. Instead of a preplanned mission, it's a spastic dance with constant correction going on. But what it's trying to do is maximize sunlight capture, right?
- 6.57** OK, and the same thing with bacteria, when they're to find the center of food source. Let's say you drop some sugar. Bacteria are really simple one-celled organisms, but they can find the sugar. How? They just run an extremely simple and elegant rule, which is, you randomly select a direction and you move in a straight line. If things are getting better, you keep going. If things are getting worse, then you randomly change direction by tumbling. That's it. That's the rule. I just keep going. If things are getting better and better, I just keep going. And if things are getting worse, I say, OK, I'm going to randomly choose another direction. In other words, the strategy is to lock things down when they're improving and to dump it when it's not working. This is how bacteria can very quickly and efficiently work their way to the densest point of a food source.
- 6.58** I'm proposing that there's a similar principle at work in the brain, which is, instead of maximizing sunlight or food, it's working towards maximizing information. And it does this by always recalibrating the system so it's maximally sensitive to new information that it can pick up on. So I call this infotropism, and it suggests that the neural circuitry constantly shifts to maximize the information it can extract. And if we think back on the previous lectures, we saw the way that brains come to employ their sensory organs, whether they're capturing photons or electrical fields or scent molecules or whatever.

We saw that the brain learns how to steer its body, whether it has fins or legs or robotic arms or whatever.

- 6.59** What the brain is doing is, it's fine-tuning its circuitry to maximize the data it can eat from the world. This fine-tuning, as we saw, is helped along by rewards, which cause brain-wide broadcast through the circuitry to let the whole circuitry know that something worked. In this way, with a minimum of preprogramming, the system works out how to maximally function in this world. OK, so for example, we saw that the way that these neural landscapes carve themselves in Hayato and William, allowing them to distinguish different sounds. We saw reward-based modification in the last chapter, but we can understand this as a higher level of infotropism, which is that brains adjust to maximize the data that matters to them.
- 6.60** And so there's all kinds of examples of infotropism that we see. And all of these are the same thing, which is the brain saying, "OK, look, I already got this, and now I want to make sure that I'm maximally set up to detect new things." The brain is always maximizing its resources to best interpret whatever it has. And this is consistent with what we learned in the last lecture about cortical remodeling, which is that brains reflect not simply their world, but specifically your world, your outside world. And so, just as the plant seeks sunlight and the bacteria seek sugar, the brain seeks information. It's always, constantly changing its circuitry to maximize what it can draw out of the world. I mentioned earlier, what it's doing is, it's building an internal model, which it hopes is going to match the world. Obviously, as we get older and better at this, we're like, OK, I mostly match the world. But the world's always full of surprises.
- 6.61** Having an internal model, that equates to having predictions. We can phrase it in that way. You've got good predictions about what's going on. If the world proceeds as expected, the brain saves energy. Remember, I talked in the very first lecture about soccer players. The amateur player has a whole ton of brain activity and is burning lots of energy, whereas the pro is burning very little energy. This is because he's got predictions of the world that match the world well. He already knows what to do and where to go, and so he just doesn't have to burn energy, whereas the amateur is still trying to reach that point of good prediction.
- 6.62** So fundamentally, the brain is a prediction machine, and that is the driving engine behind its constant self-reconfiguration. Everything we've been talking about, about brain plasticity and why it's changing in circuitry and so on, it's because it's trying to maximize its predictions. So by modeling the state of the world, the brain shapes itself to try to have good predictions, and therefore, if it has good predictions, it's maximally sensitive to the unexpected.
- 6.63** OK, so let's wrap this lecture 6. We saw various illusions like the motion aftereffect, and we came to understand that this happens because your brain tries to make the expected invisible. If it knows it's coming, then it tries to cancel it out. We also learned about the importance of surprise. So surprise is what causes the brain to pay attention

to something and say, "Oh, my internal model wasn't perfect there. I didn't know there was a teddy bear sitting on that chair. So now I'm going to make an adjustment to my internal model." Finally, I proposed this model of infotropism, which says that brains are always adjusting so they can maximize the information from the world, so they can discount the things that it already knows and be maximally sensitive to new data. So thank you for your attention. I'll take any questions.

- 6.64** You said that the brain is always seeking information, right? And it's a predictable machine. So as a normal human being that likes to learn, like whether it's myself or my children, what's your recommendation? Should we be always learning something new, seeking to learn something new, every single day or once a week? Not once a week, but on a weekly basis, on a regular basis, on a second-to-second basis? And how often or far apart is that element of surprise, is going to help me learn and captivate what I'm learning?
- 6.65** Yeah, great question. It's so important. So I talked in the last lecture about why I think the gamification of education is so useful. It's because you can learn lots of things, but in a context that you care about, in a context that's going to force you to pay attention. So in other words, if you showed me a list of dates in Mongolian history, and let's say I don't care about Mongolian history, then I'm not even going to have surprise because I'm not invested in it. It doesn't matter to me at all. I categorize that as like, oh, a bunch of dates that I don't know or care about. But if you show me something I care about, then I'm really going to pay attention, and I care because I've got the internal model. So this is why I'm so jazzed about the opportunities for individualized education, for exactly this reason.
- 6.66** In answer to your other question about how often should you force yourself to learn new stuff? All the time. This is the most important thing that you can do, the thing that you have control over, and to some extent, over your children, too, is making sure that there's always new things to learn because every new thing that is new and interesting causes your brain to say, oh, I'm going to update my internal model a little bit. You can imagine, if you're not challenging your internal model, it just stays at that level. It's not doing much of an expansion.
- 6.67** This is actually the most important thing for people in your life who are older or maybe retired. The most important thing you can do is make sure that they are staying cognitively active because, otherwise, as their brains degenerate because of physical problems like Alzheimer's or whatever, they're not building new bridges and roadways, but instead, that's just going away. So the most important thing is to make sure that they're constantly challenged with things that matter to them.
- 6.68** Actually, to piggyback off of that question, do you think there is a time that we need to absorb information or a time to learn something new and then take a break so your brain can really calibrate? Because, for example, TikTok, a lot of people, you can get a ton of information that's interesting to me. But then I'll realize, 10 videos earlier, I don't

even remember what I watched, even though it was interesting to me at that time. Whereas maybe if I took a break, I could have absorbed that information more. Do you think there is that time frame period?

- 6.69** Yeah, I do. But I think it's not necessarily a time frame thing. It's about, do I have the opportunity to do something with it? So let's imagine, whatever, on TikTok, when I surf it, I see lots of parkour videos. For some reason, that's what the algorithm serves up to me. But I love those things because I say, "Oh, I didn't know you could do that with your body and do this thing." And so then maybe I'll go and try it. But it's the thing of, like, not just—and we talked about this in terms of the 10,000-hour rule and actually implementing knowledge. If I'm seeing something and then trying it, or seeing something about a math problem and then going out and doing something with it, I'm actually building on my internal model. I take your point that if you're just watching video after video but not doing anything with it, it's all input. I'm not really getting a chance to nail any of it down.
- 6.70** Is there a social brain speed that can be increased or decreased? How do you do it?
- 6.71** What did you say, brain speed?
- 6.72** Yeah, like the way of processing things.
- 6.73** No, it's just a matter of forcing yourself to learn and do new things all the time, because when you do that, what you come up with is more and more compressed ways to encode information. So, just take the example of the amateur soccer player and the expert soccer player. For the amateur, he doesn't get what's going on. There's just stuff, and people are running or whatever. But for the expert, he thinks, "Oh, I get it. This is this maneuver where these guys are screaming for me and these guys are coming around this way, and so I'm going to go this way." But it's like he has a more compressed representation of what's going on where he can say, "Yep, I got this," in just a sentence.
- 6.74** And so that is what maybe we mean by brain speed. Certainly this is true for physicists, mathematicians, or whatever. They say, "Oh, yeah, this is this kind of thing." This is just like Heisenberg's uncertainty principle, but it's just a different flavor of this or whatever. So that's how you can get faster at things. But it's all the same hard work of chopping wood and carrying water. And you got to do the work to get there and say, "OK, now I've spent a lot of time learning these things. I've really improved my internal model such that I've got a very fast and efficient model."
- 6.75** Is it true that the brain seeks the path of least resistance? As I understood that it does like to conserve energy. But if it always likes to learn and it's wanting to be active, then how do those two things reconcile?
- 6.76** Yeah, I'm so glad you asked that question. Yes, the brain is almost always trying to take the path of least resistance because it's energy efficient. And this is why you have to

force yourself off the path. And that's really what learning is about. So when you sit down and force yourself to learn and look for the things that are interesting questions and so on, you're forcing yourself off the path because, otherwise, your brain can convince itself, "Oh, yeah, I know the answer. This is easy. I got this." And look, people do this with politics all the time. People think, "Oh, yeah, I know the truth. The truth is clear to me. I don't know why it's not clear to them and them. If I could just shout loudly enough in all caps on Twitter, everyone would see that I know the truth, and they don't." But this is all about path of least resistance and not challenging ourselves to do that extra work. And that's why education is so important.

- 6.77** I'm trying to see where the reticular activated system fits in in all this. Can you tell us a little bit more about that?
- 6.78** The which system?
- 6.79** The reticular activation system. How does that—
- 6.80** That's an interesting question. I don't know that it fits into this so much. I usually think about that in terms of being involved in sleep and other issues. The main thing that's involved with brain plasticity are things like acetylcholine, which open up parts of the brain and say, "Look, I don't know why you're making the wrong prediction there, but this didn't match up with what I saw from the world. So, be plastic." And then the system gets to reform there. And what we're going to talk about in the next lecture is—actually, I won't give the punchline yet, but it's about how that changes through time. The acetylcholine system that I told you about acts very differently in children and adults. So we'll talk about that.
- 6.81** When you were saying that, if it stays unchanging for too long, the brain ignores it, is that just for visual things or can it also be maybe for thinking more like trauma and childhood, and things where parents are treating you the same way and you just stop paying attention to it or growing from it?
- 6.82** Yes, but for a slightly different reason. So it doesn't just happen in a visual system. It happens across all senses. For example, if you're in a room where an air conditioner is on and then the air conditioner goes off, and you say, "Oh, I didn't even realize it was on, but I clearly heard it disappear," so it happens in other senses. But with something like how you're being treated by the world, that's a very deep thing about our internal model, about our understanding of what the world is. And so you, as a child, think this is what parents do. This is how the world operates. And whatever spot on the planet you grow up on, you think like, "Oh, I get it. This is what the deity that controls this is." Then you grow older and you realize, "Oh, wow, there's actually lots of different ways to look at this," whatever.
- 6.83** But the point is that what you express is just an issue of your learning of the world. When you drop into the world, you don't understand anything about cultures and people and neighborhoods and how parents should treat you or whatever. So you just

get used to what's going on, but for a slightly different reason than this active recalibration thing.

6.84 In terms of attraction, if your partner is unchanging and he becomes predictable—

6.85 No, that's—I can take it from here. Yes. It turns out this is one of the challenges of relationships, is that people start to ignore each other more and more, just like you ignore the Troxler colors or anything that your brain is getting used to, and you see less and less of the cosmic rain or whatever. And that's because your brain is constantly trying to make predictions about your partner. And the better it gets at that, the more it's like, all right, nothing surprising there. This is why it's important, of course, in relationships to make sure that you keep injecting surprise and new things like that. But yes, it's also why people will hear you less and less, and pay attention less and less.

6.86 In terms of neuroplasticity, what's the role of pharmacology in it? If you say, like, acetylcholine can sprinkle your brain and you get more open to learn, is there something in pharmacology that we can use to get better at it?

6.87 Great question. So, what we're going to see in the next lecture is, again, I don't want to give away the punchline, but it has to do with how this works in adult brains. So, if you're wondering, can I just get acetylcholine in my brain? That actually won't be a very useful thing, and you'll understand why. But there are, people suspect that, for example, psychedelic drugs are just opening up plasticity broadly so that you have a different way of seeing things because the brain always tries to take the path of least resistance.

6.88 And what happens with psychedelics is that everything is open in this different way, and the thoughts can take different pathways. And so when people come out of that, they say, "Wow, I didn't even realize that I could think that, that I could be that person. I could think those sorts of thoughts." And so, the idea is just the experience of having that might be the thing that reopens up all kinds of plasticity. That's what you can do pharmacologically.

Lecture 7: Adapting with Age

At a glance — Adapting with age — how plasticity changes over a lifetime.

- 7.1** OK, welcome back, everybody. So I've been speaking this whole time as though brain plasticity is something that just happens in the brain all the time. But in fact, it changes through the course of a lifetime. That's what we're going to talk about today. In the 1970s, there was a psychologist at MIT named Hans-Lukas Teuber, who got interested in this question about brain damage from soldiers in World War II. He noticed that all these tens of thousands of soldiers had obtained brain damage, and he wanted to know how they had fared, what had come out. He looked through records and found people's outcomes.
- 7.2** What he found is that some people did better than others, clinically. He scoured the paperwork and he looked at stuff, and he finally realized there was a very simple answer, which is the younger the soldier was when they got the brain damage, the better they came out. The older they were, the worse their recovery. Why does that happen? It's because brains have diminishing flexibility. I used in the first lecture the analogy of the globe and the different countries. And young brains are like the globe was 5,000 years ago when country borders could be anything. You could have had the countries end up anywhere.
- 7.3** But today, it's less likely that country borders are going to shift. So if you look at America and Canada and Mexico, those borders have settled into place, and it's really unlikely that they're going to move around. But 250 years ago, there was a lot of flexibility. There was the French territory and the British territory and the Spanish territory, and all these things were constantly in flux. So the way to think about the brain is that a young brain is like the Earth a long time ago, and now an adult brain has settled into place. As a result of all the wars and the clanging of swords and whatever, the border settled into place, whereas the young brain is still at the dawn of its wars and can re-imagine its territories.
- 7.4** I'll give you an analogy here, which is that, if you look at the beginning of any nation, let's say you look at the Constitution, find that there are lots of changes at the beginning, and then it slows down. So for example, with the American Constitution, 12 of the amendments took place in the first 13 years. And after that, there were a maximum of four changes over any 20-year period, and most periods had no changes at all. And the latest change, the 27th amendment, was in 1992. And the Constitution has been at a standstill ever since then. And that's just like brains. That's what happens. Brains figure out all the stuff, make changes, and then they settle into place.
- 7.5** Last analogy on this is that, if you look at the IRS tax code, what you find is that it's grown and grown, and now it's super complicated, but now it's really hard to make changes to it. So this is what brains do. They learn the rules of the world, and that's

great, but it's at the expense of further change. Actually, I do have a couple more analogies. One analogy is that I live in Silicon Valley, and all around me, there are these young startup companies that are very dynamic and scrappy, and they're always doing stuff. And what happens, if you watch any of these long enough and they become successful, is they become an established company, and they're entrenched, and what happens is they become bloated and unable to move around flexibly and nimbly.

7.6 This is the story of virtually every company that you see in Silicon Valley. Google started this way, and now Google's got this giant rulebook of how they have to do their business and so on. It's very different. And this is why big companies spin off little startups out of the company because that's their strategy to deal with this problem. And by the way, this is what humans do. So we have spin-offs. We get inflexible, and so then we have progeny of our own.

7.7 Let's return to Hayato and William, who we met at the beginning when we were looking at how language shapes your brain. So Hayato is born in Japan. William is born in Palo Alto, California. And if we look at them across the lifetime, what we find is that they both start off extremely flexible. So recall what we talked about last time, which is, in terms of language, both babies can hear all the sounds of language. They can hear all the sounds of each other. But with time, really, by 10 to 12 months, the landscape is already forming, such that they are unable to hear the sounds that the other one can hear.

7.8 But it's more than that. It's more than picking up the rules of their culture and their neighborhood and their world, their social interactions, everything. Depending on the age that they are, depending on the generation that they fall into, they might do this by reading a scroll or reading a book or flipping through an iPad. It doesn't matter. But the point is all this information is going in, and it starts making them less flexible. So by the time they get older, then they become adults.

7.9 Hayato belongs to a particular political party, and he's unlikely to change. And William plays the piano reasonably well, but he doesn't really have interest in other instruments. Hayato likes to cook, but he uses essentially the same 14 ingredients for everything he does. William spends his time with a vanishingly small fraction of all the available websites that he could go on to. Hayato commands a respectable golf game, but he doesn't really care about any other sports. William lives in a city of eight million people, but he only has three close friends, and so on and so on.

7.10 What these life trajectories illustrate is a general point, which is that babies are born with very few built-in skills, but lots of flexibility, whereas adults have developed real skills. They are experts at things, but they don't have too much flexibility left. And so the way we capture this is we talk about babies having fluid intelligence, and adults have crystallized intelligence, because a baby can't run a corporation or deliver a lecture or whatever because the baby just doesn't have those skills. The adult has

developed those skills, but it's a trade-off. They've developed expertise at the expense of flexibility.

7.11 And so this is why I used that quotation from Heidegger a few lectures ago. I just think it's so insightful about the trajectory that humans follow. And this is like the story that I mentioned in a different lecture about Itzhak Perlman, the violinist, where the concertgoer said, "I'd give my life to play like that," and he said, "I did." Because to get good at something means to close the doors on other things. You only possess a single life, and so what you devote yourself to sends you down certain roads, and all the other roads remain unused for you.

7.12 OK, so this underscores this general point that babies have few skills, but lots of plasticity. Adults have expertise, but less plasticity. So there's this trade-off between adaptability and efficiency. So remember, we talked about how the brain burns jobs down into the circuitry, and that's what makes it efficient. That's what makes it massively good at what it does. So, as your brain becomes good at particular things, it's less able to tackle others. Now, why does this happen in the brain? Well, it's for at least two reasons.

7.13 The first one has to do with pruning. So it turns out that the brain starts off massively interconnected, and we'll talk about that in a second. I'll show you a picture of that. But everything is connected to everything, essentially. I'll give you a sense of how you can measure that, which is, let's imagine you hook up an EEG to an adult and to a baby, and then you play a loud beep. What you see in the adult is that the auditory areas respond to the beep. But in the baby, you have the auditory and the visual areas both respond to that. And with time, it gets squished down just to the auditory areas responding to it. Why? Because in the baby, there's this massive cross-connectivity. Everything is connected to everything, and that gets pruned like a garden.

7.14 So if we look at what happens over time, when neurons start off, they actually don't have that many connections, but they start massively growing connectivity to the point that they're actually making as many as 2 million new connections every second in a baby's brain. It's shocking, the speed of this connectivity that's going on. But by age two, it reaches its peak. You have neurons with about 15,000 connections each at age two, reaches a peak, and from there it starts pruning. From there, all you're doing is losing connectivity. I shouldn't say all you're doing. Most of what you're doing is just losing connectivity. You're figuring out the roads and the pathways that are the ones that maximally resonate with your world, and everything else is going away.

7.15 Now, the upside is that you end up with lightning-fast ways of solving problems and doing things like walking or putting food in your mouth or giving a lecture or whatever. You're finding all these really efficient ways to do it in the brain. The downside is that it's hard to attack problems inventively. Here's my analogy for this. Think about two towns where people need to get from one to the other. So people interested in going from one to the other might take different paths. Some people walk along the scenic

route, along the ridgetops. Some people go down to the river and hike along the slippery rocks. Some people build bridges, whatever.

- 7.16** But with time, what happens, we've all seen this, whether it's on a college campus or between two towns or whatever, with time, certain paths become the standard, and eventually, the government builds the bridge and then paves it. That becomes the standard route between. So things start with broad optionality and eventually get reduced to the standard. And again, what that means is that the interaction between the two cities is much more efficient now, but the optionality has gone away. So this is the same thing with brains. They begin with many possible routes, and then with time, the practiced pathways are the ones that become difficult to exit.
- 7.17** So beyond the diminishing optionality of the pathways, there's a second reason why older brains are less flexible. When they change, they only do so in small spots. So what we talked about last time is the way that you've got these neuromodulatory systems, like acetylcholine, that release acetylcholine. They broadcast to the whole brain, saying, "Hey, something interesting just happened. I want you to be plastic. I want you to allow change to happen here." So we talked about that system last time. If I play a note on the piano and then stimulate your acetylcholine system, you'll get changes in the brain.
- 7.18** But what's interesting is that, when you look at this in a baby, you find that these acetylcholine systems are splashing this neuromodulator everywhere, and you have big, broad changes. But when you become an adult, it's very targeted. The system says, "Oh, OK, I'm going to release acetylcholine just right there and just right there, because those are the parts of the internal model that didn't do a good job predicting. So I'm just going to make little changes here and here." And so it's not a broad thing.
- 7.19** I think of this, my analogy is like a pointillist artist, who is mostly done with his painting and says, "It needs a little dab of blue there. It needs a little dab of red over there." That's what these systems are doing. It's the same system, but now, instead of broadcasting broadly, it broadcasts very specifically. And so your painting is essentially finished, but you're making little changes here and there when you need them.
- 7.20** OK. So because young brains, because they allow for this global change, this acetylcholine that gets splashed everywhere, my colleague, the scientist Alison Gopnik, describes babies as the research and development department of the human species, which is to say, they're the ones who are trying to absorb everything and really understanding new ways of doing things. And I might have mentioned this, but one of the really fascinating parts is the way that we, as babies, learn everything that the whole species, the whole Homo sapiens species, has done before us, and then we leap off from there. And that's why our species is so successful, because we're not just redoing the same thing over and over.
- 7.21** We have this brain plasticity that allows us to absorb everything and say, "OK, I get it. I'm in a world with laptops and cell phones that let me call across the world and Zoom

meetings so I can instantly teleport myself to anywhere on the planet. And there exist airplanes where I can be 30,000 feet up in the air." We have a million humans in the air at any moment. That's weird. A hundred years ago, no one would have predicted that. But if you're a baby born now, that's just the world you're born into. And they say, "Great, now I'm going to go from there."

7.22 Now, sometimes you might wonder, well, what is it like to be inside the head of a baby? I mean, we were all there, but you can't remember it. But there are ways, I think, that we can get close to understanding by seeing other situations where we have a lot of plasticity going on. One of them is what I mentioned before. When you're traveling, when you're in a new place, everything's new, you're having to pay lots of attention to the outside world, as opposed to when you're at home and you're saying, "I got it. My couch is there, the kitchen is over there. I don't have to pay any attention to what's going on." But when you're doing new things, you are like a baby in that way, where you're making lots of changes and discovering new things. You're drinking in the new sights of the new land. You're experiencing more novelty, more learning, more distributed attention.

7.23 So babies start off as very plastic, and that's how they build an internal model of their world. And once they have a good internal model, then they're like adults, and they're not making so much change anymore because they don't need to. What we're seeing so far is that brains are most flexible at the beginning, but it's not a linear decrease like that. It's not that you just have this slow diminishment of it. There's something called the sensitive period, what used to be called the critical period, which is where certain things need to happen or else it's too late. During this course of lectures, we've seen lots of examples of this so far.

7.24 For example, you recall Matthew, who I talked about in the first lecture, who had a hemispherectomy where half his brain was removed. You can only do a hemispherectomy until a patient is about eight years old. Matthew was six at the time. Otherwise, what you find, I saw one surgeon describing it as, if you do this as an adult, they have to change the task to meet their brains, as opposed to changing their brains to meet the task. So in other words, with Matthew, he just figures out, "OK, what do I need to do in terms of brain plasticity to do this task?" Other people need to say, "Wow, this is never going to work unless this thing's over here and the handle's over here, and I can pick it up that way and so on." You have to do the hemispherectomy early. It's this incredible thing where you can remove half the brain, but you can't do that in an adult.

7.25 OK, we saw the case of Genie, where this was the girl who was found. She had been beaten and starved and restrained and so on. Then we talked about how she had physical problems. She couldn't chew food. She wasn't toilet trained and so on. But the key about language is what I want to zoom in on, because you remember what happened is these psychologists all came in and everyone was heartbroken about

Genie and wanted to give her lots of attention and train her and teach her language and so on, but they couldn't. It was too late. She had missed the critical period.

7.26 By the way, there was nothing genetically wrong with Genie. There was no reason that she should have had a problem learning language. When children like this are found, these are called feral children. When feral children are found, this is always the case, that they can't learn language because if they've missed that critical period of just not getting spoken to. By the way, does anyone know, with a deaf baby, a deaf baby, how does that work? Does a deaf baby end up being unable to learn language? What's that?

7.27 It can.

7.28 It can? But how?

7.29 I don't know. Helen Keller?

7.30 OK. Yeah. Helen Keller can. That's right. But what about deaf children in general? Visual. Exactly right. So what deaf babies do is they do motor babbling. So when someone's speaking to them with sign language, they're doing this with their hands and figuring that out. And as long as they are exposed to language, which can be sign language, perfectly fine. Then they're able to pick it up and no problem. Right, but the child needs to be exposed to language by a certain age. Otherwise, it is too late for them to get language. As long as there's input to pick up on, the baby will do so. That's the key. But after that door swings shut, it's too late to learn the fundamentals of communication.

7.31 So again, I just want to be clear. What we're not talking about is learning a second language or things like that. What we're talking about is the fundamental of what language is. In other words, I can put a word on an item, and that identifies that item for someone else. The concept of what language is, that's the part that needs to be learned at the beginning, or else it's too late. There's also windows for more subtle aspects of language, too, like accents.

7.32 So what's the difference between Mila Kunis and Arnold Schwarzenegger? As it turns out, Mila Kunis speaks American English with no accent, and Arnold Schwarzenegger does have a thick Austrian accent. And so because Mila doesn't have an accent, most people don't know that she was born in Ukraine and didn't speak any English until she was about seven years old and moved here. But Schwarzenegger, on the other hand, lived in Austria and didn't move here until his early 20s. And as a result, he is unable to get rid of his accent in American English, but Mila was perfectly fine at getting rid of her accent.

7.33 Why? It's because it depends on the age that you move to a new country. So the general story is, if a child moves before the age of 13, they won't have any accent in whatever country they move to. If they're around age 13, they might have a little bit, but if they move after, by their teenage, late teenage years, they're unable to get rid of

their accent. So this is something you can observe anywhere. And what this demonstrates is a window where it's important to move if you want to get rid of an accent. Now, by the way, I talked last time about motivation, and obviously, that plays a role here as well, because Hollywood actors figure out how to shake accents. Just as an example, Hugh Laurie, who played House on television, has a British accent. But when he played House on television, he spoke perfect American English. Why? Because he's an actor and his job depends on it, and it was motivating for him to do that.

- 7.34** So it's not that it's not possible to shake an accent. It's that for someone like Schwarzenegger, he was massively successful with an accent, and so you don't really need it. You don't really need to get rid of it. But here's the thing. What this demonstrates is that your ability to sonically morph into a new country has a window to it, just like many things that we're going to see in today's lecture. There are many windows that we're going to see.
- 7.35** OK, I'll give you just one more clinical example. If a child is born with misaligned eyes—you can have eyes that are misaligned, what's colloquially called wall-eyed or cross-eyed—it turns out there's a very straightforward way to fix this, which is, in analogy to what we saw with stroke, where you pinned down the good arm, what you do is you patch the good eye and you force the other eye. Actually, what you do is you surgically correct the extraocular muscle. But then what you do is you force vision through the weak eye. And the reason that's important is because your visual cortex is essentially sharing the territory between the two eyes. But if one eye is dominant and the other eye is always looking off not where you're intending, your visual cortex starts diminishing the territory that that eye has. So what you do is you fix it and then you cover the good eye, so this guy has a chance to fight for territory again.
- 7.36** OK, so it's a very cool method. It's very straightforward and works great, just like constraint therapy. But the point is, this is a very successful method for children, but it has to be done by about six years old. Because as we're going to see, the visual cortex is one of the earliest areas to start locking down. We'll come to this in a few minutes, but some parts of the brain are very flexible your whole life, but the visual cortex in particular really locks down. And that's why you need to make sure you do this early in a child.
- 7.37** I'll give you another example of this visual cortex locking-down thing, which is a few times in earlier lectures, we talked about this issue that for a blind person, their visual cortex gets taken over by hearing and by touch and by other things. But it turns out, if you look at things, people are born blind or go blind when they're young or go blind later in life, the takeover of the visual cortex is almost complete for people born blind, less so for people who go blind when they're young, and even less for people who go blind later. So in other words, all I told you was that visual cortex gets taken over, but it depends on when, because some parts of the brain, like the visual cortex, become

firmer and firmer in their hold on the territory, and then they're not capable of getting taken over. So timing matters.

- 7.38** And I'll give you an example of this. I made a television show on PBS called *The Brain*, and one of the people that I spoke with at length is a guy named Mike May, who was normally sighted, but at the age of eight, there was a chemical explosion that scarred his corneas, and he was blind after that. Mike became a very successful businessman and became a champion downhill skier, blind downhill skier, actually, which is quite extraordinary. He functions by having beeps, so there are beeps on the trees and things like that, and he can navigate accordingly. Mike was living a perfectly fine life, but sometime in his 30s, and he had some kids.
- 7.39** Sometimes in his 30s, he heard about a surgery where you could repair the corneas to get rid of this scarred chemical damage so he could see again. So he thought, "Great, I'm going to do this." So he got the surgery, and there was a television camera, a bunch of cameras there, to have this first moment where he can see again. Because what happened is his cornea was repaired such that light was finally going through there and getting to the back of his brain. And I saw this footage when the bandage was first taken off and his kids are there and his wife's there. And he says, "Holy cow." And the television crew thought that was great and so on. But his facial expression was such that it was clear there was something a little bit wrong. He was taken home and he was there with his kids, and he said, "Let me keep looking at you," and so on. But there was something clearly not right going on.
- 7.40** What happened was, Mike was not actually seeing. His corneas were now fine, such that there was visual information going to the back of his brain, but his visual cortex was unable to interpret what the heck these signals meant because his visual cortex had locked down through his adulthood here. And so everything to him didn't make sense. So for example, a hallway he was used to when he walked down a hallway, knowing that the walls stay the same distance apart, when you look at a hallway, it visually looks like that. It didn't make any sense to his brain. Or when his wife was driving him home from the hospital, there were these overhead highway signs, and he kept thinking that they were going to get hit by these signs because he just saw some kind of looming object that looked like it was getting closer and closer.
- 7.41** Anyway, Mike, as it turns out, has never really regained vision, even though his eyes are perfectly fine now. It's because his visual cortex locked down, his brain lost the ability to understand how to use that part of the brain. It's a very important insight about flexibility diminishing with age. Everything is easier in young brains, like learning how to see and stuff like that. But also with Perlman, if you listened to him playing the violin and thought, "I want to be a violinist, too, and I'm going to put in more hours than he is and become better than Perlman," you couldn't. Why? Because he got a head start. Because he has just been doing this from the time he was a kid. And if you don't play

violin, you are way too late. It doesn't matter if you put in the same number of hours. The point is that his young brain was doing this from an early age.

- 7.42** And just like Mila Kunis with her unaccented speech in American English, she's been doing that from a young—everything is easier to morph into or become expert in for the young. So acquiring vision and language and violin proficiency, it depends on normal input from the world. We saw that some lectures ago, that the brain drops into the world half-baked and expects a particular kind of input. And if a child like Genie does not get that input, it's too late. She's not going to be able to learn it. So the ability to learn language or possess vision or interact socially or walk normally, have normal neurodevelopment, this is all limited to the years of young childhood. This is the important point. And after a certain point, you just can't get it anymore. Those abilities are lost. So that's why I emphasize this point about the importance of making sure that babies have the right input while they're babies.
- 7.43** Now, what's interesting is the way that we're so influenced by things in our childhood because of this diminishing flexibility. So childhoods are massively important. As an interesting example, I just thought this was a fascinating thing that I ended up following these studies on. It turns out that there's a correlation between how tall a man is and how much salary he commands, on average. And so in America, every height of inch translates to about 1.8% increase in take-home pay. Now, why is this?
- 7.44** The assumption is that this results from some height discrimination issue where everyone wants the tall guy because he seems commanding or something. But actually, that's not it. It turns out the best predictor of a man's salary correlates with his height at the age of 16. Now, that's nothing to do with nutrition or anything, because if you correlate with his height at age 11 or 7 or something, there's not a strong correlation. It has to do with how strong they were when they were 16. Now, what does that have to do with it? It's because the teenage years, this is where you develop your social status. This is where you figure out how to interact with people in the world and develop your skills at this thing. And so who you are now is strongly dependent on who you were at the age of 16.
- 7.45** And by the way, one thing that's consistent with this is that you find that this height-pay correlation goes most strongly with very social careers, and it's not so much with blue-collar jobs or other things or artists, whatever. But in socially oriented careers, like salesmen and whatever, this goes along with it because who you were at 16 determined how you fit into the world, your comportment in the social domain. So not surprisingly, how people treated you in your formative years has a great deal of impact on who you are in the world in terms of your self-esteem, your confidence, and your leadership.
- 7.46** And one of the things I found interesting, actually, which I was shocked when I happened to see this little article, was that Oprah Winfrey is worth about \$2.7 billion. So what I was surprised when I read is that she has a deep-rooted fear of becoming

homeless and penniless. It's a deep fear that she can't get rid of. And so it's because of the path that got her here. Before she was an empress of the media, she was an impoverished child in Mississippi, born to a teenage single mother. And the roots that we lay down when we're young are always with us. So she had this fear when she was young. And even now that it's so improbable that could ever happen to her, that is what sticks with her.

- 7.47 OK, so one of the things, just as a side note, I've got kids, and I've been thinking about this is, I think sociologists and psychologists are going to be studying this for the next 100 years about the effect of the COVID lockdown and missing a year or a year and a half of school on kids of different ages. What is the effect of this? And I don't know the answer to that, but I suspect that it's going to become clearer in the fullness of time. I mean, it might take 50, 70 years, whatever it is, before we finally come to conclusions about what is different about that generation and the different ages that kids were locked down. But the point is, what happens to us when we're young, sticks with us our entire lives.
- 7.48 Now, to capture this idea of a sensitive period, I've been using the analogy of a door swinging shut. There's one sensitive period. But in fact, what I want to make clear, and I hinted at this with the issue of the accents versus language and so on, is that it's not one door, it's many doors. There are many doors, there are many sensitive periods with everything that we're talking about. I'll give you an analogy, which is, the brain is so impressionable in its early days that it can get into hot water. Some of you may know this guy in the 1930s who would just be present when geese would hatch from the eggs and the geese would imprint on him, say, "Oh, that must be my mother." Then he could walk around and these geese would grow up and follow him. What he found is that geese formed this relationship with the first animate object that they see.
- 7.49 That's a perfectly sufficient strategy. It usually works. Usually, the first animate object you see is your mother, but it can get fooled in the wrong circumstances. But here's the thing, even though people said, "Wow, look, these geese have a very tiny window of plasticity where they just imprint on that thing," the fact is, a goose can figure out lots of things later in life, too, right? Like, where the river is and where to best look for food or the identity of other geese that they meet in their adulthood and so on. So it's not that there's a single door. There's multiple doors.
- 7.50 And I'll give you a surprising result in neuroscience, which is some colleagues of mine did a study where they looked at the adult visual cortex. So you have a perfect—this was done in monkeys, and they looked at a perfectly sighted monkey, and they damaged a little part of the retina at the eye. And then they measured, how does the visual cortex change when you're not getting vision from a little patch of the world here? And what they found really surprised them, and it surprised everybody, which is that, in adult monkeys, there was no change in the visual cortex. Even though the

monkey wasn't getting any vision from there, just didn't change anything about the visual cortex.

- 7.51** But that was unexpected because there are changes in, let's say, the somatosensory cortex or the motor cortex in adults. We talked about this guy with the reversed steering. I didn't talk about this guy, but have you seen these jetpack things where you can actually blow air out of your arms and do this thing like Ironman? So this guy actually built this. It's very hard to learn how to do it, but he's gotten quite good at it. I've had other friends who have started to learn, and at first they crash a bunch of times. But the point is, adults can learn how to do this or any new thing. I mean, I've talked extensively about whatever, pogo sticks or surfboards or whatever. And I'm sure all of you, as adults, have taken up new things that you hadn't done before, let's say in the motor domain.
- 7.52** So it was surprising that there was no plasticity in the visual domain, even though your motor and sensory systems can do so much flexible learning at any age. What this tells us is that some regions, like the visual system, lock down quickly, but other regions stay flexible. Now, that's weird. And so what it means is, it's not just one thing happening even across regions in the brain. So what some people have suggested is, OK, well, maybe it has to do with the different learning strategies that the brain needs to deal with. So, some things in life, you have to stay flexible on, like learning a new word, a new vocabulary word, or the ability to learn new directions, how to get to something, or visual recognition of people's faces that you hadn't met before, and now you need to know how to recognize their faces and so on.
- 7.53** Other areas, you can lock it down because the building blocks of vision, as in lines come in different orientations, there's a certain spread of angles, how to chew food, general rules of grammar, whatever, that can get locked down. So this was suggested in the literature that maybe the brain areas are flexible predicated on whatever the learning strategy was there. But my question when I read this was, well, how would the brain know that in advance? How could the brain know what's supposed to happen? So anyway, I proposed a new hypothesis, which is that it has to do with the stability of the incoming data. So let me explain this.
- 7.54** If the incoming data is really stable, the system can harden around that and say, "OK, I think I got this." And if the data is changing all over, it says, "OK, I'm going to remain flexible." So here's the thing. Something like the phonemes, the sounds of language that you're hearing around you, that's really stable in your childhood. You're hearing thousands of hours of language spoken by your parents and other people, and you get the sounds of language. But something like, let's say, Hayato and William decide that, as adults, they want to meet up and go windsurfing. They can figure out how to do it. Why?
- 7.55** Because your body plan is always flexible, in the sense that, at times you gain more weight or you lose more weight, you hurt your foot and you have to limp, you ride a

bicycle, you ride a skateboard, you surf. Your brain is really used to, "OK, there's lots of ways that I can locomote, and it's not one thing," as opposed to the sounds of language are really one thing going on. You have lots of ways that your body changes all the time. You sit in a car seat, whatever. You sit on a recumbent bike, you go to the gym, and you do weird new things. So that's very flexible in terms of the input from the world. And so my assertion is that that system then stays flexible. So the primary auditory cortex locks down, the primary visual cortex locks down, but not your somatosensory cortex, not your somatomotor cortex.

7.56 Now, what's really interesting is if we zoom into a single area like vision, what you find is that, in low-level areas, so what we call primary visual cortex, that's the red area here, that's just encoding for edges and angles and colors, and that's it. Whereas more high-level areas of the visual cortex, like the area I'm showing in blue there, that encodes for the recognition of objects and faces and stuff like that. So I say, "Oh, this is a remote controller for a computer." Obviously, I never saw this in my childhood, but as an adult, I can learn this and figure it out and store that. And so if you ask me a month from now, "Hey, what was on the podium with you?" I say, "Oh, yeah, there was a thing. It was about this big. It had whatever."

7.57 So there are parts of my brain, interestingly, that stay flexible, like that blue part, but the red part locks down. So what I want to point out here is there's a hierarchy of flexibility going on here. Whereas the stuff that never changes in the world, the edges and angles and colors, that gets stabilized and does not change. But there's flexibility with object recognition. So that part of the brain is staying more flexible. My entire life, I can learn new things. I can learn in 25 years from now what the newest object is that we can't even imagine now. So the point I'm making here is that there's a hierarchy of flexibility, even within a system like vision, where the parts that say, "OK, I got this," they lock down, and the parts that say, "Whoa, there's new stuff all the time," they stay flexible.

7.58 Because I love analogies, I'll just give one more here, which is, if you're building a library, what you do is you say, "OK, I'm going to define the space. I'm going to set where the bookshelves are," and that essentially stays stable. But the inventory on the shelves, that changes the whole time the library exists. You're always swapping stuff out and putting new displays and blah, blah, blah. That stuff changes. But the fundamental architectural stuff, that gets locked down when you're first building the library. And so that's a good— so building the shelves of the library, that's the red part of the brain there, and swapping out the inventory and having new displays, that's the blue part of the brain there.

7.59 You guys have now taken this brain plasticity class, and someone's going to ask you at some point, "Hey, is the brain plastic as we get older?" What you need to tell them is, it depends on the brain areas that we're talking about. So some parts of the brain are no longer flexible as you get older. Your primary visual cortex, primary auditory cortex,

these things are locked down. There's no single answer to the question. Plasticity declines with age, but across the brain, it declines differently, sometimes steeply, sometimes shallowly, depending on the function.

7.60 By the way, I proposed a hypothesis in one of my books, *Livewired*, that there's an analogy with genetics here. This is something that science is still working to understand, but genomes seem to lock in their nucleotide sequences, some more than others. In other words, some sequences of the gene are all folded up and they just can't be gotten to, and there's no changes happening there. Whereas other parts of the genome are really, they're constantly mutating and there's flexibility that's happening there. I proposed a hypothesis that the variability of a genetic sequence actually mirrors features in the world.

7.61 For example, humans live at all kinds of different latitudes, and they have different exposure to sunlight. So something like skin color is very easy to change, very easy for that to change. That's extremely variable among humans. But other things like how to digest sugars and how to make your heart beat and stuff like that, it's fundamental stuff. You don't want to change that at all. And so that stuff is really locked in in the genome, and that's not variable at all. I think my speculative hypothesis is that future research might be able to quantify the variability of all kinds of things in life, mental and social and behavioral functions in human life, and put to the test this issue that the most flexible circuits in the brain mirror the most variable parts of the environment.

7.62 Let's come back to this issue about adult plasticity. So older brains have more closed doors. That's what we saw. And that's why Teubers' World War II veterans fared worse if they were older when they got brain damage. And this is why Arnold Schwarzenegger retains a thick accent and so on. In the same way, the older a city is, the more that its infrastructure becomes resistant to shifting. So for example, who's been to Rome? Has anyone been to Rome? OK, great. And who's been to Manhattan? OK, so Manhattan is a very nice grid-like system. Why? Because Manhattan is a very new city as compared to Rome, which is a very ancient city. Rome had these walking paths and certain things that got built and whatever. And as a result, when you go to Rome, you still got all these snaking paths. You don't have some nice grid layout. Rome is unable to untangle its windy roads because too much history has glued these snaking routes into place.

7.63 Just like developing humans. What happens is, just like humans in cities, you get these deepened tracks along early roads. So I've read several quotations from physicists who are maybe in their 30s or 40s who are already lamenting that they can feel that they're losing brain plasticity. And many of you probably know that most, pretty much every physics Nobel Laureate won the prize for work that they did in their 20s. And they can feel that they're already losing flexibility for thinking new things. But despite that, there's plenty of brain plasticity in adults, too. Neural configuration is something that lasts throughout our lives. You're always formulating new ideas. You're accumulating fresh information. You're remembering new people and events.

- 7.64 So despite decreased flexibility, Rome still evolves. So everyone thinks of Rome like this, but in fact, here's the city's business district. The city of Rome right now is not what the city of Rome was 20 years ago. It's got Internet cafés, and it's ringed with cell phone towers, and it's got all the accoutrements of modern life. And so although the basic architecture is difficult to change, the city nonetheless advances all of its finer points to accommodate new business opportunities and circumstances. So that's the analogy, is that there's still plenty of adult plasticity going on, even though the architecture, the deep infrastructure, is hard to change.
- 7.65 So we've seen this throughout this course. For example, when an adult takes up juggling or takes up a new musical instrument or studying for medical exams, any of that stuff, there's adult plasticity to the degree that you can see changes with the naked eye. When the black cab drivers in London learn all the routes of London, there's changes that you can see with the naked eye on their brain scans. And there's a really stunning one. I don't think I've mentioned this yet, but there's a study that's been going on for many decades now called the religious order study, where in Chicago and surrounding areas, there's a bunch of nuns, like thousands of nuns who agreed to donate their brains upon death, so it could be autopsied and examined and so on.
- 7.66 And has anyone heard of this study? This has been going on for a long time. OK, great. So it turns out, a number of the nuns, at autopsy, it was discovered that they had Alzheimer's disease, which means that their brains were physically degenerating. The tissue was physically being ravaged by Alzheimer's disease. And yet nobody knew that when they were alive because cognitively they were fine. Now, this is very weird, right? I mean, no one ever expected you could find someone's brain degenerating, yet they were fine. What was going on here?
- 7.67 What happened is these women lived in the convents till the day that they died. So they had social responsibilities. They had chores. They played games with their fellow sisters. They got in fights with their fellow sisters. They were doing things all the time. They were singing together. They were praying. They were doing all kinds of stuff till the day they died. As opposed to, if you look at people who retire, this is unfortunately not an uncommon story, where people retire and their social lives shrink, and they are at home and they sit on the couch and watch Jerry Springer, and they're not doing anything anymore.
- 7.68 If both people have Alzheimer's disease, what happens is the brain is physically degenerating. In these people, you see real cognitive deficits. But in the nuns, they were constantly building new roadways and pathways because they had all these responsibilities in life. Stuff was happening. And so even as things were falling apart, they were saying, oh, well, I got to build this new one here and here and here. And so it's this extraordinary study, this extraordinary picture that has emerged, about the importance of being cognitively active. That's how you maintain brain plasticity.

- 7.69** And so I want to tell you guys, if you have parents or grandparents that are of a certain age, you have to make sure that they are staying cognitively active, that they're facing new challenges, that they're—I think I might have mentioned this earlier in the thing, but just that they're constantly dealing with challenges that are between frustrating but achievable. And so should they do Sudoku? Sure, until they get good at Sudoku, and then they should drop that and do something else. They should always be doing things where their brain has to do work to figure out what to do.
- 7.70** And by the way, the reason I mentioned about the social lives of these nuns is because it is said that there's nothing harder that brains do than other people, right? Because you never know what the other person is going to say and how they're going to react and so on. So it keeps your brain really active if you're with other people. Unfortunately, a lot of people, when they retire, they stop having social lives. OK, so make sure your parents don't sit around, and your grandparents. This is the single most important lesson that has emerged for cognitive fitness.
- 7.71** Learning can happen in any age, but why is it slower as the brain matures? We've all seen that, if you want to learn a second language at your current age, it's probably going to take you longer to do it. And is this because of the swinging doors shut? Maybe. But I actually think there's another way to look at this about why it takes slightly longer to learn things as you mature. And this has to do with the internal model, which I've mentioned several times before. Your brain locked in silence and darkness is trying to make a model of the outside world out there. And what we talked about last time is that it only makes shifts and changes when something is unpredicted and it's relevant. That's the only time it's going to change its internal model.
- 7.72** So what happens is, as you age and figure out how to operate in the world, your brain is less challenged with novel stimulation because you get it. You know how to do your job. You show up, you do your work, your boss says, "Yeah, you did a fine job," and so on. And so you don't need to make changes. So when you first started, I don't know, your high school, everything was new, and you're looking around, you're trying to figure all kinds of things, social things, where the classrooms are, everything like that. Lots of plasticity going on. But by the time you go to college, you go to this, and you get a job, and whatever, you pretty much have the rules of the world down. You know how it operates.
- 7.73** And that's actually a really good thing because your brain's solidification reflects its success in understanding the world around it. I mean, here's the weird part, is that the fact that you're less flexible means that you're good at what you do. You know how to speak language, you know how to walk, you know how to get food to your mouth, you know how to interact with people to whatever degree, you know how to drive a car, you know how to do all these extraordinary things. You know how to use a cell phone, which, try showing that to somebody 200 years ago. I find these things amazing. I flew

in an airplane over here, and the airplane is making 100,000 corrections in a second based on beach sand that we've melted down and made silicon chips.

7.74 I was thinking about, what if you took a super smart genius from hundreds of years ago, like Isaac Newton, and tried to show him this world, it would blow his mind. He wouldn't have any capacity to even understand the kind of thing that we take for granted. Anyway, the point I want to make here is just that you have become successful at understanding this world and operating in it. You say, "Oh, I think I want to fly to California, so I'm going to go on my cell phone and get a ticket, and I'm going to sit on one of these amazing machines and go over there and whatever." You are able to operate in this world.

7.75 And so, one way you can look at this diminishing plasticity is to think about it as success in understanding the world. And the way I want to really pose this question to you is, would you actually want the plasticity of a child again? Most of us think we would. It would be great to learn a second language so fast and so on. But the fact is, if you had the plasticity of a child, it means you would lose everything you know, all the skills that you have, but also all the people you know, all the memories that you've had. All that would be gone if you were able to just absorb things freshly again like a child.

7.76 You wouldn't be able to read a book. You wouldn't be able to have a meaningful conversation or walk by yourself or get food for yourself. You would be helpless again, just like a little infant. So it's something interesting to think about here. And on this note, one of my favorite quotations from Robert Frost is, "The afternoon knows what the morning never suspected."

7.77 OK, so I'm going to summarize today's lecture. What we saw is that young brains are more flexible than older brains. Young brains have many more connections in the neurons in between, in between neurons, in between regions. And what happens is your experience with the world whittles that down. The world ends up pruning the garden in your head until what you end up with is something that exactly resonates with your world. And so that's how we go from a fluid intelligence where we're able to take in—I mean, you could drop a baby anywhere on the planet or in the future on any planet and maybe you figure it out and figure out how to function well in that world. So that's fluid intelligence. But that's how we go from fluid intelligence to a crystallized intelligence, by whittling down the pathways where we say, "Got it, this is how to operate in this world."

7.78 We also saw that plasticity is not one thing, as in it diminishes at a particular age, but instead, in different areas, it's diminishing at different rates. It gets locked down at different times. Even within something like language, you have the rules of grammar and what language is, all the way to things like accents. And finally, we saw that plasticity, although it diminishes generally, it never comes to an end, which is why you're able to continue to reconfigure your networks. You're here in this class, you're learning new things, you're exposed to new ideas, and this is why you're able to

accumulate fresh information and remember people and events. So thank you for your attention. I'll take any questions.

- 7.79** So you said that when asking for a baby-like mind, we are basically asking to forget everything that have been acquired in the lifetime. People with amnesia, what is happening with their brain? Are they basically forgetting everything that they have a current life or just a neuropath way is just closing?
- 7.80** Great question. It happens that we'll talk about that in the next lecture. But when somebody gets amnesia, what they forget are their autobiographical details, as in, what is my name? Where do I live? But they can still walk and get food to their mouth and operate a cell phone and get on an airplane and things like that. So what you're forgetting is just a little part of it. And hang on to that question until the next lecture, because we'll talk about the different aspects of memory. But the point is, their whole brain hasn't become plastic. They've just lost a very particular way of retrieving information that is in there. And by the way, it remains in there. They're just not able to get at it.
- 7.81** OK, so when it comes to IQ and different types of intelligence like that, what does that mean in neuroscience, in your brain? Is that the fluidity of your brain, or is that more how quickly your brain can crystallize?
- 7.82** Great question. We don't know how to define intelligence in neuroscience. There are lots of hypotheses about it. Some people suggest it's about your ability to squelch distractors and just concentrate on one thing. Some people feel like it has to do with the number of things that you can keep in working memory and manipulate at any time. Some people think it's related to how well you can retain memories to do things with it. Some people think it has to do with how well you can simulate possible futures and evaluate them.
- 7.83** It's likely that the word intelligence is a word with too much semantic weight on it, meaning we can't summarize all these things with a single word. So even though we all feel like, "oh, yeah, that's an intelligent person, and that's not," it's actually a very subtle issue about that. And it may have something to do with brain plasticity, but not in a way that's obvious.
- 7.84** All right. At the beginning of the lecture, you brought up the relationship between the child brain or the recently born child and then the adult. I'm curious, have you guys ever tracked parents and their child when they're born? Because I'm not a parent, but you are. When you had kids, did the world open up to you through the eyes of them? And have you guys ever tracked that, between that duality?
- 7.85** Yeah, very cool question. I can report subjectively that it is a time of lots of learning because you're figuring things out on the fly, even things that you thought maybe you knew from the parenting book or something, but everything's different when you're actually doing it. So it is a novel experience, and in that way, your brain is much more

plastic. And as you're suggesting, you're getting to think about, "Hey, what is it like to see the world that way?", as opposed to this internal model that I've gotten used to.

7.86 People have done these experiments in rodents, for example. And you find that, for example, when a female rat has pups, there's all kinds of changes that happen in her brain hormonally and so on, but also in terms of the cognitive networks. And she becomes better able to do things like multitasking and so on, because this is what parents have to get better at doing, is multitasking. So anyway, there is animal study on that, and I can report subjectively that it is a time when there's just a lot of new things happening.

7.87 Otherwise, we settle into life. We say, "OK, I've got my job. I know how to do my job on the assembly line." And if any of you ever find yourself in a position like that, where you feel like, "Wow, I'm just really doing the same thing over and over, and this year for me was pretty much like last year," figure out what to do differently. I mean, have kids maybe, I don't know, but do something different.

7.88 I wanted to know your opinion on if hypnosis can help us undo an unwanted, practiced pathway?

7.89 I don't know. There are claims about hypnosis. It's not clear to me that it is all that it's cracked up to be. Some people can find themselves in a state of consciousness that's in between your normal consciousness and sleep. This is the state that we would summarize as hypnosis. There's a lot of study about whether the memories that people pull up during, let's say, hypnotic state, just like during a dream state, are real and actual memory. Some people say, "I went in the hypnotic state and I remembered this thing happening to me." Sometimes it's true, sometimes it's not true. Yeah, I don't know. I would say there's a lot of study on hypnosis, and one of my colleagues at Stanford studies this. So it's not as fringy as it once was, but I would say it's not established in the center either.

7.90 I was just wondering if you could give us two or three practical things or examples or ways that things we could do regularly to make sure we maximize our brains' flexibility, although it's decreasing over time.

7.91 Great. OK, so who's wearing a watch here? OK, so take your watch off and put it on your other hand. OK. And I want you to keep it on the other hand. And whenever you want to look at the time, instead of being automatized about just doing that, you think, "Oh, wait. Oh, I have to do something new." The world has a little bit of novelty in it. If you're not wearing a watch, tonight, when you go home, brush your teeth with your other hand. It's not that hard to do, but you're getting out of your routine.

7.92 One thing I've noticed and interviewed some people on is that people towel themselves off after a shower the exact same way every time. It's like a routine that people— try switching that up. Do something new. You know, wherever you work at home, like where your desk is, whatever, it's easy to switch things. Just move your desk over to

the other side of the room. Take two paintings on the wall and just swap them. I mean, it'll literally take you like 30 seconds to do. But do this kind of thing every month or two so that you're constantly changing things up. Because I'm married, it drives my wife crazy when I do this stuff, but she's gotten used to me doing this. This is the stuff that's important.

7.93 The last piece of advice I'd give on this is try to drive a different route home every day. So I do that from work. Every time, I just try to find new roads and new ways of doing it, because you might have noticed that the first time you go, let's say, to your new place of work, it seems like, wow, this is a long drive. And then on successive days, it seems like the time shrinks and shrinks to get there. The time hasn't changed. It's just that you have become an automatized zombie. And so you're just doing life unconsciously at that point. By driving a different route home, I'm not telling you how to live longer. I'm telling you how to make it seem as though you've lived longer, which is that you've got novelty. There's things happening. You're challenging your internal model. You're expanding it. You're expanding your maps of your city and so on. These are the things I would do. Seek novelty.

7.94 I have a question. You've mentioned language several times. And of course, we understand that the younger the person is, the more susceptible they are to learning different kinds of languages. Do you have any recommendations for adults as in learning different languages?

7.95 Yeah. The whole thing comes down to motivation, which is to say, you need some reason why you're doing it. And maybe you can stimulate the reason enough by saying, "Look, I want to have a good, healthy brain," and maybe that's enough for you. But for some people, it's that they get a girlfriend or boyfriend who speaks a foreign language and they want to learn how to do it. And so that's a really successful way to get it. I mean, unless you're already married or something, but that's a really successful way to learn a language.

7.96 Whatever it is, maybe you'll get a job promotion if you speak this new language. Whatever it is, there's got to be something that motivates you. Just like I mentioned with Hugh Laurie, learning how to speak English with a perfect American accent, he was motivated because he was getting paid a bunch of money to have this role on this hit television show. So with proper motivation, you can do anything. That's the key.

Lecture 8: Learning and Memory

At a glance — Learning and memory as core expressions of plasticity.

- 8.1** OK, welcome back. In this session, we're going to talk about one of the main issues about brain plasticity that we haven't touched on yet, which is learning and memory. The key is that you are keeping track of your entire life, who you are, what your name is, where you grew up, where you went to elementary school and high school and whatever. You've got so much memory in your brain, and that is entirely because of brain plasticity, because everything that goes in, your brain is reconfiguring the network and retaining what happens. So that's what we're going to unpack today.
- 8.2** One of the oldest, actually, I think the oldest rule in neurology is called Ribot's law. I'm going to tell you about how that works. But how many of you have ever had, let's say, a grandparent who is losing memory? Who's having problems with memory? OK. What you might have noticed is that they can remember their past quite well. They remember all kinds of things about their childhood and things that happened to them when they were young adults and so on, but they can't remember what they did yesterday or last week or last month.
- 8.3** So somehow the older memories are more stable than the newer memories. And this is Ribot's law, is that older memories are more stable. But this is very weird because it's not how anything else operates. I mean, if you look at any company or university or whatever, what they remember is, OK, here's the changes that happened over the last month, last year, but they can't remember what happened five years ago or 10 or 15 years ago. No other organization works this way, but somehow brains work this way, where older memories stick around longer. So the question is, why do brains do this backwards? Why are older memories more secure?
- 8.4** So it turns out this is a critical clue to understanding what's happening under the hood. And so now we're going to turn to this phenomenon of memory. Just before we get there, does anyone know what Einstein's last words were? OK, turns out nobody does. And the reason is, he died in a hospital in Princeton, New Jersey, and his last words were all spoken in German, which was his native language. The nurse who was on duty didn't speak German, so nobody knows what his last words were.
- 8.5** But it turns out this is a very common thing, that people who speak, let's say, two languages or more, when they're on their deathbed at the end of their lives, they forget their newer languages, and they only remember the languages that they had when they were young. How does memory work? How does all this happen? Did anyone see the movie Memento? If you didn't, it's a good movie. You should see it. Anyway, in this movie Memento, the protagonist gets brain damage, and as a result, he can't form new short-term memories. It's called anterograde amnesia. So he's unable to write down any new memories.

- 8.6** So what does he do? He tattoos information to his skin. So his main tattoo across his chest says, "Find him and kill him." And as he gets more and more clues, he tattoos it because he knows that his window of memory is only going to last about 15 minutes, and then he'll forget. And so he needs to write it down that way. So we are all like this character in the sense that we need to retain information, but the rest of us, instead of tattooing it on our skin, tattoo it into our neural circuitry. And that's how our future selves know what our past self has been through, so that we know what to do next.
- 8.7** Now, how does this work? Well, 2,400 years ago, Aristotle made a first attempt at understanding this process, and he wrote a manuscript, *On Memory*, and he used the analogy of pressing an imprint into a wax seal. Because unfortunately for Aristotle, he was a super smart guy, but he had no data to draw on, so he was stuck with analogies like this. But the magic by which an event becomes a memory in your neural forest, that would remain a mystery for a couple thousand years until very recently.
- 8.8** Here's what we now know. When you learn a new fact, there are physical changes in the structure of your brain. I sometimes think about this as, like when a rock hits the windshield of your car, your car remembers the rock because there's a change in the car as a result of that interaction. This is what happens. Your brain, with everything that it's interacting with, changes and has evidence of what happened. And so for a long time, neuroscientists have been working to figure out what are those changes, how do you orchestrate the stuff across vast seas of neurons? What does this have to do with the neuron shape, the connection strengths, the internal chemicals, all this stuff? How do you embody knowledge this way, and how do you retrieve it decades later? So that's what we're going to talk about today.
- 8.9** So simple forms of learning and memory. People have studied this, for example, in this sea slug. And this sea slug called the aplysia is the subject of lots and lots of labs. Does anyone know why? Because they have big neurons, and so it's easier to get a little electrode on there and see what's going on. So what they do is they study simple things like, you poke the slug gently with a stick, and it withdraws its gill and its siphon. It does this little withdrawal reflex. But if you do this every 90 seconds, the sea slug stops doing the withdrawal because it gets used to it. It remembers that there's nothing harmful about the stimulus, and so it stops reacting to it.
- 8.10** So it's a very basic form of memory. And for example, Eric Kandel won the Nobel Prize for this, for his work on sea slugs, in 2000. But while these experiments have taught us a great deal about the internal chemicals and what can change, how neurons can change in the molecular level, animals that arrived later to the evolutionary game, like mammals, for example, have memory capacities that are way more far-reaching than those seen in invertebrates.
- 8.11** We can remember the details of our autobiographies. We can remember what we have dreamed and imagined. We can remember the spatial details of vast regions of geography. We can acquire complex skills that allow us to navigate economic and social

and academic worlds and climactic conditions and so on. And conveniently, we also have the ability to forget irrelevant minutiae, such as the location of where I parked my car at the airport two trips ago. It just doesn't matter to me, or the exact wording of a conversation and so on. So we balance, actually, remembering almost everything and forgetting things that are irrelevant.

- 8.12** One of the first investigations of how memory really works in mammals began in the 1920s, a guy named Karl Lashley. What he did is, he trained rats to run a maze. So when the rat learns how to run the maze, it has remembered. It has somehow stored in its head exactly where to go and how. Then what he did, he trained a bunch of rats on this, and then he made little cuts in the brains of a number of these rats in different areas, because what he wanted to find was the location of the engram, which is where is the memory stored.
- 8.13** So the question is, where could you find, where could you disrupt that engram, such that the rat forgets how to do it? And it turns out that—he then gave the rats time to recover and then put them back on the maze. And it turns out, all the rats did the maze just fine. So the experiment was a failure, but its failure was its lasting success because what everyone realized was, wow, memory is distributed. It's not in a spot. There's not some spot for memory. It's distributed around the whole brain.
- 8.14** So there's no such thing as a dedicated memory organ, which is what he was looking for. It's not like a filing cabinet. Think of it more like cloud computing, where when I have an email on my Gmail, it's stored somewhere. It's not on my laptop, it's stored in the cloud. And what that means is it's actually distributed on servers all over the planet. And that's what memory is like. How do you write a memory? Let's say I go on a ski trip with my friends and I remember things about it, but how does that get written into neurons? This is the question, is how do you have a programming language that translates the realm of experience into the realm of the physical?
- 8.15** A long time ago, let's say in the 1800s, you guys may or may not know this, but it used to be thought that the brain was what's called a syncytium, which is a continuous network like subway tubes. So think of blood vessels. Your blood vessels are a continuous network, right? And it was assumed that the brain was like that, too. Why? Because if you just look at the brain and the spinal cord, it looks like that. It looks consistent with this thing, like the blood vessels, in that sense. And until the invention of good microscopy, that was always the assumption.
- 8.16** And what happened is this Spanish neuroscientist, Ramon y Cajal, came along, and I mentioned on the first day, he had invented these staining techniques, and he was able to figure out that, actually, it's not a continuous network. It's a bunch of individual neurons, these individual cells, 86 billion of them, that all communicate with each other. And he called this the neuron doctrine, and he won one of the first Nobel Prizes. And so the doctrine ushered in a new question, which is, "OK, that's cool. If brain cells are separate, how are they communicating with each other?" And the answer was

quickly determined that it was these synapses, these connections between the neurons.

- 8.17** And what happened is, as soon as that was discovered about synapses, these connections, Ramon y Cajal was one of the first to suggest that maybe memory is stored in the connection strength between the neurons. By the way, I just want to say as a historical side note, Ramon y Cajal was a photographer, which is how he had these stains. These were just chemical photographic stains, and that's how he stained the tissue and came up with all these neurons and so on. But one thing that I thought interesting that I didn't know is that this was a selfie. He set up his two video. The reason I thought that was interesting is because I always thought selfies were a recent thing, but it turns out since the advent of photography, people have loved doing them.
- 8.18** Ramon y Cajal said, "Look, it's the connections. Maybe that's how you store data." I mentioned a few lectures ago that then, by the late '40s, a guy named Donald Hebb said, "Hey, yeah, maybe you could do that. Maybe you could actually change the connection strength." And that's how you get all these neurons that are responding to the banana, and these all are responding to apple. You strengthen the connection between them, and that becomes the banana assembly and the apple assembly.
- 8.19** And then I mentioned that, by the early '70s, people discovered long-term potentiation and long-term depression, which is a way of saying, you can strengthen the neuron, you can weaken the neuron, you can do it to different degrees. And so all this felt like, wow, this is coming together as a story about how the synapses between neurons can store information. And then by the early '80s, people started coming up with neural networks, artificial neural networks.
- 8.20** So there was a physicist named John Hopfield who was working on this and said, "Look, if I just can change the connection strength between these different units, then you can do things." You can actually train it to recognize a letter of the alphabet. So the letter T gets recognized by a particular constellation of neurons. You train it up on the letter T, you strengthen the connections in the right way. And then what's interesting is that if you present it with a corrupted version of a T, it'll still converge to the T because that's what you've trained it on. That's how you've strengthened these things.
- 8.21** So even if you give it a stimulus that's close, it'll say, "OK, I think that's what I'm recognizing there." And so these networks can be surprisingly robust to degradation, and distributed memories will still be retrieved. I'll just give you an example of some of this early work. So you train it on some pictures, then you give it a degraded version of those things, and it reconstructs it pretty well. These are called attractor states. So the original thing that you've trained it on and the thing that it reconstructs is the attractor state. And even if you give it something, this is just like what I was talking about with, where you have a mountain valley and anything along here, any raindrop is going to flow down to the middle. That's the same idea here.

- 8.22** And this is what humans are really good at, right? So if you're at the beach and you see a shark fin, you don't have to see the whole—you just see the shark fin, you're like, "Oh, that's a shark." I can imagine the whole rest of this visual there. So that's what we're good at doing. And so people got really excited about this stuff. So in general, a Hebbian rule is really good at making associations between events. And I mentioned in an earlier lecture about associative neural networks, which is to say that everything is connected by other experiences in the brain.
- 8.23** So when I go on that skiing trip, my visual system is seeing a lot of snow and white and trees and whatever. And I tasted some hot chocolate, and I had the feel of the boots on my feet, and my friend was there. And there's all these very particular things that happen, and those all get associated with each other. They strengthen the connection between them such that if I think of any part of that, the whole thing comes back together. That's what the power of an associative neural network is.
- 8.24** If you think of your best friend in your best friend's house, all these things can come out of just thinking of any part of that because these are associated with each other. And so even if I draw a very simple picture of your friend's house, that'll generate, oh, the house, the friend, this, the conversation we had, and so on. OK. So that's how you think about associations.
- 8.25** Now, what's happened in recent, really, in recent years, I mean, although this has got a long history to it, but people have been building bigger and bigger artificial neural networks predicated in this idea. The idea is, OK, look, biology is complicated, but let's just assume you've got units and you've got connections between these units and you can change the strength of these. In a sense, artificial neural networks are like a cartoon of biology that have taken off but become very successful. I just want to mention also the rise of artificial neural networks. This isn't because of new theoretical ideas. It's actually quite old theoretical ideas. It's just because of the computing power that we've developed. That's it. That's the whole reason that AI is taking off now and didn't 50 years ago. We have more compute.
- 8.26** Anyway, so people were able to show, even several years ago, you can do incredible things like beat the world champion chess player or the world champion Go player at these games that are extraordinarily complicated, in the sense of, Go has billions and billions of possible moves, but it's trivial to beat the best human in the world at it. But what's interesting is artificial neural networks, while they are massively impressive, they also make weird mistakes. I don't know if you guys have spent a lot of time playing with ChatGPT and others, but you find that, even though sometimes you are blown away by how intelligent it is, sometimes it does things that are just dumb, that a human would never do.
- 8.27** And by the way, people have shown things like, let's say I say, what animal is this? And it says, that's a panda bear. It turns out you can add a very specific set of pixels to that, so that, to a human, it would still look like a panda bear, but now the network will say,

oh, that's a gibbon or something. So you can fool these networks in very funny ways. And by funny, what I mean is, a human would never fall for that. The panda plus those colored pixels still looks like a panda to us, but not to the thing because it's doing these very complicated gradient descent things, and you're choosing those pixels in a particular way.

8.28 The other thing is, although this is getting better, but traditionally, when artificial neural networks are asked to change tasks and do something else, they traditionally fail catastrophically at that sort of thing. People are getting better and better at putting together what's called a mixture of experts, where you have all kinds of things going on. But a three-year-old child can get food to her mouth and walk and socially manipulate adults and do all kinds of things in ways that it's very hard to get artificial neural networks to do.

8.29 And also, I'll just mention that, even though AI does all kinds of things like beat the world champion Go player, it really does best on tasks like this that I've listed. They're cheap and easy to simulate and fully observable and easy to score. It's great at this stuff. If you give it other sorts of video games to play that don't have clear rules and so on, it doesn't do very well on that stuff. So, to understand the magic of what's happening in the brain, that is, to do, the ability for the brain to do things that artificial neural networks cannot, we require a clear-headed look at the challenges and the tricks of real biological memory.

8.30 The first thing we're going to start with is that the enemy of memory in the brain, how you store memories in the brain, it's not about time, it's other memories. What I mean by this is, you've got a long life, and you've got a lot of memory to store in there. And so the question is, how do you do that? This is what's known in the field as the stability-plasticity dilemma. So the problem facing brains is they've got years of stuff that they have to encode, and we have to take on new information over the years and the decades. But the new learning has to be able to protect the old data while it's taking on the new data.

8.31 So in artificial neural networks, what you have is a training phase where you train the whole network, and then you have the recall phase. But real animals don't have that luxury. We don't get to operate that way. Instead, we have to learn everything on the fly through a lifetime. And so what you find in artificial neural networks is if you keep putting more data in, the earlier memory gets all blurred out, and sometimes the whole thing can turn to memory mud. So this is the stability-plasticity dilemma that real brains face, is how do you retain plasticity so you can memorize new things? Oh, that new movie, that new actor, that new book I read, whatever, while still remembering, Oh, yeah, this is where I grew up, these are my parents' names, and so on. How do you change a system constantly while retaining everything?

8.32 Again, you're not protecting against the ravages of time. You're protecting it against other memories that are flowing in. OK, so that's a problem that artificial neural

networks face, but real brains don't face that problem. OK, so why? So the fact that real brains don't have that problem indicates something to us, which is that it's probably more than just strengthening and weakening of synapses. That's probably not the whole picture. Something more is happening, and that's what we're going to see now.

- 8.33** The first solution to the stability-plasticity dilemma involves this thing that I mentioned in the last lecture about turning plasticity on and off in spots where it's relevant. So instead of saying, OK, the whole thing is plastic all the time, you say, OK, look, I've learned this. This is in place, but I do need to change that little piece there, and I do need to change that little piece. And so that's what you get out of something like acetylcholine being released in very specific places. This specificity slows the descent into memory mud because you're not changing the whole thing. You're only changing where something important is happening. That's the only thing you're doing.
- 8.34** OK. And so as a result, you're not treating the brain like a blank slate on which you're writing everything. Instead, experiences turn into memories only when they have relevance to you, especially when they're connected to high emotional states such as fear or pleasure, things like that. So that reduces the chances of overwhelming the network because not everything is getting written down. But this doesn't solve the stability-plasticity dilemma by itself, because there's still plenty of salient memories that you have in your life. You have to worry about storing that. So there's something more that the brain does.
- 8.35** So, as a second solution, it doesn't always hold memories in one place, but actually it passes the information around to other places for more permanent storage. So, one place we know about that is with, for example, a patient named Henry Molaison, who many years ago had this operation where both of his hippocampuses on both sides of his brain were removed. It's a part of the brain, and he was having epilepsy, and so the surgeons removed both of those. Now, the reason they did that is because, in the past, they had done surgeries where they removed one on a patient. The patient was perfectly fine, so they thought there was no problem removing both.
- 8.36** But what happened with Henry was he became unable to form new memories. So this was a big deal where they realized, Oh, my gosh, you really need both hippocampuses to be present in order to form new memories. But here's the important part I want to emphasize. All of his memories from before the surgery were perfectly fine. Instead, he was just like the guy from Memento who simply couldn't form new memories, but everything before was fine.
- 8.37** So what does this tell us? It tells us that the hippocampus is really important for forming new memories, but it's not where memories stay. Instead, somehow the memories are getting passed off somewhere else. And so this is one example of many where things aren't just happening in one spot and staying there, but instead they move along. Just another example is when you learn a skill, you have changes in the area of the brain called the basal ganglia, but that ends up going to the prefrontal

cortex or areas in basal ganglia going to the cerebellum. Details don't matter. All I want to point out here is that you've got things where you have parts of the brain that are absolutely necessary to learn something, but they don't retain the thing over time.

8.38 OK, so what we see here is that parts of the brain teach other parts of the brain. OK, and so you can think about this with the analogy of a warehouse, where the first warehouse doesn't get all filled up, you're moving boxes constantly to other places so you can maintain the space. But if it's simply going to a second warehouse, then that one eventually is in trouble. So I think that's not a full solution to the stability-plasticity dilemma. And that brings us to the trailhead of a deeper solution that I want to suggest here, which is that everybody hoped that looking at synaptic changes would be the Rosetta Stone. Starting with Ramon y Cajal suggesting that, to Donald Hebb suggesting theoretically that, to discovering that synaptic changes can happen, everybody thought, great, that's probably the answer to the whole thing.

8.39 But it now seems clear that's only part of the picture. This is because a cell is as complicated as a big city, and it's got an enormous life going on in the inside. And so people have speculated that maybe the changes in synaptic strength are just ways that cells have to do things to keep the system from blowing up into epilepsy or sinking down into depression. You have to keep things just right so that all the cells can operate without running into trouble at a network level. But there are tons and tons of changes that are happening inside of cells, too, and that's what we're going to look at in just a moment.

8.40 Because inside of cells, you find whole other worlds. And I'm not going to go into detail on this, but if you look at any biology textbook, you'll find, wow, it's really complicated in there. And everywhere we look in biology, what we find are plastic parameters, meaning, adjustable parameters where you say, OK, I can change the number of receptors on the membrane or the way this chemical cascade is going and the way the amount of stuff coming out here and whatever. Everywhere you look, let's say on the inside of a cell, you find the same principles of plasticity where something can change and hold on to that change.

8.41 So if you are an alien, a space alien, discovering humans for the first time, you would be baffled by this complexity from the growth and insertion of newly born neurons into the system, to all the changes all the way down to gene expression changes happening. Plasticity is taking place everywhere along here. And synapses might be the last thing you would notice. With so many degrees of freedom in biological systems, the possibilities are vast for storing memory. So, three reasons to think that synapses might not be enough. One of them is, I mentioned all these things with the jugglers and the violin players and blah, blah, blah. If it were just changes in synapses, we wouldn't expect to see these physical changes that you can see with the naked eye. So something else is going on beyond just that.

- 8.42 Number two is, there appears to be, for sure across mammals, but it appears in humans, too, that there is neurogenesis, which means the insertion of new neurons, the birth of—it's a very small stream, but the birth of these new neurons and insertion into the network. And that would mess everything up if you were just trying to keep the synaptic strength done.
- 8.43 And then the third thing is what's called epigenetics. And we don't need to go into too much detail here, but it turns out that experiences in the world can actually work their way down and change your gene expression. So the genes that you are expressing can be changed based on what happens in your world. But again, without going into details, I just want to say, you wouldn't expect all this kind of thing if you were just tweaking synaptic strengths.
- 8.44 So why does the field concentrate essentially entirely on synapses? And the answer is because that's what we can most easily measure. And so all the rest of the action in the cell is pretty much too tiny for our current technology to get a good picture of the fast dynamics of the living brain. And so that means that, as a field, we're like the drunk looking for the key under the spotlight because that's where the best light is. And the rest of the action going on in the brain is generally, we can't measure it well because it's just so extraordinarily, it's a whole universe going on inside each cell. And so generally, we ignore it and just pretend synapses is the answer.
- 8.45 So I have a suggestion. I have a suggestion which is, how can we think about the interaction of all these pieces and parts? What are the principles by which you can have all these degrees of freedom, and it doesn't just spin out of control? And so I propose that the important issue is not what all these biological pieces and parts are, but instead the time scale on which they operate. And I'll explain what I mean here in the next slide, which is a friend of mine, Stewart Brand, proposed that if you want to understand, let's say, a city, you have to look at multiple layers functioning simultaneously at different speeds. So the fashion in a city changes rapidly, while the businesses in an area, they change a little bit more slowly.
- 8.46 And the infrastructure, like the roads and the buildings, that evolves more gradually. And the rules and laws of the society, the governance, that evolves even more slowly. And culture runs on an unhurried timetable of its own, resting on these deep foundations of story and tradition. And at the slowest scale, nature plods along at the time scale of centuries or millennia. So although it's not always noticed, all these scales interact with each other. And the fastest layers instruct the slower layers with accrued innovations, and simultaneously the slower layers provide checks and structure to the fast layers.
- 8.47 Anyway, the power and resilience of a city doesn't arise from any single level of the system, but instead on their interaction. It's a very cool way of looking at this. He called this pace layering. So I was inspired by this, and I suggested the same thing going on with the brain, which is that brains learn the rules of the world, and then that new data

is in these successively, in the fastest layers, and it works its way down into these layers. And so instead of ranging from fashion to governance to nature, what we can do is we can view the brain's pace layers as ranging from the really fast biochemical cascades all the way down to changes in gene expression.

8.48 And so, if all these flows are linked up correctly, what happens is, you get fast changes going on, and eventually those work their way down into the system. And this is, I think, the way to understand Ribot's law, which is that older memories are getting more and more carved into the system at lower and lower levels, where they become more stable with time. So I'm asserting this, but I want to now give you lots of evidence to explain why this seems to be the case.

8.49 So think about when you write code for an operating system, like what underlies your laptop when you run that. Some of the code consists of really simple basic functions. Let's call that ring zero, that's the kernel, like how you add numbers or how you write something to memory. And that gets established really early on in code. And some of the codes are more sophisticated things, and that's always changing. How do you accommodate input through a USB-C port or something like that? And if you're a company, and then all the way out to the applications, the thing that's on your phone, you have someone enter their name in here, into this form and press the button.

8.50 So, a code base has all this new stuff that rests on top of very old basic stuff. And if you're a company like Microsoft or Apple, there are parts of your code that are so embedded that no one ever looks at them, and you would never want to touch that. Whereas all these other parts of the code on the outside, these are hacked on by programmers all the time and everywhere in between. So let's return to this idea that we talked about in the last lecture, about different time scales of plasticity in different areas. There are basic functions of the brain, like the visual recognition of angles and edges. Think of that like the kernel in programming language. And the applications is all the stuff that remains plastic, all the stuff that's changing, so that parts of the system are more flexible.

8.51 So the idea is that you've got these plastic parameters, by which I mean some areas are super changeable and some are much more conservative, and they take a long time to change. And these are all chained together according to their speed. So let me give you several examples of what I mean here. So first look at this. So the question is, what's going on here? Is this guy drunk? Does anyone have a guess about what's going on here? Is this someone with a disease?

8.52 OK, so what this is, is an astronaut that just returned last year from 197 days in space. And what happens is, when an astronaut returns from a long journey in orbit, they don't just step out and walk around. They've forgotten how to walk. They have to learn how to walk in Earth's gravity almost as though they're learning afresh. But they relearn very quickly. They don't have to recapitulate their infancy. They figure it out. Within an hour, they're walking again. And so his performance right after the flight and his speed

of getting back to walking normally reveals what we call savings in the brain. There are savings that happen, which means, even if you have totally forgotten something, you don't know how to do it anymore, you can relearn it very quickly.

8.53 I'll just give you an example. I knew a guy who had a deaf girlfriend when he was maybe in high school, and so he learned how to speak sign language. And then they broke up and he totally forgot all the sign language. He completely forgot it all and couldn't remember how to do anything. But then 15 years later, he got another girlfriend who was deaf, and he realized he was pretty good at sign language because he thought he had forgotten entirely. Maybe some of you had this experience with a language, where you learn something and you thought, "Well, I've really forgotten that," but you go back to that country, you're like, "I'm pretty good at it." OK, that's the notion of savings, which is to say, somehow it's still stored in the brain. Even though you think, "oh, yeah, I haven't had—" whatever, you're not relearning from scratch when you go back to that second country.

8.54 And this is related to this issue of schema that we talked about earlier. So you remember Destin and his trick bike. He was pretty quickly able to learn, "OK, when I'm on the trick bike, I do this, and when I'm on a normal bike, I do that," and he was able to ride them both. OK, so I mentioned that last time, that he had a schema for each, but we can now understand schema at a deeper level. It's not that you have short-term learning, and the short-term learnings overwrite each other. It's that the programs get written into successively deep layers, depending on the context of what's happening in the brain.

8.55 So after training, Destin has two programs built into the long-term circuitry, this deeper long-term circuitry. And the context of "Which bike am I on?", that just steers the correct path through the circuitry. But if you just were thinking about, how do I build a system with units and connections between the units, then as soon as you learn the trick bike, you would have forgotten and overridden this other thing. But it is written down deep in the circuitry in real brains. In the end, by the way, I think that we can think about plasticity all the way down to the level of species, by which I mean, exceptionally useful programs get burned all the way down to the level of the DNA.

8.56 So consider instincts, which are the behaviors that we don't have to learn. These come about from what I'm calling plasticity at a longer time scale, which is the Darwinian plasticity of species. The idea is that things that prove really useful end up all the way down in the genome so they can get passed along. Let me say one thing, which is that I think the worst semantic mistake that we have made is using the same word memory for both what we do in computers and what we do in brains, because the reason this matters is because it's hard to shake metaphors even when they're badly mismatched.

8.57 And so the key is that human brains don't store memory, anything like computers do. Instead, brains retain and retrieve the memory of a movie, like the last movie you saw, you're not doing it pixel by pixel. You're not saying, "Oh, yeah, this is exactly what

happened." Say you just got some sense of the movie, the gist of the movie, something you liked about it. You remember your favorite stories, not word by word. Many of you are bilingual here. If I tell you a joke in English and you turn to your friend and say it in another language that you speak, you're not storing the joke, phoneme by phoneme. You're storing the gist of the joke. That's what memory is. It's storing something about the meaning of it, but nothing about the specific details of it. And so this is the sense in which I hate the fact that we talk about computer memory because it's a very different sort of thing.

- 8.58** So one of the things that's interesting is that everything you encode is in relation to everything that you've learned before you—concepts, historical, physical, social. Two people can look at a list of dates in Armenian history. But if you have a rich model of Armenian history, then it means something different to you than it does to the other person because new facts get incorporated into your network. For the other person who doesn't really know anything about it and doesn't care, there's no scaffolding on which to attach it.
- 8.59** By the way, I'll just say as a side note, I teach medical students, and often medical students say, "My God, if I learn one more thing, something's going to fall out." But actually, that's not at all how memory works. What we're doing is building scaffolding. We're building ways that new details can hang in to structures, rather than doing, like, the way a computer program would just have to store a file for each thing.
- 8.60** Recall that in the pace layer structure that I proposed, these slow layers provide a framework for the fast layers. So, as a result, early experience becomes foundational. The early experience, as we talked about in the last lecture, essentially provides the architecture on which everything subsequent is built, just like the city of Rome. That provides the architecture, and then you're building things on top of that and on top of that. But the way that you understand the world has a lot to do with the way that you experience the world as a child. OK, but everything new is understood through the filter of the old.
- 8.61** So, what does this mean? Well, one thing, I ended up, I mentioned this during the Q&A on a previous lecture, but one thing this means is that this idea that, in the Matrix, where Trinity calls Tank and says, "I need a pilot program for a B212 helicopter," and he says, "OK, hold on." He looks it up and he sees this whole thing with the helicopter and he uploads it to her brain—that's not going to be something that's plausible in any way that we can see because your brain and my brain, and your brain and your brain and your brain, are completely different because of the architectural foundations, because your brain is like Rome and yours is like Manhattan and so on.
- 8.62** So there's no such thing as like, oh, yeah, here's a little file that tells you how to ride a B212, because for you, you think about an analogy to, OK, it's like a motorcycle and you're doing these things, and you think about it like, oh, it's more like a horse when I

learn how to ride this thing. Everyone's got a totally different foundation upon which they're building the notion of how to fly that helicopter.

- 8.63** The key thing that I want to emphasize about this idea of pace layers, where you've got things that change rapidly and things that are more conservative, and more conservative, and more conservative, and so on, is the interaction between them. I suspect that, as the field moves forward, we're going to discover lots of things that can be understood in this context. So let me give three examples. For example, I mentioned with Lord Nelson who lost his forearm, the primary somatosensory cortex changes rapidly. But downstream from that, it changes more slowly. And downstream from that, it changes even more slowly and so on.
- 8.64** And that's why, I mentioned, we get phantom limb pain. Because what happens is these systems downstream that are changing more and more slowly are saying, "Wait a minute. Doesn't the information from there relate to the hand?" So the primary somatic sensory cortex knows there's no hand there. It doesn't care. It's forgotten about the hand. But the downstream areas say, "Wait a minute, I'm getting information." And that's why with someone with phantom limb pain, let's say someone has lost an arm, you touch their face and they say, "Oh, yeah, I feel that on my thumb. I feel that on my back of my hand," and so on. Because the downstream areas are getting this information, and they are more conservative to change. And so that's why you get these confusions.
- 8.65** Another example is something called hyperthymesia. Has anyone heard of this? OK, cool. It's where a person has essentially a perfect autobiographical memory. They don't forget anything. So the actress Marilu Henner, it turns out, has this. You can ask her any date, and she'll say what the weather was on that day and what she was wearing and who they saw and what she was doing. And so I think when the field of neuroscience gets to a point where it can really understand this phenomenon all the way down, what they're going to see is that this is really about an interaction between the layers, meaning, the interfacing of different layers at an unusual speed.
- 8.66** Because in most brains, what happens is, you learn something and it gets encoded to the next layer only if it's really important. If its message is really important and it keeps saying the same thing, then the more conservative areas say, "OK, fine, I'll take that on." Then if it keeps saying that, then the even more conservative areas say, "OK, fine, I'll take that on," and so on, and it works its way down the system. But my clinical suspicion about hyperthymesia is that, for some reason, her areas at the top are speaking to the more conservative areas sooner than they should, so everything's getting written down in a way that wouldn't normally happen. And it's interesting because, as a side note, it might sound great to remember everything, but as the writer Balzac said, "Memories beautify life, but only forgetting makes it bearable."
- 8.67** OK. Or take synesthesia. How many people have heard of synesthesia? Anyone? OK. OK. So synesthesia is where there's a mixture of the senses. So for example, you might

look at letters or numbers, and that triggers a color experience. And this is something I've studied in my laboratory for about 20 years now. One of the things I did many years ago is, I built online the synesthesia battery so I could quantify—instead of doing this with one or two or three people, I can quantify tens of thousands of people on this. So the way this works is, you see a letter or a number, and you've got a full color palette, 16 million different colors you can choose, and you identify the color that best matches your synesthetic percept. Because for a synesthete, when they see the letter V, it triggers a color experience on the inside. It's just self-evidently true that, for them, V is purple or blue or red or whatever. OK, and by the way, it's different across people.

- 8.68** So what happens is, you find exactly the color that best matches your synesthetic percept, and then you enter that and you get the next letter. And the key is, I give all the letters and numbers at random, three times each. And what I'm looking at is how consistent is your report about your synesthetic color? This is the way to very sensitively separate real synesthetes from people who are trying to fake it, because a real synesthetes always gets triggered the same color, whereas somebody who's—so I ran lots of control subjects where I said, "Look, use free association, use your memory, whatever you can to try to pass this test." People can't pass the test.
- 8.69** OK, so the only point I want to make here is that, what I did is, I quantified lots and lots of synesthetes. Now, so we found something really weird in this data, which is that we found that, although most people's alphabets are totally different colors, there were a lot of people born between the late 1960s through the 1980s who had the same colors. They had the same colors going on. And this was very weird. We examined this in great detail to understand why hundreds of people had the same colors, but only if they were born within a certain time window of each other.
- 8.70** And it turns out, this was because of the Fisher-Price magnet set, which became very popular around that time. And so people were imprinting on the colors of the Fisher-Price magnet set. And by the way, so nobody who was born before the Fisher-Price magnet set came out had that set of colors and then it faded out after the set became not popular anymore. So what that tells us is that it wasn't what caused synesthesia, but the synesthetes were imprinting on that.
- 8.71** So synesthesia, like hyperthymesia, is probably something about the pace layers interacting faster than they should, because for the rest of us, we see a letter A when we're a kid and it's of some color, but then we see another letter A of a different color and another letter A of a different color, and your brain says, "Oh, fine. I guess A isn't associated with any particular color. I'm going to think that that's not colored at all." But for a synesthete, one hypothesis is that they see the colored letter and that sticks all the way down and they say, "OK, got it. As are always purple and Bs are always yellow and Cs are always red," and so on.
- 8.72** So in terms of societies, it would be like the fashionistas got, obtained too much power and pushed their latest fads all the way to the governance layer, or something like that.

So, while hyperthymesia and synesthesia, these are not considered diseases or disorders, they're statistically unusual, though, suggesting that the speed of interaction of the pace layers for most brains has evolved to be optimized in a particular way.

8.73 I'll just mention one other thing, which is not a pathology, but something special, which is that I suspect that breakthroughs occur for people, or they have a breakthrough where they realize, "Oh, you know what? My relationship with my mother, I completely misinterpreted that, or what this person said to me at work, for three years it hurt my feelings because I thought this, and now I realize there's a totally different way to look at that," I think that happens when you have a change at a deeper layer, something that you always believed to be true. And when you discover that's no longer true, that requires revision to deeper layers, which feels very different.

8.74 So now I want to zoom back out to the issue of memory, just talking about memory. I've been speaking as though memory is sort of one thing, but in fact, memory has many different faces. And given the variety of anatomical structures in different regions of the brain and things going on inside cells, it's perhaps no surprise that memory, that there are many different kinds of it. So take a case like Jody Roberts, who in 1985 worked in Washington as a journalist, and one day she vanished, and her loved ones searched everywhere for her. And after many years, they finally resigned themselves to the conclusion that she was probably dead. But she wasn't. It turns out, she had simply forgotten everything about her life, about who she was, where she lived, so on.

8.75 So she ended up adopting a new identity and moved to Alaska and ended up having a life there. It wasn't until 12 years later that someone recognized her, and she was returned to her family, even though she had no recollection of them. But the important thing is, Jody was able to—she was a waitress in Alaska. She started a web development business, stuff like that. She was still able to function fine. She had all these memories of other things. It was just her autobiographical memory that had gone away.

8.76 And by the way, this is a not-uncommon thing. One of the first very famous cases about this was the guy named Ansel Bourne. This was one of the first documented cases. He was a preacher living in Rhode Island. And in 1887, he moved to Pennsylvania as someone else with a different name. And that's because he woke up, he didn't know who he was, he didn't know where he was. He had no memory of his life. And what happened is, two months later, he woke up and remembered everything, and he ended up getting returned home.

8.77 So he was studied by William James at Harvard at this time. William James is one of the great early psychologists on this stuff. But interestingly, Ansel's story was most likely an inspiration for something else. Can anyone guess? The Bourne Identity, the movie that came out, about this kind of amnesia. Anyway, the thing I want to emphasize about stories like Jody's or Ansel's is that they could still remember other things, like how to speak, how to walk, how to socialize. That was all fine. They just couldn't remember

their autobiographies. And these sorts of cases and thousands of others has given us the understanding that there are many kinds of memories.

- 8.78** So on the broadest level, there's short-term and long-term memory. Short-term memory is if you say, "Hey, can you remember this phone number?" And then I go over there, or you say, "Hey, could you get me a coffee? And I want soy milk and not too much," whatever. And I remember it just while I go over there, and then I forget it afterwards. That's short-term. But long-term memory, there are many different types. So long-term memory, I'm thinking about what I did on my birthday last year, that stuff.
- 8.79** We can distinguish all kinds of things within long-term memory, like facts and events, explicit memory. And that, by the way, is what counts as autobiographical memory. Sorry, the events part. The events is the autobiographical memory. And then there's all the implicit memories, your skills, your habits, everything you've learned how to do, walking, riding a bicycle, all these things that you can do, but you can't describe how you do it. So, within the implicit memory, there's lots of different subtypes.
- 8.80** OK, so the simple principle here is that memory is not one thing. There are many, many different aspects to memory. And I'll just give you one notion of one of the challenges that artificial neural networks are facing, which is, sometimes you need to know general things like apples are fruits, and other times you need to know specific things like, oh, there was one red apple in that bunch. So how do you get a single brain to learn both kinds of facts? I mentioned this the other day about long-term statistical learning versus one-shot learning. Artificial neural networks have a very difficult time doing both of these. You can't set it to do both, where it learns generalizations over time and it learns specific facts.
- 8.81** And this apparent incompatibility of goals yields an important clue, which is that the brain needs to have different mechanisms for different time scales of learning. So I won't go into detail here, but there are different parts of the brain, like the hippocampus and the cortex, which are involved in different speeds of learning things. We can go into that in questions if anyone wants. But the trick is, some parts of the brain do rapid learning from individual experiences, like that apple, there's one red apple in the bunch. Other times, it's a slow extraction of statistics like, OK, apples taste good, and they're edible, and so on.
- 8.82** As activity passes through the brain—this is the summary of what we've just seen—it changes the structure. Now, that's a simple summary, but it's incredible that the brain has evolved to be so good at this sort of thing. So you've got this massive system that is physically changing itself to optimally reflect the world that it's embedded in. This is the whole game of brain plasticity, is figuring out, OK, whether I'm playing soccer or I'm learning a fact or whatever, I am changing my structure over this giant forest of 86 billion neurons to reflect the world out there. And these changes must be done in such a way that they can then be retrieved when you need it, and say, "Oh, yeah, that's the

answer to that thing." Sometime in the distant future, maybe 10 years from now, you need that answer.

- 8.83** So one thing that I've asserted in my writing is that progress in the field of memory is going to require the maximally realistic view of what we're trying to explain. So even though artificial neural networks have become very successful just having units and connections, there's much more to be done here. The changes that are underlying actual memory in humans, with all the properties that we get out of human memory, this is probably distributed widely over titanic numbers of neurons and synapses and molecules and genes, and with the temporal linking between these.
- 8.84** That's how I'm going to wrap chapter 8. But what I want to do now is summarize just the main principles that we've learned here. So principle number one that we saw is that brains are not really anything like computers. I gave the example of the way you can squish liveware, stuff like that, or you can remove an entire hemisphere of the brain as long as this is done in the young. And this is just totally different from the way that our technology, that we know how to build technology, where you can damage it or tear half the circuitry out. That would never work. Somehow it works with liveware.
- 8.85** OK, principle two, brains come to the table half-baked and they absorb the world around them. One of the consequences of this is that you need the right experiences. So in a normal environment, neurons grow to look a certain way. If the environment is deprived, the neurons themselves are shrunk. They're not making the connections. In an enriched environment, the neurons are making bigger connections. This is why it really matters to have the right sorts of environment. I gave examples like the Romanian orphanages where the children were not touched or talked to, and they ended up with cognitive deficits.
- 8.86** Principle number three is that brains match themselves to their inputs. So we looked at things like the homunculus, the map of the body and the brain. And what we see is that every bit of real estate, just like in any big city, every bit of the real estate gets used. Nothing sits dormant. Everything gets used. And so this applies to, of course, more than the motor system. Any sensory system, like if you go blind, if you're born blind, the whole visual cortex is taken over. And so, I also suggested, is that this takeover is much faster than we had ever previously suspected, which led me and my student to a whole new hypothesis about why we dream because the visual cortex needs to defend itself against takeover from the other senses when the planet rotates into darkness.
- 8.87** OK, principle number four we saw is that the brain can wrap around new senses. So even though we're used to eyes, ears, nose, and fingertips, you can do all kinds of new things. You can push information into the body via unusual sensory channels. For example, sound pushed in through touch or visual information through sound. And we also saw examples of sensory expansion, where you increase the range of the light frequencies that you're taking in or the sound frequencies, or sensory addition, where you can add completely new senses, like where north is or how to fly a drone.

- 8.88** Principle number five is that the brain figures out how to control whatever body it's in. So we saw lots of examples of things like Matt Stutzman, who's an archer without arms, who has a world record, or Faith, the dog who can walk around, or any of these surfboarding or skateboarding dogs. So what this means is that genes do not prescript how brains talk to bodies. Instead, brains figure it out on the fly. And we saw how someday this will allow us to build extensions of our bodies at a great distance and expand our notion of self.
- 8.89** Principle number six, which I mentioned throughout, is that the cortex is a one-trick pony, which means that, everywhere you look at cortex, it's doing the same thing. And that's why I mentioned this one study where you could take, let's say, visual input and reroute it so it goes in the auditory cortex, and then that becomes visual cortex. What that tells us is that this is a very fluid system where things aren't prescribed, like that has to be visual cortex. Instead, it's just a matter of where the data cables are plugging in and the cortex wraps itself around that.
- 8.90** Principle number seven is that the brain changes based on relevance. You have lots of experiences in the world, but you don't encode everything. You don't encode the number of cracks in the sidewalk or what the person was wearing at the other side of the restaurant or whatever. You're not keeping all that stuff. What you're doing is, you're just encoding what is relevant to you. And one of the examples that we've touched on a lot is this thing about taking on new skills that you have some reason to take on.
- 8.91** So when you take on juggling because you want to impress your girlfriend or boyfriend, or you take on violin because it's important to you to be good at some new thing, whatever, we can see the changes in the brain with the naked eye in brain imaging. So what you do actually changes the structure of your brain. So, just think about this. When you're deciding, "What is it I want to put in there?", think about it as carefully as you might think about diet. Like, OK, do I really want to put in the ice cream or do I want to put in some good protein? Think about what you're taking in the next time you're surfing TikTok or something, if there's something better you could be putting in there.
- 8.92** And by the way, we can see this with great detail in brains when we look at them as we saw with, let's say, a string player versus a pianist. And we saw the clinical relevance of this, which leads to these very simple but probably counterintuitive things, which is, OK, what you do is you take the good arm and you constrain it so you're forced to use the bad stuff. Why? Because it's relevant to use the bad arm. It's relevant to get the food to your mouth or do the things you need to do, make the phone call on your cell phone. And so if it's relevant, then your brain will make the appropriate changes.
- 8.93** Principle number eight is that plasticity lasts through life, but there are several ways in which it diminishes. We looked at sensitive windows in the brain, and we saw things like learning language and then accents and so on. The fact that these different doors

diminish with age and that there are many doors. And one of the lessons that emerged is the way that things get laid down in the brain is like a city which always remains pinned to its original streets and architecture. So this is where I live. And when you look at, 100 years ago, what San Francisco looked like, it's the same layout of the streets. It's just that everything's new and there's all this Silicon Valley high-tech stuff. But fundamentally, the architecture depends on what came before it.

8.94 Principle number nine, we saw that one of the things that makes brains very special is that we write down almost everything in our lives, the events, the facts, the skills, the statistics of what we've experienced. And somehow this involves changing the structure of what's going on in there so you can retrieve that information later, but still keeping the whole thing functioning and not overwriting what was in there before. So instead of tattooing information on the skin, we are essentially tattooing it on the inside. And we saw that changes in the strength of connections is going to play a big role, but we actually don't know if that's the whole story, because the truth is that's all we can measure at the moment. And so stay tuned as the field of memory expands.

8.95 Principle number 10 is the one I haven't talked about, but I'll just talk about this for 60 seconds, which is, I think, why does any of this matter? One of the reasons is, we can actually build a whole new generation of devices predicated on the principles that you've learned here in these last eight lectures. So what we see is that the brain is doing something so differently than the way we've ever even thought about building our technology. And so I'm very enthusiastic about this idea of figuring those principles out and building this.

8.96 I'll give you an example. This is the Mars rover Spirit, which is a multibillion dollar device that went up to Mars, did a great job, rolled around, did all kinds of stuff. But eventually it got its right front wheel stuck in the Martian soil, and it couldn't get out, and it died there. It beamed its swan song to Earth and then shut itself off, and that's the end of it. It's a billion-dollar piece of space trash now. But if you contrast that with what happens with an animal who, let's say, gets its leg caught in a trap, what does it do? It chews its leg off and then it figures out how to walk on three legs. Now, it wasn't meant to walk on three legs. It wasn't preprogrammed to know how to do that. Just figures it out. Why? Because it's relevant for it. It has to get back to its pack. It has to get to its food. It has to get to its mate, all this stuff.

8.97 And so it just figures out how to rewrite its circuitry to get where it needs to go. And I would say the wolf carries on with the limp because animals don't shut down with moderate damage, and neither should our machines. So I think we're about to enter an era where we can rethink the whole foundation of Silicon Valley, where you have trim and efficient hardware, and trim and efficient software, and that's what you've got going, and instead make something that is liveware.

- 8.98** So that's all I'm going to say. That's what I wanted to tell you about brain plasticity. And there are literally tens of thousands of papers in this field, and my goal has been to distill this down to principles for you, and I thank you for your attention.
- 8.99** Can you explain a little bit more about why it's not optimal to have an increased speed between those pace layers?
- 8.100** The reason it's not optimal is because what you want is not to burn things into the system unless they're really important. So you see a gazillion things a day that just don't matter. As I'm talking to you guys, there's a garage door over here, there's a chair there, there's a person there, there's whatever. None of that stuff needs to be remembered by me a year from now. And so what I want to do is, all that stuff gets retained at the first layer that says, well, there's all kinds of changes and things I should know. But in order for the more slow-changing layers to take that on, it needs to be proven that that's important.
- 8.101** And then there's the next layer that says, OK, yeah, that really is important because these guys have been talking about that for a while. So I'm going to code that. And then these next guys say, well, these guys have been talking about it for a while. That's the idea of having pace layers and only pushing down the stuff that matters. Essentially, it's a filtration system. Think about it that way.
- 8.102** I'm just wondering if it's possible to rewrite or replace your memory on conscious level, for example, by maybe replaying an alternative scenario of how events happen, like every night before going to sleep for several months. Is there a way to trick your brain to believe this alternative scenario is the real one?
- 8.103** One is that memory is actually not that good. And we see this, for example, with eyewitness testimony. One of the areas that I spend a lot of time is at the intersection of neuroscience and the legal system. And eyewitness testimony is famously terrible because people misremember stuff for lots of reasons. In fact, one of my colleagues did an amazing study right after September 11, 2001, where she interviewed a bunch of people right after about what they saw. And she was clever enough to also ask them to tell her memories about September 10, what they had done that day. And then a year later, she tracked down all those people again and interviewed them again.
- 8.104** And it turns out that all the memories had drifted and they're thinking about, "So I'm standing on the corner, I saw this," and blah, blah, blah. It had changed from what they'd said a year ago. And by the way, it was true with the banal memories, too, from September 10. So all memory is equally lousy, whether it's about a big important event or something unimportant. Memories drift. Memories get changed by new information that comes in. For example, let's imagine you had dinner with some married couple that you know, and then a few months after that, they get divorced. And then when you think about that dinner a year later, you think, "Oh, yeah, I saw some signs there."

But maybe it's true or not true. It's just you have new information now, and so you interpret things in that context. OK, so memory is far from perfect.

- 8.105** Now, as for your second question about whether you could change a memory on purpose, I don't know that anyone's ever tried to do that, so I don't have the data on that, except that I would say people often want to get rid of traumatic memories. And there's a very weird footnote here, which is that it turns out that some memories are actually erasable, meaning, if you remember—let's imagine that I asked you to remember the name of your fifth-grade teacher, and then, God forbid, let's say, something fell off the ceiling and hit you in the head. You would actually forget that memory because somehow it was stored in deep cold storage, and you've now brought it out to make it active, and you need to restore it. And if you get hit in the head while it's in this active form, it's gone. You didn't reconsolidate the memory.
- 8.106** Now, that's a very weird thing, but true, that you have to reconsolidate memories for them to be in there. So you can lose memories that way. It turns out, though, that that's not true with traumatic memories, which is the thing that everyone would want to lose. Those get stored essentially using a secondary memory track involving the amygdala such that those are unerasable.
- 8.107** So in terms of trying to remember, let's say you're in this particular moment. It's not necessarily a skill that you want to remember, but let's say you're in a particular moment and you think to yourself like, "Wow, I really want to be able to remember this and recall this in the future." Is there a way to do that?
- 8.108** Yes. Relevance. You just have to figure out why you care about that moment and what it means to you. So obviously, let's imagine that this is a crime scene where we're standing right now. I might really think, OK, I need to remember who's sitting where and blah, blah, blah, because let's say I'm a detective and that's my job, or because I'll get a promotion in the police agency if I'm the one who cracks the case. There's going to be some relevance to it. But yes, you can pay attention.
- 8.109** The thing we don't do most of the time is pay attention, because as I've said in several places in these lectures, we've got this internal model. And all you're generally trying to do is get by in the world with your internal model. Like, yeah, it's good enough. I know where I am. I don't need to worry about all the details around here. But we all notice different things based on our internal model. So if I'm an HVAC repair man, I'll notice the ducts up there. And if I'm a lighting specialist, I'll notice the lights. If I'm a camera person, I'll notice, "Oh, that's the XJ95 camera," whatever. And we all are noticing different things based on our expertise and based on what is relevant to us.
- 8.110** But in answer to your question, you can do better at seeing the world by forcing yourself to pay attention. All you need to do is come up with a trick of why it's relevant to you, why it matters. Maybe it's as simple as you want to be the kind of person who remembers this moment. But that's what you need.

- 8.111** A lot of people claim that it is better to study in the morning than to study any time in the day. Does learning a new skill or a new idea depends on the time in the same context? A lot of people struggle with speech compared with understanding and listening to anything. What is the problem with that? Why they struggle with it?
- 8.112** OK, so two things. The issue about studying in the mornings, I would tell you the answer to that, but I studied the answer in the afternoon, so I've forgotten. The answer, it's not actually true. You can study anytime. It's based on when you are motivated and relevant and awake and whatever. Some people are clearly night people, some people are morning larks and everywhere in between. I don't think that's a truism generally about studying in the morning. It's just a matter of what matters to you and how you write stuff down that way. There's all kinds of claims like this about creativity, about whatever. Most of them are totally made up with no scientific basis to them.
- 8.113** OK. As far as why—were you asking a question why some people have difficulty understanding speech?
- 8.114** So they can understand the concept so clearly, but they cannot retrieve it in verbal terms. They cannot explain it.
- 8.115** Oh, they can't express it. Oh, well, OK. So that's true of so many things in life. And that has to do with this distinction between implicit memory and explicit. Explicit simply means you can articulate it. Implicit memory means, I can't really explain it, like riding a bicycle. I'm really good at it, but I really couldn't tell you how to do it. In fact, there's a lot of learning that happens like that. I'll just give you one quick example, like plane spotting, which was a big thing in World War II, where you had British who would stand on rooftops and look at a distant plane and say whether that was a German aircraft or British aircraft. And so the government really wanted to find more of these plane spotters, but it turns out you can't really train somebody on this.
- 8.116** And so the only way that they realized they could train people is if you have the plane spotter and the person who wants to learn, and the person who wants to learn makes a guess, and the plane spotter says yes or no. And it's just this feedback, but it's impossible to explain, well, you listen for this, and it's the shape of the—it's too hard. But if you just give lots of feedback to the student, the student will get it and be able to do it. There are lots of examples like that in the world. I would say probably most of your skills in life are inarticulable. You could not explain to me how you walk.
- 8.117** I don't know if you guys have seen all the robot videos. For decades, it was really hard to get a robot to walk without falling over and stuff. And that's because, although we do it all the time, it's a really super complicated thing that we're doing. So that's why we can often understand something but not articulate it.
- 8.118** In the case of Jody that you were talking about, how do these individuals make sense of the common sense information that they're missing? So if someone were to ask them

about their family, do their brains fill in the blanks with false information? Do they deem it irrelevant? How would that work?

8.119 Such a good question. No, instead, they just feel like, "You know what? I can't remember. I just don't know." And so, Jody ended up getting remarried to somebody in Alaska, and she had a family. But I don't know what sort of social decision she made about that. If the husband asked her, "Oh, so where'd you grow up?" And she thinks, "I really don't know," she might, for social reasons, fib about it and say something, but as far as her understanding of it, it would just be sort of a blank.